T2K 6.3933×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis

Costas Andreopoulos[a], Thomas Dealtry[a,b], Lorena Escudero Sánchez[c], Nick Grant[d], and Alfons Weber[a,b]

[a] *Particle Physics Dept., Rutherford Appleton Laboratory, STFC*
Harwell Oxford Campus
Oxfordshire OX11 0QX, UK
[b] *Subdepartment of Particle Physics, University of Oxford*
Denys Wilkinson Building, Keble Road
Oxfordshire OX1 3RH, UK
[c] *Instituto de Fisica Corpuscular*
CSIC and University of Valencia
Valencia, Spain
[d] *Physics Dept., Lancaster University*
Lancaster, LA1 4YB, UK

Abstract

We report the results of the 3-flavour ν_μ-disappearance analysis, performed by the *VaLOR* (Valencia-Lancaster-Oxford-Rutherford) group, on the combined Run 1+2+3+4a dataset. This analysis is an update of the work presented in Refs. [1], [2], [3] and [4] with improved methods and an enlarged dataset. The Run 1+2+3+4a dataset corresponds to an integrated J-PARC neutrino beam exposure of 6.3933×10^{20} POT.

This analysis predicts 421.76 ± 23.33 (syst) 1 μ-like ring events in SuperK in the absence of any oscillation, but only 115 were observed. The observed deficit has a strong energy dependence; the ratio of observed to expected, under the no-oscillation hypothesis, is ∼27% < 0.5 GeV, ∼10% between 0.5 and 1 GeV and ∼73% > 1 GeV.

A ν_μ-disappearance analysis was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter. The observed reconstructed energy spectrum of 1 μ-like ring events was fitted allowing $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ to float δ_{CP} was fixed at 0 and a mass hierarchy was assumed[1]. The other *non-23-sector* oscillation parameters, and all 41 systematic parameters considered in this analysis, were also allowed to float in the fit.

The 3-flavour ν_μ-disappearance fit to the combined Run 1+2+3+4a dataset, assuming the normal mass hierarchy, yields $|\Delta m^2_{32}| = 2.463\times10^{-3}$ eV^2/c^4 and $\sin^2\theta_{23} = 0.514$ as the best-fit values (p-value = 0.88). The fit assuming the inverted mass hierarchy yields $|\Delta m^2_{32}| = 2.530\times10^{-3}$ eV^2/c^4 and $\sin^2\theta_{23} = 0.511$ as the best-fit values (p-value = 0.89).

Confidence regions in $\sin^2\theta_{23}$ - $|\Delta m^2_{32}|$ space are shown using both the constant-$\Delta\chi^2$ method and the method of Feldman and Cousins. Using the profile $\Delta\chi^2$ distributions obtained from the fit the 90% (68%) CL allowed regions for $|\Delta m^2_{32}|$ and $\sin^2\theta_{23}$ can be summarized as 2.30×10^{-3} (2.35×10^{-3}) eV^2/c^4 < $|\Delta m^2_{32}|$ < 2.63×10^{-3} (2.57×10^{-3}) eV^2/c^4 and 0.433 (0.456) < $\sin^2\theta_{23}$ < 0.594 (0.572) for the normal mass hierarchy, and 2.37×10^{-3} (2.42×10^{-3}) eV^2/c^4 < $|\Delta m^2_{32}|$ < 2.70×10^{-3} (2.64×10^{-3}) eV^2/c^4 and 0.431 (0.454) < $\sin^2\theta_{23}$ < 0.591 (0.569) for the inverted mass hierarchy.

[1] Two independent fits were performed. The first assumed the normal mass hierarchy. The second assumed the inverted mass hierarchy.

Contents

1 Introduction **9**

2 Summary of changes compared with the Run 1+2+3 analysis in T2K-TN-141(v8) **10**
2.1 Changes in construction of the nominal single-ring μ-like reconstructed energy spectrum p.d.f 10
2.2 Changes in the systematic effects considered by the analysis 10
2.3 Changes in neutrino oscillation fitting 10

3 Summary of systematic parameters **11**

4 Updated predictions of single-ring μ-like event reconstructed energy spectrum **15**

5 Updated estimates of the effect of systematic uncertainties in the prediction of the single-ring μ-like event reconstructed energy spectrum **23**

6 Results of 6.3933×10^{20} POT fitter validation **27**

7 Results of the ν_μ-disappearance analysis with the Run 1+2+3+4a (6.3933×10^{20} POT) dataset **35**
7.1 Best-fit oscillation parameters and systematic parameter pulls 35
7.2 Goodness-of-fit tests 35
7.3 Best-fit single-ring μ-like reconstructed energy spectra 35
7.4 Confidence regions 47
7.5 Confidence region comparisons 55

8 Summary **58**

List of Figures

1 Predicted number of 1-ring μ-like events, as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, for an exposure of 6.3933×10^{20} POT. The number of events was calculated using the BANFF-tuned MC templates. Values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} and δ_{CP}=0. The numbers shown were generated assuming the normal hierarchy. The T2K Run 1+2+3 90% CL region is superimposed for reference. 18

2 Predicted reconstructed-energy spectrum of 1-ring μ-like events, and contributions from various true neutrino reaction modes, for no oscillations and for an exposure of 6.3933×10^{20} POT. The 32 components of the spectrum are calculated separately in the actual analysis, but, for this plot, are grouped into just 5 categories: ν_μ CCQE, ν_μ CCnonQE, $\bar{\nu_\mu}$ CC, $\nu_e/\bar{\nu_e}$ CC and NC. The spectrum was generated using the BANFF-tuned MC templates. 18

3 Predicted reconstructed-energy spectrum of 1-ring μ-like events, and contributions from various true neutrino reaction modes, for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4 and for an exposure of 6.3933×10^{20} POT. The spectrum was generated using the BANFF-tuned MC templates. Values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} and δ_{CP}=0. The spectrum shown was generated assuming the normal hierarchy. Note that the vertical axis is zoomed in by a factor of 4 compared with Fig. 2. 18

4 Fractional difference in the predicted number of 1-ring μ-like events as a result of applying the NA61 flux tuning to the nominal MC templates. This number is shown as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ for an exposure of 6.3933×10^{20} POT. Values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21},and δ_{CP}=0. The numbers shown were generated assuming the normal hierarchy. The T2K Run 1+2+3 90% CL region is superimposed for reference. . . . 19

5 Fractional difference in the predicted number of 1-ring μ-like events as a result of applying the BANFF flux and cross-section tuning to the NA61 flux tuned MC templates. This number is shown as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ and for an exposure of 6.3933×10^{20} POT. The values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The numbers shown were generated assuming the normal hierarchy. The T2K Run 1+2+3 90% CL region is superimposed for reference. 19

6 Reconstructed-energy spectrum of 1-ring μ-like events, for an exposure of 6.3933×10^{20} POT, both with and without the effect of the NA61 flux tuning of the nominal MC templates. The spectra are shown both for no oscillations (left) and for oscillations with $\sin^2\theta_{23}$=0.5, $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4, the 2012 PDG values for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0 (right). Results shown for the normal hierarchy. 19

7 Reconstructed-energy spectrum of 1-ring μ-like events, for an exposure of 6.3933×10^{20} POT, both with and without the effect of BANFF flux and cross-section tuning on the NA61-tuned flux MC templates. The spectra are shown both for no 23-sector oscillations (left) and for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4 (right). In all spectra, the 2012 PDG parameters are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} and δ_{CP}=0. Results shown for the normal hierarchy. 22

8 Maximum value of the fractional difference in the predicted number of 1-ring μ-like events between varying each of $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21} and δ_{CP} within their allowed ranges and their nominal values (these are the values shown in Tab. 3 for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0). The fractional difference is shown as function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, for an exposure of 6.3933×10^{20} POT. The number of events was calculated using the BANFF-tuned MC templates. The normal hierarchy was assumed. The T2K Run 1+2+3 90% CL region is superimposed for reference. 22

9 Fractional difference in the predicted number of 1-ring μ-like events between assuming the normal hierarchy and assuming the inverted hierarchy. The fractional difference is shown as function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, for an exposure of 6.3933×10^{20} POT. The number of events was calculated using the BANFF-tuned MC templates. The values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The T2K Run 1+2+3 90% CL region is superimposed for reference. 22

10 Total error envelope of the 1 μ-like ring event reconstructed energy spectrum, for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4. The values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The normal mass hierarchy is assumed. The numbers shown were calculated for an exposure of 6.3933×10^{20} POT. BANFF-tuned MC templates were used. The error envelope was calculated as the $\pm1\sigma$ spread of bin contents using an ensemble of 100k toy MC experiments generated with randomized systematic parameters. All systematic parameters were considered and their correlations were taken into account. 25
11 As in Fig. 10, but considering only SuperK detector + FSI + SI systematics. 25
12 As in Fig. 10, but considering only BANFF systematics. 25
13 As in Fig. 10, but considering only uncorrelated cross-section systematics. 26
14 As in Fig. 10, but considering only *non-23-sector* oscillation parameter uncertainties. 26
15 Residual distributions for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23}$ = 0.37 and $|\Delta m^2_{32}|$ = 2.4×10^{-3} eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 28
16 Residual distributions for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ 5.1k toy MC experiments at 7.8×10^{21} POT (the T2K ultimate POT goal), generated at $\sin^2\theta_{23}$ = 0.37 and $|\Delta m^2_{32}|$ = 2.4×10^{-3} eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 28
17 Residual distributions for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23}$ = 0.513 and $|\Delta m^2_{32}|$ = 2.4×10^{-3} eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 28
18 Residual distributions for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ 5.1k toy MC experiments at 7.8×10^{21} POT (the T2K ultimate POT goal), generated at $\sin^2\theta_{23}$ = 0.513 and $|\Delta m^2_{32}|$ = 2.4×10^{-3} eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 28
19 Residual distributions for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23}$ = 0.65 and $|\Delta m^2_{32}|$ = 2.4×10^{-3} eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 28
20 Residual distributions for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ 5.1k toy MC experiments at 7.8×10^{21} POT (the T2K ultimate POT goal), generated at $\sin^2\theta_{23}$ = 0.65 and $|\Delta m^2_{32}|$ = 2.4×10^{-3} eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 28
21 Residual distributions for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21}, and $f^{banff}_{0;t,r}$-$f^{banff}_{8;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23}$ = 0.37 and $|\Delta m^2_{32}|$ = 2.4×10^{-3} eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 29
22 Residual distributions for $f^{banff}_{9;t,r}$-$f^{banff}_{20;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23}$ = 0.37 and $|\Delta m^2_{32}|$ = 2.4×10^{-3} eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 29
23 Residual distributions for $f^{banff}_{21;t,r}$-$f^{banff}_{22;t,r}$, and the first 10 uncorrelated cross section parameters, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23}$ = 0.37 and $|\Delta m^2_{32}|$ = 2.4×10^{-3} eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 29
24 Residual distributions for $f_{CC\bar{\nu};t}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{0;r}$-$f^{SK+FSI}_{5;r}$ computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23}$ = 0.37 and $|\Delta m^2_{32}|$ = 2.4×10^{-3} eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 29
25 Pull distributions for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21}, and $f^{banff}_{0;t,r}$-$f^{banff}_{8;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23}$ = 0.37 and $|\Delta m^2_{32}|$ = 2.4×10^{-3} eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 30
26 Pull distributions for $f^{banff}_{9;t,r}$-$f^{banff}_{20;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23}$ = 0.37 and $|\Delta m^2_{32}|$ = 2.4×10^{-3} eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 30
27 Pull distributions for $f^{banff}_{21;t,r}$-$f^{banff}_{22;t,r}$, and the first 10 uncorrelated cross section parameters, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23}$ = 0.37 and $|\Delta m^2_{32}|$ = 2.4×10^{-3} eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 30

28 Pull distributions for $f_{CC\bar{\nu};t}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{0;r}$-$f^{SK+FSI}_{5;r}$ computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.37$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 30
29 Residual distributions for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21}, and $f^{banff}_{0;t,r}$-$f^{banff}_{8;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 31
30 Residual distributions for $f^{banff}_{9;t,r}$-$f^{banff}_{20;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 31
31 Residual distributions for $f^{banff}_{21;t,r}$-$f^{banff}_{22;t,r}$, and the first 10 uncorrelated cross section parameters, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 31
32 Residual distributions for $f_{CC\bar{\nu};t}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{0;r}$-$f^{SK+FSI}_{5;r}$ computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 31
33 Pull distributions for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21}, and $f^{banff}_{0;t,r}$-$f^{banff}_{8;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 32
34 Pull distributions for $f^{banff}_{9;t,r}$-$f^{banff}_{20;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 32
35 Pull distributions for $f^{banff}_{21;t,r}$-$f^{banff}_{22;t,r}$, and the first 10 uncorrelated cross section parameters, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 32
36 Pull distributions for $f_{CC\bar{\nu};t}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{0;r}$-$f^{SK+FSI}_{5;r}$ computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 32
37 Residual distributions for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21}, and $f^{banff}_{0;t,r}$-$f^{banff}_{8;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 33
38 Residual distributions for $f^{banff}_{9;t,r}$-$f^{banff}_{20;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 33
39 Residual distributions for $f^{banff}_{21;t,r}$-$f^{banff}_{22;t,r}$, and the first 10 uncorrelated cross section parameters, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 33
40 Residual distributions for $f_{CC\bar{\nu};t}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{0;r}$-$f^{SK+FSI}_{5;r}$ computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 33
41 Pull distributions for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21}, and $f^{banff}_{0;t,r}$-$f^{banff}_{8;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 34
42 Pull distributions for $f^{banff}_{9;t,r}$-$f^{banff}_{20;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 34

43 Pull distributions for $f^{banff}_{21;t,r}$-$f^{banff}_{22;t,r}$, and the first 10 uncorrelated cross section parameters, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 34
44 Pull distributions for $f_{CC\bar{\nu};t}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{0;r}$-$f^{SK+FSI}_{5;r}$ computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. 34
45 Systematic parameter pulls for the fit to the Run 1+2+3+4a dataset. The normal mass hierarchy is assumed. 37
46 Systematic parameter pulls for the fit to the Run 1+2+3+4a dataset. The inverted mass hierarchy is assumed. 38
47 Distribution of χ^2_{gof}, the goodness-of-fit (gof) χ^2, from 1k toy MC experiments generated at the Run 1+2+3+4a best-fit oscillation point. The χ^2_{gof} value for the Run 1+2+3+4a data (χ^2_{data}) is highlighted. The normal mass hierarchy is assumed. 39
48 Distribution of χ^2_{gof}, the goodness-of-fit (gof) χ^2, from 1k toy MC experiments generated at the Run 1+2+3+4a best-fit oscillation point. The χ^2_{gof} value for the Run 1+2+3+4a data (χ^2_{data}) is highlighted. The inverted mass hierarchy is assumed. 39
49 Best-fit reconstructed energy spectrum (top) obtained from the fit to the combined Run 1+2+3+4a dataset. The Run 1+2+3+4a dataset and the prediction obtained from the no-oscillation hypothesis are also shown. The spectra are shown using the 73 bins used in the fit (see Sec. 2.2.5 in Ref. [4]). The ratio of the data and best-fit distribution to the no-oscillation hypothesis is shown in the bottom plot. The characteristic energy-dependent deficit can be clearly seen. The normal mass hierarchy is assumed. 40
50 Best-fit reconstructed energy spectrum (top) obtained from the fit to the combined Run 1+2+3+4a dataset. The Run 1+2+3+4a dataset and the prediction obtained from the no-oscillation hypothesis are also shown. The spectra are shown using the 73 bins used in the fit (see Sec. 2.2.5 in Ref. [4]). The ratio of the data and best-fit distribution to the no-oscillation hypothesis is shown in the bottom plot. The characteristic energy-dependent deficit can be clearly seen. The inverted mass hierarchy is assumed. 41
51 Best-fit reconstructed energy spectrum (top) obtained from the fit to the combined Run 1+2+3+4a dataset. The Run 1+2+3+4a dataset and the prediction obtained from the no-oscillation hypothesis are also shown. The spectra are shown using the coarse binning scheme discussed in the text (see Sec. 7.3). The characteristic energy-dependent deficit can be clearly seen. The normal mass hierarchy is assumed. 42
52 Best-fit reconstructed energy spectrum (top) obtained from the fit to the combined Run 1+2+3+4a dataset. The Run 1+2+3+4a dataset and the prediction obtained from the no-oscillation hypothesis are also shown. The spectra are shown using the coarse binning scheme discussed in the text (see Sec. 7.3). The characteristic energy-dependent deficit can be clearly seen. The inverted mass hierarchy is assumed. 43
53 Best-fit reconstructed energy spectrum in the 73-bin scheme used in the fit (see Sec. 2.2.5 in Ref. [4]) and its components obtained from the fit to the Run 1+2+3+4a dataset. The 32 components, each of which is calculated separately, are grouped into just 5 categories: ν_μ CCQE, ν_μ CCnQE, $\bar{\nu_\mu}$ CC, $\nu_e/\bar{\nu_e}$ CC and NC. The normal mass hierarchy is assumed. . . . 44
54 Best-fit reconstructed energy spectrum in the 73-bin scheme used in the fit (see Sec. 2.2.5 in Ref. [4]) and its components obtained from the fit to the Run 1+2+3+4a dataset. The 32 components, each of which is calculated separately, are grouped into just 5 categories: ν_μ CCQE, ν_μ CCnQE, $\bar{\nu_\mu}$ CC, $\nu_e/\bar{\nu_e}$ CC and NC. The inverted mass hierarchy is assumed. . . 44
55 Comparison between best-fit reconstructed energy spectra obtained from the normal and inverted mass hierarchy fits to the Run 1+2+3+4a datasets. The spectra are shown using the 73 bins used in the fit (see Sec. 2.2.5 in Ref. [4]). Also shown are the absolute difference between the normal and inverted mass hierarchy best-fit spectra and the ratio between them. 45

56 Comparison between best-fit reconstructed energy spectra obtained from the normal and inverted mass hierarchy fits to the Run 1+2+3+4a datasets. The spectra are shown using the coarse binning scheme discussed in the text (see Sec. 7.3). Also shown are the absolute difference between the normal and inverted mass hierarchy best-fit spectra and the ratio between them. 46
57 $\Delta\chi^2$ surface for the fit to the Run 1+2+3+4a dataset. The normal mass hierarchy is assumed. 49
58 $\Delta\chi^2$ surface for the fit to the Run 1+2+3+4a dataset. The inverted mass hierarchy is assumed. 49
59 Profile $\Delta\chi^2$, as a function of $|\Delta m^2_{32}|$, for the fits to the Run 1+2+3+4a dataset assuming normal and inverted mass hierarchy. The 68% and 90% constant-$\Delta\chi^2$ values are shown as horizontal dotted lines. 50
60 Profile $\Delta\chi^2$, as a function of $\sin^2\theta_{23}$, for the fits to the Run 1+2+3+4a dataset assuming normal and inverted mass hierarchy. The 68% and 90% constant-$\Delta\chi^2$ values are shown as horizontal dotted lines. 50
61 68% CL and 90% CL allowed region (including systematics), constructed with the constant-$\Delta\chi^2$ method, for the fit to the Run 1+2+3+4a dataset. The best-fit point is also shown. The normal mass hierarchy is assumed. 51
62 68% CL and 90% CL allowed region (including systematics), constructed with the constant-$\Delta\chi^2$ method, for the fit to the Run 1+2+3+4a dataset. The best-fit point is also shown. The inverted mass hierarchy is assumed. 51
63 Monte Carlo estimation, according to the method of Feldman and Cousins [31], of $\Delta\chi^2_{critical}$ values $\Delta\chi^2_{68}$ (left) and $\Delta\chi^2_{90}$ (right) as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, for calculation of the 68% and 90% CL confidence regions for the 6.3933×10^{20}-POT 3-flavour ν_μ-disappearance analysis. The surfaces were generated from an ensemble of toy MC experiments including both randomized systematic variations and statistical fluctuations. As expected from [31], the values of $\Delta\chi^2_{critical}$ decrease as the limit of the model at $\sin^2\theta_{23}\approx0.513$ is approached. The normal mass hierarchy is assumed. It should be noted that $\Delta\chi^2_{critical}$ is not calculated for every combination of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, due to us being efficient with computing time. . . 52
64 Monte Carlo estimation, according to the method of Feldman and Cousins, of $\Delta\chi^2_{critical}$ values $\Delta\chi^2_{68}$ (left) and $\Delta\chi^2_{90}$ (right), as a function of $\sin^2\theta_{23}$ for 3 different values of $|\Delta m^2_{32}|$ (2.2×10^{-3} eV$^2/c^4$, 2.4×10^{-3} eV$^2/c^4$ and 2.7×10^{-3} eV$^2/c^4$). The $\Delta\chi^2_{critical}$ values were calculated from an ensemble of toy-MC experiments, for an exposure of 6.3933×10^{20} POT, including both randomized systematic variations and statistical fluctuations. For comparison, the canonical value used in the constant-$\Delta\chi^2$ method is also shown as a horizontal line. The normal mass hierarchy is assumed. It should be noted that $\Delta\chi^2_{critical}$ is not calculated for every combination of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, due to us being efficient with computing time. 53
65 Monte Carlo estimation, according to the method of Feldman and Cousins, of $\Delta\chi^2_{critical}$ values $\Delta\chi^2_{68}$ (left) and $\Delta\chi^2_{90}$ (right), as a function of $|\Delta m^2_{32}|$ for 3 different values of $\sin^2\theta_{23}$ (0.40, 0.50 and 0.60). The $\Delta\chi^2_{critical}$ values were calculated from an ensemble of toy MC experiments, for an exposure of 6.3933×10^{20} POT, including both randomized systematic variations and statistical fluctuations. For comparison, the canonical value used in the constant-$\Delta\chi^2$ method is also shown as a horizontal line. The normal mass hierarchy is assumed. It should be noted that the *holes* in the surface are due to us being efficient with computing time. 53
66 68% CL and 90% CL allowed region (including systematics), constructed with the method of Feldman and Cousins, for the fit to the Run 1+2+3+4a dataset. The best-fit point is also shown. The normal mass hierarchy is assumed. Smoother FC contours we be prepared for the final result. 54
67 Comparison between the 68% CL and 90% CL allowed regions obtained using the constant-$\Delta\chi^2$ method for the fit to the Run 1+2+3+4a dataset, for normal and inverted hierarchy assumptions. 55
68 Comparison between the 68% CL and 90% CL allowed regions obtained using the constant-$\Delta\chi^2$ and Feldman-Cousins methods for the fit to the Run 1+2+3+4a dataset. The normal mass hierarchy is assumed. Smoother FC contours we be prepared for the final result. 56

69 Comparison between 90% CL allowed regions from MINOS [36], SuperK atmospherics [37], T2K Run 1+2+3 [8], and the 90% CL regions obtained from the VaLOR fit to the Run 1+2+3+4a dataset. Contours obtained using both the constant-$\Delta\chi^2$ and Feldman-Cousins methods are shown. The normal mass hierarchy is assumed for all results. 56
70 Comparison between 90% CL allowed regions from MINOS [35], SuperK atmospherics [37], T2K Run 1+2+3 [8], and the 90% CL regions obtained from the VaLOR fit to the Run 1+2+3+4a dataset. Contours obtained using both the constant-$\Delta\chi^2$ and Feldman-Cousins methods are shown. The normal mass hierarchy is assumed for all results. 57

List of Tables

1 Summary table of systematic parameters included the VaLOR Run 1+2+3+4a ν_μ-disappearance analysis. The nuisance parameter symbols used in this paper are shown in the first column. An index r in the nuisance parameter symbol denotes dependence on reconstructed energy while an index t denotes dependence on true energy. An index i is used where multiple systematic parameters, typically correlated between themselves, are grouped together. 11
2 A ✓symbol denotes that the given MC template is modified when the given systematic parameter is tweaked. The table lists all 44 systematic parameters considered in this analysis (first column) and all 32 MC templates used for constructing the single-ring μ-like event reconstructed energy spectrum p.d.f (first row). 14
3 The best-fit values and 1σ range for the measured *non-23-sector* oscillation parameters [26, 38]. 17
4 Calculated numbers of 1-ring μ-like events without oscillations using the nominal MC templates, the NA61-tuned MC templates and the BANFF-tuned MC templates. The total numbers of events and the numbers of events from each mode considered in this analysis are shown. These numbers were calculated for an exposure of 6.3933×10^{20} POT. 20
5 Calculated numbers of 1-ring μ-like events using the nominal MC templates, the NA61-tuned MC templates and the BANFF-tuned MC templates. The total numbers of events and the numbers of events from each mode considered in this analysis are shown. These numbers were calculated for an exposure of 6.3933×10^{20} POT and oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4. Values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} and δ_{CP}=0. The normal hierarchy is assumed. 21
6 Effect of 1σ systematic parameter variation on the number of 1-ring μ-like events, computed for no oscillations and for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4. For the oscillation scenario, the values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The normal mass hierarchy is assumed. The numbers shown were calculated for an exposure of 6.3933×10^{20} POT and BANFF-tuned MC templates were used. 23
7 Effects of 1σ systematic parameter variations on the number of 1-ring μ-like events, computed for no oscillations and for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4. Correlations are ignored for this table. For the oscillation scenario, the values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The normal mass hierarchy is assumed. The numbers shown were calculated for an exposure of 6.3933×10^{20} POT and BANFF-tuned MC templates were used. 24
8 Summary of best-fit parameters for the oscillation fits to the Run 1+2+3+4a dataset. The quoted best-fit χ^2 values (χ^2_{bf}) were computed from Eq. 5 in Sec. 4.1.5 in Ref. [4] with the reconstructed energy binning used in each fit. The table also shows the p-values obtained from the goodness-of-fit test described in Sec. 4.2 in [4] for all fits. 35
9 Summary of one-dimensional limits on $\sin^2\theta_{23}$ and Δm^2_{32}, for the oscillation fits to the Run 1+2+3+4a dataset. The limits are calculated using the constant-$\Delta\chi^2$ method. 50

1. Introduction

In this technical note, we report the results of a 3-flavour ν_μ-disappearance analysis, performed by the Valencia-Lancaster-Oxford-Rutherford (*VaLOR*) group, on the combined Run 1+2+3+4a (6.3933×10^{20} POT) dataset. This dataset was collected in four run periods: in January - June 2010 (Run 1, 0.323×10^{20} POT), in November 2010 - March 2011 (Run 2, 1.108×10^{20} POT) and, following the recovery work from the damage of the 2011 Tohoku earthquake, in March - June 2012 (Run 3, 1.579×10^{20} POT) and in October 2012 - 11th April 2013 (Run 4a, 3.3833×10^{20} POT). This analysis is an update of the work presented in Refs. [1], [2], [3] and [4] with improved methods and an enlarged dataset.

This note is meant to be read in conjunction with Ref. [4] since this analysis uses the same methods as our Run 1+2+3 analysis described in that reference, with only a few of the analysis inputs being updated. A summary of these updates is given in Sec. 2.

In this analysis, the oscillation parameters are determined by fitting the predicted reconstructed energy spectrum of 1-ring μ-like events at SuperK to the observed spectrum. The method of calculation of this predicted spectrum is presented in Sec. 2 of Ref. [4]. This analysis uses flux tuning based on recent NA61 data [10][25]. It also uses an extrapolation from ND280 to SuperK, which is implemented via a fit of flux and cross-section parameters to ND280 measurements of muon momentum-angle distributions of ν_μ CC-0-pion-like, CC-1-pion-like and CC-Other-like events whose vertices are in FGD1 [20]. Updated nominal predictions of the reconstructed energy spectrum of single-ring μ-like events in SuperK are presented for Run 1+2+3+4a in Sec. 4; the changes to the analysis summarized in Sec. 2 are incorporated into these predictions.

We consider a total of 44 energy scale, neutrino flux, neutrino cross section, efficiency, initial-state nuclear environment and final-state rescattering, and oscillation parameter systematics. These systematic parameters and their effects on the extrapolated single-ring μ-like event reconstructed energy spectrum are described in Sec. 3 of Ref. [4].

The oscillation analysis is performed using neutrino oscillation probabilities computed in a 3-flavour framework including constant-density matter effects. Details of our oscillation probability calculations, estimates of the numerical accuracy of our calculations and comparisons with calculations used by the alternative T2K oscillation analyses are given in Appendices A and B of Ref. [4].

As in our previous analyses, the oscillation fit is made using the likelihood-ratio method [29]. The mathematical details of this method are given in Sec. 3.2 in Ref. [1]. In this Run 1+2+3+4a analysis, we consider only a fit which allows all 44 systematic parameters to float, as this was the version of our fit which was preferred by the collaboration during the Run 1+2+3 analysis. The oscillation fitting method is presented in detail in Sec. 4.1 in Ref. [4], while the chosen goodness-of-fit test, suitable for the current statistics, is described in Sec. 4.2 of Ref. [4]. The performance of the fitter was tested in detail for the Run 1+2+3+4a fit using ensembles of toy MC experiments, and a summary of the results of these tests is presented in Sec. 6.

The confidence regions were constructed using the Feldman-Cousins method [31] as in our previous analyses. Systematic errors were included in the oscillation contours using the Cousins-Highland method (Bayesian integration) [32]. Contours constructed with the constant-$\Delta\chi^2$ method are also shown for both fits for comparison. Both of these methods are summarized in Sec. 4.5 of Ref. [4], and the exact methodology used for incorporating systematic errors is explained in detail.

The results of the measurement of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ using the Run 1+2+3+4a dataset are given in Sec. 7.1 of this note, while the goodness-of-fit tests are presented in Sec. 7.2. The best-fit spectra are shown for each of normal and inverted mass hierarchy assumptions for the Run 1+2+3+4a dataset in Sec. 7.3. The Feldman-Cousins and constant-$\Delta\chi^2$ confidence regions are shown in Sec. 7.4. Comparisons of our results obtained between different methods for constructing confidence regions are presented in Sec. 7.5. In that section, we also show comparisons of our T2K Run 1+2+3+4a results with results from MINOS and SuperK.

Finally, a summary is given in Sec. 8.

2. Summary of changes compared with the Run 1+2+3 analysis in T2K-TN-141(v8)

The following changes have been implemented compared with the Run 1+2+3 ν_μ-disappearance analysis described in [4] :

2.1. Changes in construction of the nominal single-ring μ-like reconstructed energy spectrum p.d.f

- The ND280 ν_μ group updated the division of reconstructed ND280 CC events. For the 2013 analyses, these events are divided into CC0π, CC1π and CC other samples (whereas for the Run 1+2+3 analysis they were divided into CCQE and CCnonQE samples). Pion tracks are identified using ionisation energy losses in the TPC or FGD, or a pion can be identified by the presence of a Michel electron in FGD1. The CC0π and CC1π samples have exactly 0 and 1 track consistent with a π^+ respectively. The CC other sample consists of events with more than 1 track consistent with a π^+ or at least one negative track or with a track consistent with an electron [19]. These three samples were used as an input to the BANFF fit of the ND280 data to find the best-fit correlated flux and cross-section parameters [20]; these parameters (taken from `postfit_banff_2013_disappearance_v4.root`) were used in turn to calculate the nominal predicted single-ring μ-like reconstructed energy spectrum in SuperK.

- An updated SuperK MC was used in this analysis, and this MC used the 11btunedv3.2 flux and the full 13a SuperK software.

- We updated the value of $\sin^2\theta_{12}$ to be consistent with the 2013 ν_e analyses.

2.2. Changes in the systematic effects considered by the analysis

- The splines used to evaluate the uncorrelated cross-section systematics were regenerated using the updated SuperK MC. The oscillation weights used to generate this now use the nominal values of oscillation parameters used in the analysis, as listed in Tab. 3

- The covariance matrix used to evaluate the final-state and secondary interaction systematics was re-generated using the updated SuperK MC.

- The SuperK detector error matrix has been recreated using more SuperK atmospheric data and updated SuperK MC [21].

- The SuperK energy scale systematic has changed from 2.3% to 2.4% [22].

2.3. Changes in neutrino oscillation fitting

- During the review process of the T2K run 1-3 result PRL [8], it was discovered that the octant choice ($\theta_{23} \leq \pi/4$ or $\theta_{23} \geq \pi/4$) has a large effect on the contour in $\sin^2(2\theta_{23})$. Therefore, we now fit $\sin^2\theta_{23}$ as default (the ability to fit for $\sin^2(2\theta_{23})$ has been retained).

- The hierarchy has a large effect, therefore we produce two results. The first using normal hierarchy; the second in inverted hierarchy.

- In [4], we did not consider the effect of uncertainties on the *non-23-sector* oscillation parameters (δ_{CP}, Δm^2_{21}, $\sin^2\theta_{13}$, $\sin^2\theta_{12}$). In this analysis, we consider the uncertainty by fitting the *known* oscillation parameters (Δm^2_{21}, $\sin^2\theta_{13}$, $\sin^2\theta_{12}$) using the constraints listed in Tab. 3. The effect of δ_{CP} is handled by throwing it uniformly in toys used to created the $\Delta\chi^2$ critical surface.

- The 7 SuperK detector efficiency uncertainties and the 6 FSI+SI uncertainties have been combined into a single 6-parameter covariance matrix. This reduces the number of nuisance parameters in the fit, allowing the fit to run faster.

- We now include oscillated $\nu_e \rightarrow \nu_\mu$ templates by reweighting the ν_μ SuperK MC with the ν_e flux.

3. Summary of systematic parameters

This analysis considers 44 systematic parameters which can be grouped into 4 categories:

1. SuperK energy scale (1 parameter).
2. SuperK detector efficiency & intranuclear and secondary interactions (6 parameters).
3. Correlated flux and cross-section parameters tuned by the BANFF group to ND280 data (23 parameters).
4. Uncorrelated cross section (11 parameters).
5. *Non-23 sector* oscillation parameters (3 parameters).

A detailed description of the systematic uncertainties considered in this analysis and their implementation can be found in Sec. 3 of Ref. [4]. All changes in this analysis with respect to Ref. [4] are summarized in Sec. 2 of this note. For convenience, a summary of all the systematic parameters considered in the present analysis is shown in Tab. 1. These systematic parameters are applied to the SuperK MC templates as shown in Tab. 2.

Table 1: Summary table of systematic parameters included the VaLOR Run 1+2+3+4a ν_μ-disappearance analysis. The nuisance parameter symbols used in this paper are shown in the first column. An index r in the nuisance parameter symbol denotes dependence on reconstructed energy while an index t denotes dependence on true energy. An index i is used where multiple systematic parameters, typically correlated between themselves, are grouped together.

SuperK energy scale systematic

$f^{SK}_{E;r}$

Reconstructed energy scale systematic. The energy scale uncertainty is estimated to be 2.4% [22]. The effect of the uncertainty is calculated by scaling the bin edges of the MC templates and, assuming a uniform distribution of events within the bins, calculating the number of events gained from / lost to neighbouring bins.

SuperK detector efficiency & intranuclear and secondary interaction systematics

$f^{SK+FSI}_{i;r}$

SuperK detector efficiency (ring counting, fiducial volume, reduction chain, OD cut, PID cut, momentum cut and decay electron cut efficiency uncertainties), FSI (final state interaction / intranuclear hadron transport) and SI (secondary interaction / detector hadron transport) systematic parameters. Six correlated parameters are included (i=0,...,5) corresponding to different modes and reconstructed energy bins.

Parameter	Description	1σ fractional error
0	ν_μ & $\bar{\nu}_\mu$ CCQE,$E_{reco} <$ 0.4 GeV	0.021
1	ν_μ & $\bar{\nu}_\mu$ CCQE, 0.4 GeV $\leq E_{reco} \leq$ 1.1 GeV	0.034
2	ν_μ & $\bar{\nu}_\mu$ CCQE,$E_{reco} >$ 1.1 GeV	0.039
3	ν_μ & $\bar{\nu}_\mu$ CCnonQE	0.119
4	ν_e & $\bar{\nu}_e$ CC	1.000
5	All NC	0.600

The following covariance matrix is used:

$$\begin{pmatrix} 0.0004 & & & & & \\ 0.0006 & 0.0011 & & & & \\ 0.0005 & 0.0010 & 0.0015 & & & \\ 0.0003 & 0.0004 & 0.0006 & 0.0142 & & \\ -0.0000 & 0.0000 & 0.0000 & 0.0011 & 1.0002 & \\ 0.0003 & 0.0003 & 0.0003 & -0.0007 & 0.0004 & 0.3594 \end{pmatrix}$$

The method of calculation of the FSI+SI portion of this covariance matrix is described in Sec. 3.1.4 in Ref. [4]. It is combined with the the SuperK detector error matrix in Ref. [21].

Correlated flux and cross-section (BANFF) systematics

Flux and cross-section parameters tuned by BANFF group on ND280 data [20]. Twenty-three correlated parameters are included (i=0,...,22).

	Parameter	Description	Best-fit	1σ (pre/post)-fit fractional error
	0	ν_μ flux normalisation, E = 0.0 - 0.4 GeV	1.020	0.121/0.086
	1	ν_μ flux normalisation, E = 0.4 - 0.5 GeV	1.016	0.130/0.089
	2	ν_μ flux normalisation, E = 0.5 - 0.6 GeV	0.991	0.122/0.081
	3	ν_μ flux normalisation, E = 0.6 - 0.7 GeV	0.962	0.115/0.077
	4	ν_μ flux normalisation, E = 0.7 - 1.0 GeV	0.928	0.129/0.086
	5	ν_μ flux normalisation, E = 1.0 - 1.5 GeV	0.982	0.116/0.078
	6	ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.029	0.100/0.068
	7	ν_μ flux normalisation, E = 2.5 - 3.5 GeV	1.050	0.095/0.064
	8	ν_μ flux normalisation, E = 3.5 - 5.0 GeV	1.031	0.112/0.071
	9	ν_μ flux normalisation, E = 5.0 - 7.0 GeV	0.959	0.152/0.071
	10	ν_μ flux normalisation, E = 7.0 - 30.0 GeV	0.919	0.187/0.079
	11	$\bar{\nu}_\mu$ flux normalisation, E = 0.0 - 0.7 GeV	1.018	0.133/0.103
	12	$\bar{\nu}_\mu$ flux normalisation, E = 0.7 - 1.0 GeV	1.000	0.117/0.090
	13	$\bar{\nu}_\mu$ flux normalisation, E = 1.0 - 1.5 GeV	0.996	0.119/0.094
	14	$\bar{\nu}_\mu$ flux normalisation, E = 1.5 - 2.5 GeV	1.014	0.123/0.104
	15	$\bar{\nu}_\mu$ flux normalisation, E = 2.5 -30.0 GeV	0.994	0.122/0.107
	16	CCQE axial-mass scaling factor	1.010	0.372/0.060
	17	Resonance-production axial-mass scaling factor	0.796	0.183/0.055
	18	CCQE normalisation, E = 0.0 - 1.5 GeV	0.959	0.110/0.076
$f^{banff}_{i;t,r}$	19	CCQE normalisation, E = 1.5 - 3.5 GeV	0.902	0.300/0.102
	20	CCQE normalisation, E = 3.5 - 30.0 GeV	0.868	0.300/0.114
	21	$CC1\pi$ normalisation, E = 0.0 - 2.5 GeV	1.200	0.317/0.162
	22	$CC1\pi$ normalisation, E = 2.5 - 30.0 GeV	1.077	0.400/0.170

The following covariance matrix is used:

$$
\begin{pmatrix}
0.0074 & & & & & & & & & & & & & & & & & & & & & & \\
0.0068 & 0.0079 & & & & & & & & & & & & & & & & & & & & & \\
0.0057 & 0.0068 & 0.0065 & & & & & & & & & & & & & & & & & & & & \\
0.0049 & 0.0056 & 0.0057 & 0.0059 & & & & & & & & & & & & & & & & & & & \\
0.0044 & 0.0048 & 0.0049 & 0.0056 & 0.0074 & & & & & & & & & & & & & & & & & & \\
0.0044 & 0.0046 & 0.0044 & 0.0050 & 0.0058 & 0.0060 & & & & & & & & & & & & & & & & & \\
0.0039 & 0.0040 & 0.0038 & 0.0040 & 0.0039 & 0.0046 & 0.0046 & & & & & & & & & & & & & & & & \\
0.0032 & 0.0030 & 0.0030 & 0.0029 & 0.0026 & 0.0032 & 0.0033 & 0.0041 & & & & & & & & & & & & & & & \\
0.0022 & 0.0022 & 0.0022 & 0.0020 & 0.0020 & 0.0019 & 0.0022 & 0.0031 & 0.0050 & & & & & & & & & & & & & & \\
0.0017 & 0.0016 & 0.0016 & 0.0016 & 0.0018 & 0.0016 & 0.0020 & 0.0020 & 0.0034 & 0.0050 & & & & & & & & & & & & & \\
0.0011 & 0.0010 & 0.0009 & 0.0010 & 0.0011 & 0.0010 & 0.0016 & 0.0012 & 0.0024 & 0.0046 & 0.0063 & & & & & & & & & & & & \\
0.0080 & 0.0063 & 0.0054 & 0.0054 & 0.0044 & 0.0051 & 0.0046 & 0.0035 & 0.0020 & 0.0017 & 0.0014 & 0.0106 & & & & & & & & & & & \\
0.0060 & 0.0050 & 0.0046 & 0.0051 & 0.0045 & 0.0051 & 0.0046 & 0.0031 & 0.0016 & 0.0016 & 0.0016 & 0.0085 & 0.0082 & & & & & & & & & & \\
0.0056 & 0.0045 & 0.0043 & 0.0049 & 0.0045 & 0.0051 & 0.0044 & 0.0032 & 0.0018 & 0.0016 & 0.0015 & 0.0082 & 0.0081 & 0.0089 & & & & & & & & & \\
0.0047 & 0.0040 & 0.0037 & 0.0040 & 0.0031 & 0.0043 & 0.0048 & 0.0026 & 0.0019 & 0.0027 & 0.0030 & 0.0067 & 0.0072 & 0.0075 & 0.0109 & & & & & & & & \\
0.0025 & 0.0022 & 0.0019 & 0.0020 & 0.0012 & 0.0017 & 0.0030 & 0.0011 & 0.0020 & 0.0038 & 0.0046 & 0.0037 & 0.0043 & 0.0044 & 0.0086 & 0.0114 & & & & & & & \\
-0.0010 & -0.0013 & -0.0012 & -0.0009 & -0.0007 & -0.0005 & -0.0004 & -0.0004 & -0.0005 & -0.0002 & -0.0000 & -0.0010 & -0.0007 & -0.0007 & -0.0005 & -0.0003 & 0.0036 & & & & & & \\
-0.0002 & -0.0002 & -0.0002 & -0.0002 & -0.0003 & -0.0003 & -0.0003 & -0.0002 & -0.0001 & -0.0000 & 0.0001 & -0.0002 & -0.0003 & -0.0003 & -0.0002 & -0.0001 & -0.0005 & 0.0030 & & & & & \\
-0.0042 & -0.0047 & -0.0047 & -0.0049 & -0.0053 & -0.0045 & -0.0034 & -0.0023 & -0.0016 & -0.0013 & -0.0008 & -0.0043 & -0.0041 & -0.0039 & -0.0030 & -0.0014 & -0.0009 & 0.0002 & 0.0058 & & & & \\
-0.0023 & -0.0025 & -0.0025 & -0.0027 & -0.0024 & -0.0031 & -0.0033 & -0.0023 & -0.0007 & -0.0003 & 0.0001 & -0.0029 & -0.0031 & -0.0029 & -0.0034 & -0.0019 & -0.0018 & 0.0000 & 0.0038 & 0.0104 & & & \\
-0.0006 & -0.0006 & -0.0008 & -0.0009 & -0.0009 & -0.0005 & -0.0006 & -0.0010 & -0.0021 & -0.0023 & -0.0022 & -0.0005 & -0.0005 & -0.0006 & -0.0005 & -0.0012 & -0.0015 & 0.0002 & 0.0022 & 0.0047 & 0.0130 & & \\
-0.0054 & -0.0055 & -0.0051 & -0.0059 & -0.0075 & -0.0074 & -0.0056 & -0.0035 & -0.0018 & -0.0013 & -0.0007 & -0.0060 & -0.0060 & -0.0059 & -0.0050 & -0.0020 & 0.0011 & -0.0033 & 0.0048 & 0.0007 & -0.0032 & 0.0262 & \\
-0.0016 & -0.0014 & -0.0013 & -0.0010 & -0.0012 & -0.0013 & -0.0018 & -0.0032 & -0.0049 & -0.0052 & -0.0046 & -0.0013 & -0.0010 & -0.0012 & -0.0015 & -0.0023 & 0.0016 & -0.0042 & 0.0002 & -0.0047 & -0.0062 & 0.0112 & 0.0288
\end{pmatrix}
$$

Uncorrelated cross-section systematics

$f_{pF;t,r}$ Uncertainty in Fermi momentum for ^{16}O was taken to be 225±30 MeV [14].

$f_{bindE;t,r}$ Uncertainty in binding energy for ^{16}O was taken to be 27±9 MeV [14].

$f_{SF;t,r}$	Nuclear environment modelling systematic: switch from Relativistic Fermi Gas (RFG) to Spectral Function (SF) model (CCQE only). 0 (RFG) - 1 (SF)
$f_{\pi-less\Delta;t,r}$	Systematic due to uncertainty in the fraction of π-less Δ decays in resonance neutrino-production events. A 20% uncertainty was assumed [14].
$f_{CCothShape;t,r}$	Systematic due to the uncertainty in CC multipion, CC DIS and CC resonant η, K and photon production cross-section. A 40% uncertainty was assumed [14].
$f_{Wshape;t,r}$	Systematic due to the uncertainty in the shape of the initial pion momentum distribution (before final-state interactions) in resonance interactions. The uncertainty was taken to be 52% [14].
$f_{CCcoh;t}$	Systematic due to the uncertainty in the CC coherent cross-section. A 100% uncertainty was assumed [14].
$f_{NC1\pi^{\pm};t}$	Systematic due to the uncertainty on the NC1$\pi^{\pm}$ cross-section. A 30% uncertainty was assumed [14].
$f_{NCoth;t}$	Systematic due to the uncertainty on the NC other (all but single $\pi^{\pm}$) cross-section. A 30% uncertainty was assumed [14].
$f_{CC\nu_e;t}$	Systematic due to the uncertainty on the ν_e / ν_μ CC cross-section ratio. A 3% uncertainty was assumed [13].
$f_{CC\bar{\nu};t}$	Systematic due to the uncertainty on the $\bar{\nu}$ / ν CC cross-section ratio. A 40% uncertainty was assumed.
***Non-23 sector* oscillation parameters**	
$\sin^2\theta_{13}$	Uncertainty in $\sin^2\theta_{13}$ of 13.9%, taken from the 2012 PDG [26].
$\sin^2\theta_{12}$	Uncertainty in $\sin^2\theta_{12}$ of 5.9%, taken from a global fit [38].
Δm^2_{21}	Uncertainty in Δm^2_{21} of 2.7%, taken from the 2012 PDG [26].

	ν_μ CCQE	ν_μ CC1π	ν_μ CC coherent	ν_μ CC other	ν_μ NC1$\pi^\pm$	ν_μ NC other	$\bar{\nu}_\mu$ CCQE	$\bar{\nu}_\mu$ CC1π	$\bar{\nu}_\mu$ CC coherent	$\bar{\nu}_\mu$ CC other	$\bar{\nu}_\mu$ NC1$\pi^\pm$	$\bar{\nu}_\mu$ NC other	ν_e CCQE	ν_e CC1π	ν_e CC coherent	ν_e CC other	ν_e NC1$\pi^\pm$	ν_e NC other	$\bar{\nu}_e$ CCQE	$\bar{\nu}_e$ CC1π	$\bar{\nu}_e$ CC coherent	$\bar{\nu}_e$ CC other	$\bar{\nu}_e$ NC1$\pi^\pm$	$\bar{\nu}_e$ NC other	Osc-ν_e CCQE	Osc-ν_e CC1π	Osc-ν_e CC coherent	Osc-ν_e CC other	Osc-ν_μ CCQE	Osc-ν_μ CC1π	Osc-ν_μ CC coherent	Osc-ν_μ CC other
$\sin^2\theta_{13}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\sin^2\theta_{12}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Δm^2_{21}	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK}_{E;r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{0;r}$	✓						✓																						✓			
$f^{SK+FSI}_{1;r}$	✓						✓																						✓			
$f^{SK+FSI}_{2;r}$	✓						✓																						✓			
$f^{SK+FSI}_{3;r}$		✓	✓	✓				✓	✓	✓																				✓		
$f^{SK+FSI}_{4;r}$													✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓				
$f^{SK+FSI}_{5;r}$					✓	✓					✓	✓					✓	✓					✓	✓								
$f^{banff}_{0;t,r}$	✓	✓	✓	✓	✓	✓																										
$f^{banff}_{1;t,r}$	✓	✓	✓	✓	✓	✓																										
$f^{banff}_{2;t,r}$	✓	✓	✓	✓	✓	✓																										
$f^{banff}_{3;t,r}$	✓	✓	✓	✓	✓	✓																										
$f^{banff}_{5;t,r}$	✓	✓	✓	✓	✓	✓																										
$f^{banff}_{6;t,r}$	✓	✓	✓	✓	✓	✓																										
$f^{banff}_{7;t,r}$	✓	✓	✓	✓	✓	✓																										
$f^{banff}_{8;t,r}$	✓	✓	✓	✓	✓	✓																										
$f^{banff}_{9;t,r}$	✓	✓	✓	✓	✓	✓																										
$f^{banff}_{10;t,r}$	✓	✓	✓	✓	✓	✓																										
$f^{banff}_{11;t,r}$							✓	✓	✓	✓	✓	✓																				
$f^{banff}_{12;t,r}$							✓	✓	✓	✓	✓	✓																				
$f^{banff}_{13;t,r}$							✓	✓	✓	✓	✓	✓																				
$f^{banff}_{14;t,r}$							✓	✓	✓	✓	✓	✓																				
$f^{banff}_{15;t,r}$							✓	✓	✓	✓	✓	✓																				
$f^{banff}_{16;t,r}$	✓						✓						✓						✓						✓				✓			
$f^{banff}_{17;t,r}$		✓		✓	✓	✓		✓		✓	✓	✓		✓		✓	✓	✓		✓		✓	✓	✓		✓		✓		✓		✓
$f^{banff}_{18;t,r}$	✓						✓						✓						✓						✓				✓			
$f^{banff}_{19;t,r}$	✓						✓						✓						✓						✓				✓			
$f^{banff}_{20;t,r}$	✓						✓						✓						✓						✓				✓			
$f^{banff}_{21;t,r}$		✓						✓						✓						✓						✓				✓		
$f^{banff}_{22;t,r}$		✓						✓						✓						✓						✓				✓		
$f_{pF;t,r}$	✓						✓						✓						✓						✓				✓			
$f_{SF;t,r}$	✓						✓						✓						✓						✓				✓			
$f_{\pi-less\Delta;t,r}$		✓			✓	✓		✓			✓	✓		✓			✓	✓		✓			✓	✓		✓				✓		
$f_{bindE;t,r}$	✓						✓						✓						✓						✓				✓			
$f_{CCothShape;t,r}$				✓						✓						✓						✓						✓				✓
$f_{Wshape;t,r}$					✓	✓					✓	✓					✓	✓					✓	✓								
$f_{CCcoh;t}$			✓						✓						✓						✓						✓				✓	
$f_{NC1\pi^\pm;t}$					✓						✓						✓						✓									
$f_{NCoth;t}$						✓						✓						✓						✓								
$f_{CC\nu_e;t}$													✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓				
$f_{CC\bar{\nu};t}$							✓	✓	✓	✓									✓	✓	✓	✓										

Table 2: A ✓symbol denotes that the given MC template is modified when the given systematic parameter is tweaked. The table lists all 44 systematic parameters considered in this analysis (first column) and all 32 MC templates used for constructing the single-ring μ-like event reconstructed energy spectrum p.d.f (first row).

4. Updated predictions of single-ring μ-like event reconstructed energy spectrum

The predicted number $N_{SK;r}$ of 1-ring μ-like events in the rth reconstructed energy bin is computed as follows:

$$N_{SK;r} = \sum_{m} \sum_{t} \sum_{r\prime} P_{m;t} \cdot T_{r;r\prime;f_{\Delta E}} \cdot S_{m;t;r\prime;\vec{f}} \cdot N^{MC}_{SK;m;r\prime;t}. \quad (1)$$

In Eq. 1, $N^{MC}_{SK;m;r\prime;t}$ is the input SuperK Monte Carlo (MC) template containing the number of events in the 1-ring μ-like MC sample with true reaction mode m, and found in the true energy bin t and the reconstructed energy bin $r\prime$. The construction of the nominal MC templates was presented in Sec. 2.2 in Ref. [4] Updated SuperK MC samples are used in this analysis as discussed in Sec. 2 of this note. The following 32 MC templates[2] were used:

- ν_μ CCQE,
- ν_μ CC1π,
- ν_μ CC coherent,
- ν_μ CC other,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1$\pi^\pm$,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC other,
- $\bar{\nu}_\mu$ CCQE,
- $\bar{\nu}_\mu$ CC1π,
- $\bar{\nu}_\mu$ CC coherent,
- $\bar{\nu}_\mu$ CC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC other,
- ν_e CCQE,
- ν_e CC1π,
- ν_e CC coherent,
- ν_e CC other,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC1$\pi^\pm$,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC other,
- $\bar{\nu}_e$ CCQE,
- $\bar{\nu}_e$ CC1π,
- $\bar{\nu}_e$ CC coherent,

[2] Each of the 32 MC templates has 6132 2-dimensional bins (= 84 true energy bins × 73 reconstructed energy bins). The 84 true energy bins are arranged as follows: 6 50-MeV bins from 0-0.3 GeV, 28 25-MeV bins from 0.3-1 GeV, 40 50-MeV bins from 1-3 GeV, 5 100-MeV bin from 3-3.5 GeV, 1 bin from 3.5-4 GeV, 1 bin from 4-5 GeV, 1 bin from 5-7 GeV, 1 bin from 7-10 GeV and 1 bin from 10-30 GeV. The 73 reconstructed energy bins are the following: 60 50-MeV bins from 0-3 GeV, 4 250-MeV bins from 3-4 GeV, 4 500-MeV bins from 4-6 GeV, 4 1000-MeV bins from 6-10 GeV and 1 bin from 10-30 GeV. The binning is unchanged with respect to the Run 1+2+3 analysis, however after discussion with T2KSK, events with reconstructed energy >30 GeV are now included in the 10-30 GeV bin.

- $\bar{\nu}_e$ CC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC other,
- oscillated $\nu_\mu \rightarrow \nu_e$ CCQE,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC1π,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC coherent,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC other,
- oscillated $\nu_e \rightarrow \nu_\mu$ CCQE,
- oscillated $\nu_e \rightarrow \nu_\mu$ CC1π,
- oscillated $\nu_e \rightarrow \nu_\mu$ CC coherent and
- oscillated $\nu_e \rightarrow \nu_\mu$ CC other

(the first 28 templates were used in Ref. [4]). The nominal MC templates are reweighted using both flux-tuning weights based on NA61 hadro-production data (see Sec. 2.3 of Ref. [4]) and flux and cross-section tuning weights based on ND280 data (see Sec. 2.4 of Ref. [4]). Updated BANFF weights were used in this analysis, as discussed in Sec. 2 of this note. $S_{m;t;r\prime;\vec{f}}$ is an overall, multiplicative, systematic error factor depending on the reaction mode m, the true energy bin t, the reconstructed energy bin $r\prime$ and a vector of nuisance (systematic) parameters $\vec{f}$. A summary of the nuisance parameters considered in this analysis was given in Sec. 3 of this note. $T_{r;r\prime;f_{\Delta E}}$ is a transfer function describing the migration of events between the reconstructed energy bins r and $r\prime$ due to uncertainty in the SuperK reconstructed energy scale, expressed here in terms of the nuisance parameter $f_{\Delta E}$. Finally, $P_{m;t}$ is the 3-flavour oscillation probability applied in the true energy bin t of the SuperK MC template which corresponds to mode m. The CC MC templates constructed from the unoscillated MC samples are weighted with the corresponding survival probability: The ν_μ CC templates are weighted with P($\nu_\mu \rightarrow \nu_\mu$), the $\bar{\nu}_\mu$ CC templates with P($\bar{\nu}_\mu \rightarrow \bar{\nu}_\mu$), the ν_e CC templates with P($\nu_e \rightarrow \nu_e$) and the $\bar{\nu}_e$ CC templates with P($\bar{\nu}_e \rightarrow \bar{\nu}_e$). The CC MC templates made from the oscillated ν_e MC sample are weighted with P($\nu_\mu \rightarrow \nu_e$) (the sample was generated assuming 100% of ν_μ oscillate to ν_e). The CC MC templates made from the oscillated ν_μ MC sample are weighted with P($\nu_e \rightarrow \nu_\mu$) (the sample is created by reweighting the ν_μ events to the ν_e flux, which assumes 100% of ν_e oscillate to ν_μ). We neglect contributions from ν_τ-CC and $\bar{\nu}_\tau$-CC as their energy threshold is around 3.5 GeV and their effect is negligible[3]. The same oscillation parameters are used for both ν_μ and $\bar{\nu}_\mu$ oscillations. Oscillations are not applied to the NC MC templates since they serve as proxies for the mixtures of $\nu_e + \nu_\mu + \nu_\tau$ NC and $\bar{\nu}_e + \bar{\nu}_\mu + \bar{\nu}_\tau$ NC MC templates which are unchanged under 3-flavour oscillations. All oscillation probabilities are computed in a 3-flavour framework including matter effects in constant density matter (assuming an Earth crust density of 2.6 g/cm^3). Custom oscillation probability calculation code was developed within the VaLOR analysis framework. Details and estimates of the numerical accuracy of this code can be found in Appendices A and B of Ref. [4].

In this section, we present the expected numbers of events and single-ring μ-like reconstructed energy spectra in SuperK for various oscillation scenarios. We also present estimates of the effect of the NA61 and BANFF tunings and of assumptions made in the 3-flavour neutrino oscillation framework used in this analysis. Unless explicitly stated otherwise, the normal hierarchy is assumed, δ_{CP}=0, and the values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}. All plots were generated for an integrated exposure of 6.3933×10^{20} POT.

The expected number of single-ring μ-like events is shown in Fig. 1 as a function of the oscillation parameters $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$. The expected number of events ranges from ∼116.75, at the T2K Run

[3] A back-of-the-envelope calculation can be found in the replies to comments and revision history document of [4] that gives $\ll 1\%$ ν_τ contamination in the 1-ring μ-like SuperK sample. A SuperK MC study is required to determine whether the SuperK efficiency assumptions made there are valid.

Parameter	Best-fit	1σ error
Δm^2_{21}	7.5×10^{-5} eV^2/c^4	∼2.7%
$\sin^2(\theta_{12})$	0.306	∼5.9%
$\sin^2(\theta_{13})$	0.0251	∼13.9%

Table 3: The best-fit values and 1σ range for the measured *non-23-sector* oscillation parameters [26, 38].

1+2+3 first octant best-fit point ($\sin^2\theta_{23} = 0.50$ and $|\Delta m^2_{32}| = 2.44\times10^{-3}$ eV^2/c^4), to more than ∼138 for oscillation parameter values still within the T2K Run 1+2+3 90% CL region. This illustrates the great sensitivity of the T2K ν_μ-disappearance analysis in measuring $\sin^2\theta_{23}$ with the Run 1+2+3+4a data.

Predicted single-ring μ-like SuperK reconstructed energy spectra are shown in Fig. 2 for no oscillations, and in Fig. 3 for oscillations with $\sin^2\theta_{23} = 0.5$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4.

MC tunes derived using NA61 and ND280 data are applied to the nominal MC templates as described in Sec. 2.6 in Ref. [4]. The fractional difference in the expected number of single-ring μ-like events when switching from the nominal MC to the NA61-tuned MC templates is shown in Fig. 4 as a function of the oscillation parameters $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, In Fig. 5 is shown the fractional difference in the expected number of single-ring μ-like events when switching from the NA61-tuned MC templates to the BANFF-tuned MC templates. From these figures it can be seen that, for oscillation parameters in the T2K Run 1+2+3 90% confidence region, ∼12.4-13.0% more single-ring μ-like events are obtained with the NA61-tuned MC than with the nominal MC. In the same part of the parameter space, ∼7.0-8.2% fewer events are expected with the BANFF-tuned MC than with the NA61-tuned MC. The effects of the NA61 and BANFF tunes on the total expected number of single-ring μ-like events and on the numbers in each of the 32 component reaction modes are shown in Tab. 4 for no oscillations, and in Tab. 5 for oscillations with $\sin^2\theta_{23} = 0.5$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4. Finally, the effects of the NA61 and BANFF tunes on the reconstructed energy spectrum of single-ring μ-like events are shown in Figs. 6 and 7 respectively. In each case, spectra are shown for both no oscillations and for oscillations with $\sin^2\theta_{23} = 0.5$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4.

When making a fit to obtain the best-fit values of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, δ_{CP} is fixed to 0, and we also fit other oscillation parameters ($\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21}) using the errors listed in Tab. 3 as constraints. The best-fit values and constraints come from the 2012 PDG [26] for $\sin^2\theta_{13}$ and Δm^2_{21}, and from a global fit [38] for $\sin^2\theta_{12}$, so as to be consistent with the latest ν_e appearance fits [24, 23]. The maximum possible effect of this assumption on the expected number of single-ring μ-like events is shown in Fig. 8. In each bin of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, a large number of toy MC experiments was generated with randomized values of the *non-23-sector* parameters. The random values of $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} were drawn from Gaussian distributions with means equal to the 1σ errors, while the values of δ_{CP} were drawn from a uniform distribution between -π and +π. These four *non-23-sector* parameters were taken to be uncorrelated. The maximum fractional deviation from the default case was recorded and plotted in Fig. 8. From this figure, it can be seen that the maximum effect of the *non-23-sector* oscillation parameters is of the order of ∼1.7% in the part of the parameter space covered by the T2K Run 1+2+3 90% confidence region.

All the results in this analysis assume the normal mass hierarchy, unless explicitly stated. The effect of this assumption on the expected number of single-ring μ-like events is shown in Fig. 9 as a function of the oscillation parameters $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$. The effect is large and potentially misleading, and some clarifications are in order. This analysis can assume a given hierarchy without loss of generality of its results. Fig. 9 reveals that $|\Delta m^2_{32}|$ is a poor choice as the atmospheric mass-squared difference when both mass hierarchies need to be considered, since $|\Delta m^2_{32}|$ is the largest mass-squared difference for the inverted hierarchy but the second-largest for the normal hierarchy. The large difference shown in Fig. 9 is caused by the change in value of Δm^2_{31} when switching from the normal to inverted hierarchy.

Other choices exist for the atmospheric mass-squared difference, and these are explained in Sec. 2.3 in [3]. An example of an alternative choice is shown vividly in Fig. 5 in [3]; this technical note describes a repeat, in a 3-flavour framework, of our official Run 1+2 disappearance analysis [2] which was originally performed in a 2-flavour framework.

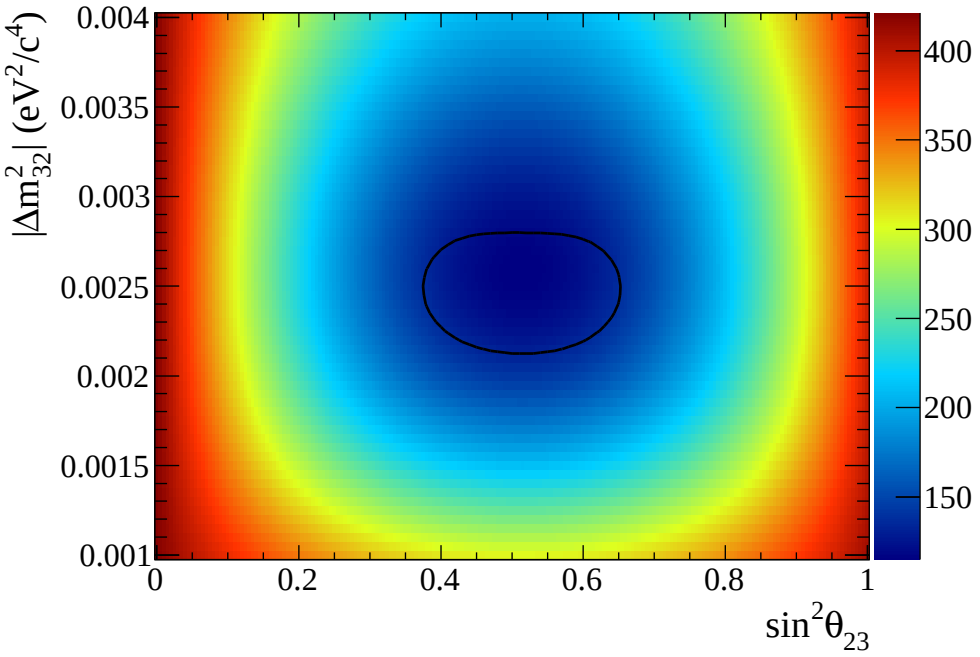


Figure 1: Predicted number of 1-ring μ-like events, as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, for an exposure of 6.3933×10^{20} POT. The number of events was calculated using the BANFF-tuned MC templates. Values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} and δ_{CP}=0. The numbers shown were generated assuming the normal hierarchy. The T2K Run 1+2+3 90% CL region is superimposed for reference.

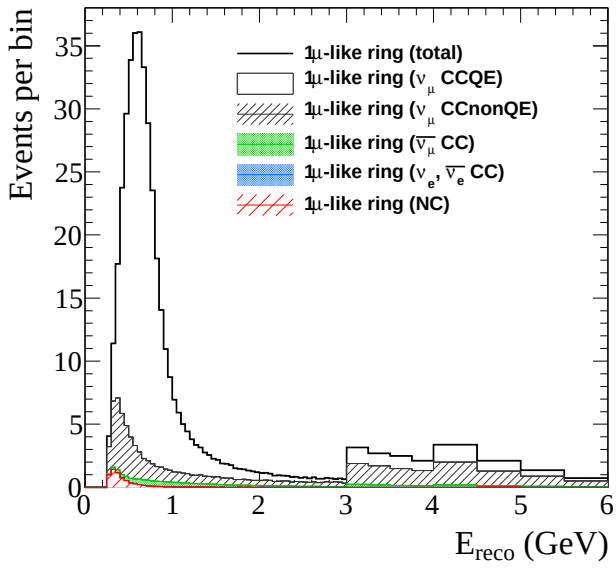


Figure 2: Predicted reconstructed-energy spectrum of 1-ring μ-like events, and contributions from various true neutrino reaction modes, for no oscillations and for an exposure of 6.3933×10^{20} POT. The 32 components of the spectrum are calculated separately in the actual analysis, but, for this plot, are grouped into just 5 categories: ν_μ CCQE, ν_μ CCnonQE, $\bar{\nu_\mu}$ CC, $\nu_e/\bar{\nu_e}$ CC and NC. The spectrum was generated using the BANFF-tuned MC templates.

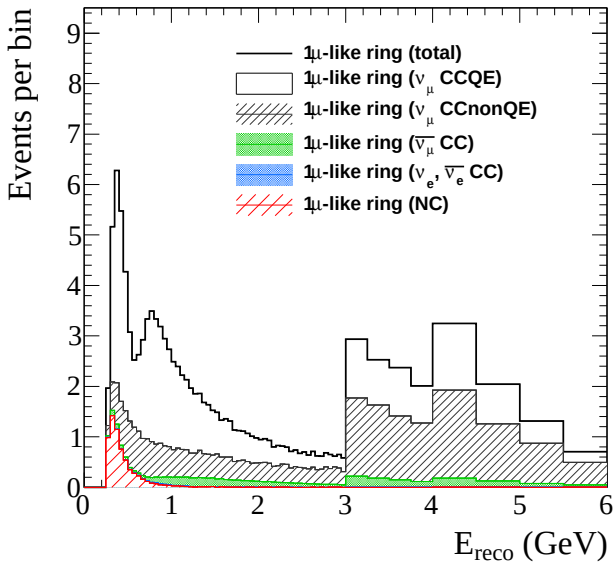


Figure 3: Predicted reconstructed-energy spectrum of 1-ring μ-like events, and contributions from various true neutrino reaction modes, for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4 and for an exposure of 6.3933×10^{20} POT. The spectrum was generated using the BANFF-tuned MC templates. Values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} and δ_{CP}=0. The spectrum shown was generated assuming the normal hierarchy. Note that the vertical axis is zoomed in by a factor of 4 compared with Fig. 2.

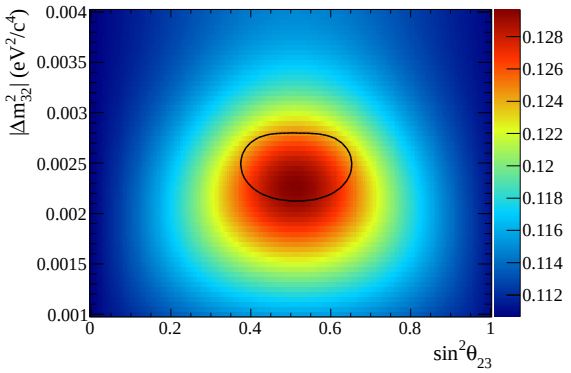


Figure 4: Fractional difference in the predicted number of 1-ring μ-like events as a result of applying the NA61 flux tuning to the nominal MC templates. This number is shown as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ for an exposure of 6.3933×10^{20} POT. Values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21},and δ_{CP}=0. The numbers shown were generated assuming the normal hierarchy. The T2K Run 1+2+3 90% CL region is superimposed for reference.

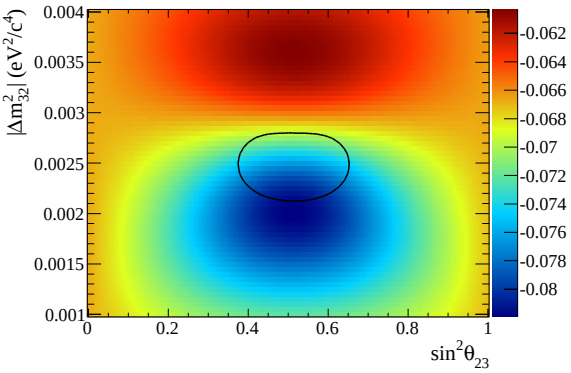


Figure 5: Fractional difference in the predicted number of 1-ring μ-like events as a result of applying the BANFF flux and cross-section tuning to the NA61 flux tuned MC templates. This number is shown as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ and for an exposure of 6.3933×10^{20} POT. The values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The numbers shown were generated assuming the normal hierarchy. The T2K Run 1+2+3 90% CL region is superimposed for reference.

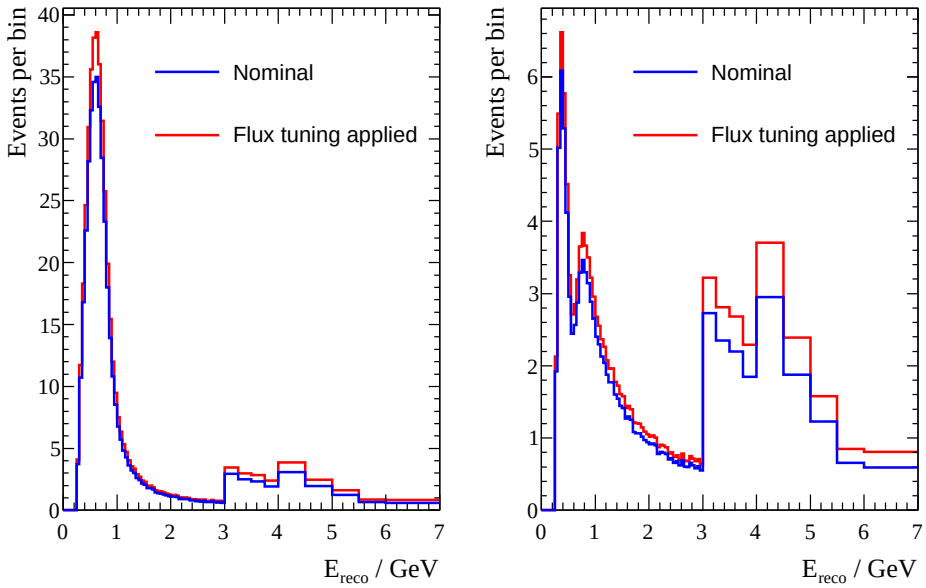


Figure 6: Reconstructed-energy spectrum of 1-ring μ-like events, for an exposure of 6.3933×10^{20} POT, both with and without the effect of the NA61 flux tuning of the nominal MC templates. The spectra are shown both for no oscillations (left) and for oscillations with $\sin^2\theta_{23}$=0.5, $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4, the 2012 PDG values for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0 (right). Results shown for the normal hierarchy.

	N_{SK} (nominal MC)	N_{SK} (NA61-tuned MC)	N_{SK} (BANFF-tuned MC)
Total	406.885590	451.896983	421.761456
ν_μ CCQE	323.303195	357.193543	333.937255
ν_μ CC 1π	54.678985	61.910123	57.152003
ν_μ CC coherent	1.845439	2.080564	2.040598
ν_μ CC other	7.748844	9.367262	9.303195
ν_μ/ν_τ NC $1\pi^{+/-}$	3.312363	3.790767	2.837281
ν_μ/ν_τ NC other	2.833863	3.344421	3.123164
Oscillated ν_μ CCQE	0.000000	0.000000	0.000000
Oscillated ν_μ CC 1π	0.000000	0.000000	0.000000
Oscillated ν_μ CC coherent	0.000000	0.000000	0.000000
Oscillated ν_μ CC other	0.000000	0.000000	0.000000
$\bar{\nu}_\mu$ CCQE	8.765816	9.448778	8.979168
$\bar{\nu}_\mu$ CC 1π	2.831107	3.055314	2.817605
$\bar{\nu}_\mu$ CC coherent	0.416568	0.449246	0.450921
$\bar{\nu}_\mu$ CC other	0.505131	0.543629	0.530826
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	0.183251	0.199530	0.138736
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	0.171682	0.185494	0.169629
ν_e CCQE	0.034418	0.037030	0.035433
ν_e CC 1π	0.020994	0.023160	0.022092
ν_e CC coherent	0.001028	0.001123	0.001123
ν_e CC other	0.005655	0.006715	0.006598
ν_e NC $1\pi^{+/-}$	0.088973	0.102385	0.073063
ν_e NC other	0.109167	0.128465	0.118603
$\bar{\nu}_e$ CCQE	0.002116	0.002127	0.001992
$\bar{\nu}_e$ CC 1π	0.001339	0.001346	0.001242
$\bar{\nu}_e$ CC coherent	0.000131	0.000134	0.000134
$\bar{\nu}_e$ CC other	0.000430	0.000434	0.000420
$\bar{\nu}_e$ NC $1\pi^{+/-}$	0.011925	0.012072	0.008187
$\bar{\nu}_e$ NC other	0.013168	0.013319	0.012187
Oscillated ν_e CCQE	0.000000	0.000000	0.000000
Oscillated ν_e CC 1π	0.000000	0.000000	0.000000
Oscillated ν_e CC coherent	0.000000	0.000000	0.000000
Oscillated ν_e CC other	0.000000	0.000000	0.000000

Table 4: Calculated numbers of 1-ring μ-like events without oscillations using the nominal MC templates, the NA61-tuned MC templates and the BANFF-tuned MC templates. The total numbers of events and the numbers of events from each mode considered in this analysis are shown. These numbers were calculated for an exposure of 6.3933×10^{20} POT.

	N_{SK} (nominal MC)	N_{SK} (NA61-tuned MC)	N_{SK} (BANFF-tuned MC)
Total	113.170201	127.802057	117.776282
ν_μ CCQE	64.362490	71.563162	67.144118
ν_μ CC 1π	26.993670	31.161684	27.453878
ν_μ CC coherent	0.756691	0.874092	0.879517
ν_μ CC other	6.882348	8.377876	8.317477
ν_μ/ν_τ NC $1\pi^{+/-}$	3.312363	3.790767	2.837281
ν_μ/ν_τ NC other	2.833863	3.344421	3.123164
Oscillated ν_μ CCQE	0.114233	0.126452	0.121548
Oscillated ν_μ CC 1π	0.026640	0.029728	0.028577
Oscillated ν_μ CC coherent	0.000799	0.000890	0.000890
Oscillated ν_μ CC other	0.002280	0.002583	0.002545
$\bar{\nu}_\mu$ CCQE	4.310464	4.655209	4.351636
$\bar{\nu}_\mu$ CC 1π	2.024431	2.180599	1.967334
$\bar{\nu}_\mu$ CC coherent	0.222639	0.240227	0.240845
$\bar{\nu}_\mu$ CC other	0.448188	0.481943	0.470039
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	0.183251	0.199530	0.138736
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	0.171682	0.185494	0.169629
ν_e CCQE	0.031971	0.034415	0.032912
ν_e CC 1π	0.019924	0.022017	0.020920
ν_e CC coherent	0.000961	0.001050	0.001050
ν_e CC other	0.005562	0.006611	0.006496
ν_e NC $1\pi^{+/-}$	0.088973	0.102385	0.073063
ν_e NC other	0.109167	0.128465	0.118603
$\bar{\nu}_e$ CCQE	0.002029	0.002039	0.001908
$\bar{\nu}_e$ CC 1π	0.001301	0.001308	0.001204
$\bar{\nu}_e$ CC coherent	0.000127	0.000130	0.000130
$\bar{\nu}_e$ CC other	0.000426	0.000431	0.000417
$\bar{\nu}_e$ NC $1\pi^{+/-}$	0.011925	0.012072	0.008187
$\bar{\nu}_e$ NC other	0.013168	0.013319	0.012187
Oscillated ν_e CCQE	0.174872	0.192606	0.180557
Oscillated ν_e CC 1π	0.057775	0.063921	0.064998
Oscillated ν_e CC coherent	0.004620	0.005090	0.004929
Oscillated ν_e CC other	0.001368	0.001539	0.001508

Table 5: Calculated numbers of 1-ring μ-like events using the nominal MC templates, the NA61-tuned MC templates and the BANFF-tuned MC templates. The total numbers of events and the numbers of events from each mode considered in this analysis are shown. These numbers were calculated for an exposure of 6.3933×10^{20} POT and oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4. Values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} and δ_{CP}=0. The normal hierarchy is assumed.

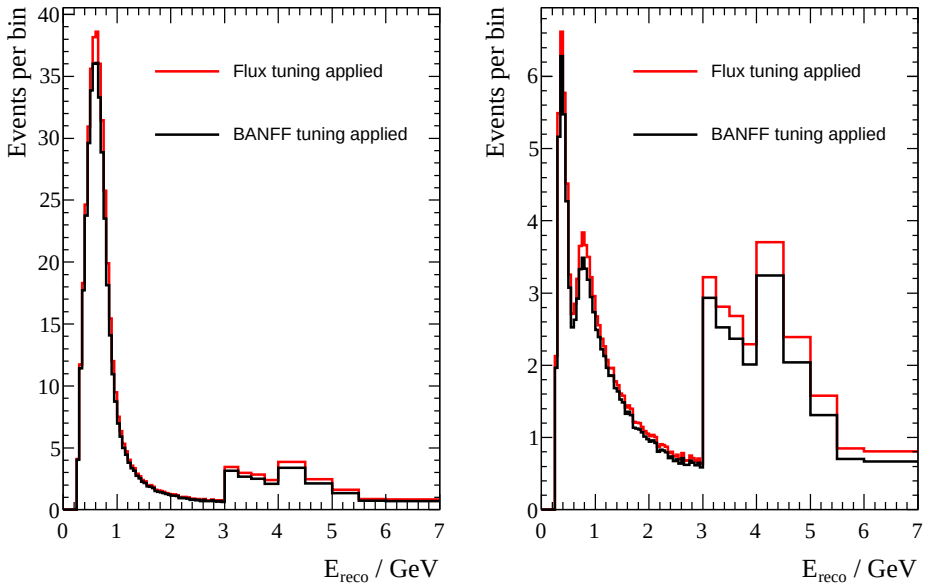


Figure 7: Reconstructed-energy spectrum of 1-ring μ-like events, for an exposure of 6.3933×10^{20} POT, both with and without the effect of BANFF flux and cross-section tuning on the NA61-tuned flux MC templates. The spectra are shown both for no 23-sector oscillations (left) and for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4 (right). In all spectra, the 2012 PDG parameters are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} and δ_{CP}=0. Results shown for the normal hierarchy.

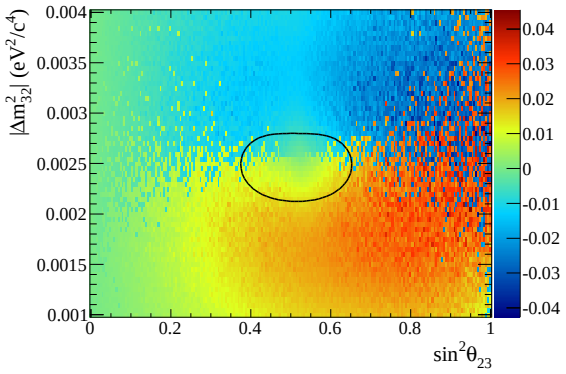


Figure 8: Maximum value of the fractional difference in the predicted number of 1-ring μ-like events between varying each of $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21} and δ_{CP} within their allowed ranges and their nominal values (these are the values shown in Tab. 3 for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0). The fractional difference is shown as function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, for an exposure of 6.3933×10^{20} POT. The number of events was calculated using the BANFF-tuned MC templates. The normal hierarchy was assumed. The T2K Run 1+2+3 90% CL region is superimposed for reference.

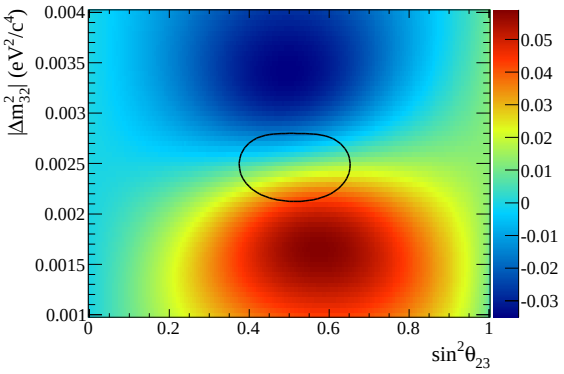


Figure 9: Fractional difference in the predicted number of 1-ring μ-like events between assuming the normal hierarchy and assuming the inverted hierarchy. The fractional difference is shown as function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, for an exposure of 6.3933×10^{20} POT. The number of events was calculated using the BANFF-tuned MC templates. The values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The T2K Run 1+2+3 90% CL region is superimposed for reference.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (no oscillation)	$\delta N_{SK}/N_{SK}$ (oscillations with typical parameter values)
SuperK detector + FSI + SI	3.63%	5.69%
BANFF (prefit)	25.97%	21.90%
BANFF (postfit)	2.74%	2.80%
Uncorrelated XSec	3.15%	5.33%
Non-23-sector oscillation parameters	0.%	0.22%
Total (BANFF prefit)	26.41%	23.25%
Total (BANFF postfit)	5.53%	8.29%

Table 6: Effect of 1σ systematic parameter variation on the number of 1-ring μ-like events, computed for no oscillations and for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4. For the oscillation scenario, the values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The normal mass hierarchy is assumed. The numbers shown were calculated for an exposure of 6.3933×10^{20} POT and BANFF-tuned MC templates were used.

5. Updated estimates of the effect of systematic uncertainties in the prediction of the single-ring μ-like event reconstructed energy spectrum

In this section, we present the effects of the systematics described in Sec. 3 on the expected number of single-ring μ-like events and the reconstructed energy spectrum in SuperK. These effects are shown both for no oscillations and for oscillations with $\sin^2\theta_{23} = 0.5$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4. For the oscillation cases, the values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} and δ_{CP}=0. The normal mass hierarchy is assumed. All results are generated for an integrated exposure of 6.3933×10^{20} POT and the BANFF-tuned MC templates are used.

The effects of systematic uncertainties on the expected number of events is summarised in Tab. 6 both for no oscillations and for oscillations with the typical parameter values given above. In this table, systematic parameters are grouped into a few categories and all parameter correlations are taken into account. Entries for both the BANFF prefit and postfit errors are provided. The effect of each parameter individually is presented in Tab. 7 in which the correlations between parameters are necessarily ignored.

The effects of the combined systematic uncertainties on the reconstructed energy spectrum of single-ring μ-like events is shown in Fig. 10 for the typical oscillation scenario used. This plot shows the total error envelope for the 1 μ-like ring event reconstructed energy spectrum; this is calculated as the $\pm1\sigma$ spread of bin contents from 100k toy MC experiments generated with randomized systematic parameters. In Fig. 10, all systematic parameters were considered and their correlations were taken into account.

Error envelopes from the effects of certain groups of systematic parameters are shown in Figs. 11 (SuperK detector + FSI + SI systematics only), 12 (BANFF systematics only), 13 (uncorrelated cross-section systematics only), and 14 (*non-23-sector* oscillation parameters only).

Detailed studies showing the effects of individual systematics on the expected number of single-ring μ-like events are shown in [6].

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (no oscillation)		$\delta N_{SK}/N_{SK}$ (oscillations with typical params)	
	BANFF prefit	BANFF postfit	BANFF prefit	BANFF postfit
$f^{SK}_{E;r}$	0.0%	0.0%	0.0%	0.0%
$f^{SK+FSI}_{0;r}$	0.074%	0.081%	0.13%	0.15%
$f^{SK+FSI}_{1;r}$	2.1%	2.2%	0.81%	0.88%
$f^{SK+FSI}_{2;r}$	0.43%	0.45%	0.98%	1.1%
$f^{SK+FSI}_{3;r}$	2.4%	2.0%	4.6%	4.0%
$f^{SK+FSI}_{4;r}$	0.017%	0.016%	0.25%	0.27%
$f^{SK+FSI}_{5;r}$	1.1%	0.92%	3.8%	3.3%
$f^{banff}_{0;t,r}$	0.36%	0.28%	0.84%	0.68%
$f^{banff}_{1;t,r}$	1.0%	0.76%	0.66%	0.52%
$f^{banff}_{2;t,r}$	1.9%	1.4%	0.12%	0.088%
$f^{banff}_{3;t,r}$	2.3%	1.6%	0.30%	0.22%
$f^{banff}_{4;t,r}$	3.4%	2.4%	2.2%	1.6%
$f^{banff}_{5;t,r}$	1.0%	0.67%	1.8%	1.3%
$f^{banff}_{6;t,r}$	0.64%	0.39%	1.7%	1.1%
$f^{banff}_{7;t,r}$	0.33%	0.2%	1.0%	0.64%
$f^{banff}_{8;t,r}$	0.41%	0.22%	1.3%	0.74%
$f^{banff}_{9;t,r}$	0.20%	0.076%	0.65%	0.27%
$f^{banff}_{10;t,r}$	0.051%	0.019%	0.17%	0.066%
$f^{banff}_{11;t,r}$	0.061%	0.051%	0.034%	0.030%
$f^{banff}_{12;t,r}$	0.061%	0.049%	0.049%	0.041%
$f^{banff}_{13;t,r}$	0.085%	0.068%	0.16%	0.13%
$f^{banff}_{14;t,r}$	0.10%	0.080%	0.27%	0.23%
$f^{banff}_{15;t,r}$	0.079%	0.061%	0.25%	0.2%
$f^{banff}_{16;t,r}$	25.0%	3.6%	18.0%	2.9%
$f^{banff}_{17;t,r}$	2.8%	1.0%	6.7%	2.3%
$f^{banff}_{18;t,r}$	7.9%	6.0%	4.2%	3.4%
$f^{banff}_{19;t,r}$	1.4%	0.55%	3.9%	1.6%
$f^{banff}_{20;t,r}$	0.36%	0.16%	1.2%	0.54%
$f^{banff}_{21;t,r}$	3.7%	1.5%	4.9%	2.0%
$f^{banff}_{22;t,r}$	1.7%	0.47%	5.4%	1.6%
$f_{pF;t,r}$	0.063%	0.06%	0.13%	0.12%
$f_{SF;t,r}$	2.8%	3.0%	0.85%	0.93%
$f_{Wshape;t,r}$	0.12%	0.079%	0.40%	0.27%
$f_{CCothShape;t,r}$	0.28%	0.31%	0.78%	0.90%
$f_{\pi-less\Delta;t,r}$	3.2%	2.5%	5.6%	4.4%
$f_{bindE;t,r}$	0.50%	0.42%	0.25%	0.19%
$f_{CCcoh;t}$	0.54%	0.59%	0.81%	0.96%
$f_{NC1\pi^{\pm};t}$	0.31%	0.22%	1.1%	0.78%
$f_{NCoth;t}$	0.25%	0.24%	0.85%	0.87%
$f_{CC\nu_e;t}$	0.00051%	0.00049%	0.0076%	0.0081%
$f_{CC\bar{\nu};t}$	1.2%	1.2%	2.4%	2.4%
$\sin^2\theta_{13}$	–	–	0.070%	0.079%
$\sin^2\theta_{12}$	–	–	0.040%	0.036%
Δm^2_{21}	–	–	0.027%	0.025%
δ_{CP}	–	–	0.21%	0.19%

Table 7: Effects of 1σ systematic parameter variations on the number of 1-ring μ-like events, computed for no oscillations and for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4. Correlations are ignored for this table. For the oscillation scenario, the values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The normal mass hierarchy is assumed. The numbers shown were calculated for an exposure of 6.3933×10^{20} POT and BANFF-tuned MC templates were used.

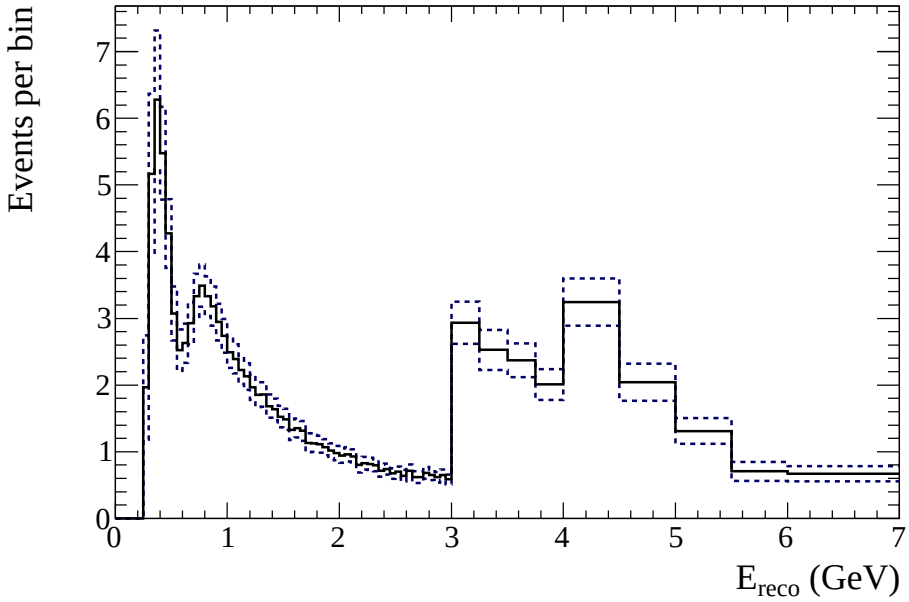


Figure 10: Total error envelope of the 1 μ-like ring event reconstructed energy spectrum, for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4. The values shown in Tab. 3 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The normal mass hierarchy is assumed. The numbers shown were calculated for an exposure of 6.3933×10^{20} POT. BANFF-tuned MC templates were used. The error envelope was calculated as the $\pm1\sigma$ spread of bin contents using an ensemble of 100k toy MC experiments generated with randomized systematic parameters. All systematic parameters were considered and their correlations were taken into account.

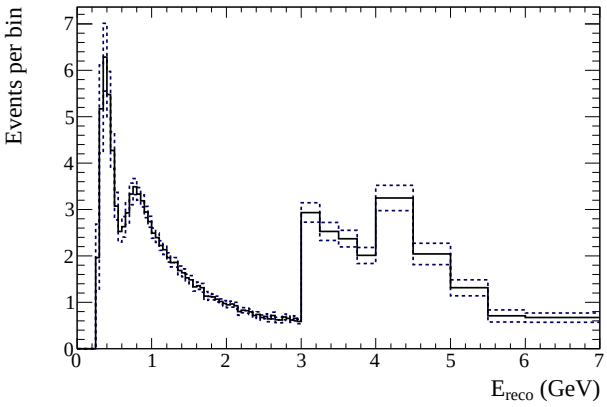


Figure 11: As in Fig. 10, but considering only SuperK detector + FSI + SI systematics.

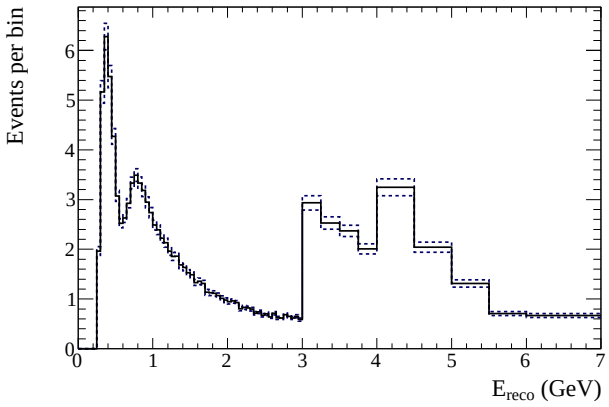


Figure 12: As in Fig. 10, but considering only BANFF systematics.

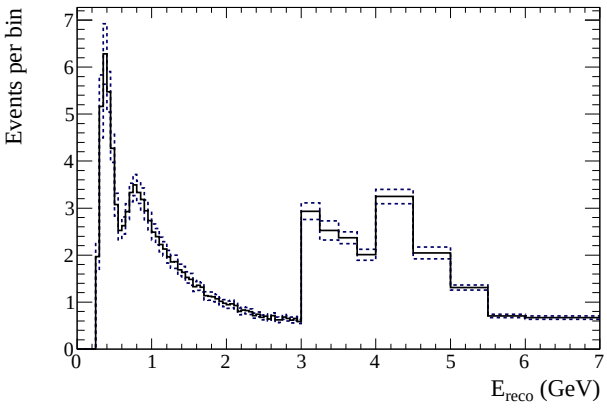


Figure 13: As in Fig. 10, but considering only uncorrelated cross-section systematics.

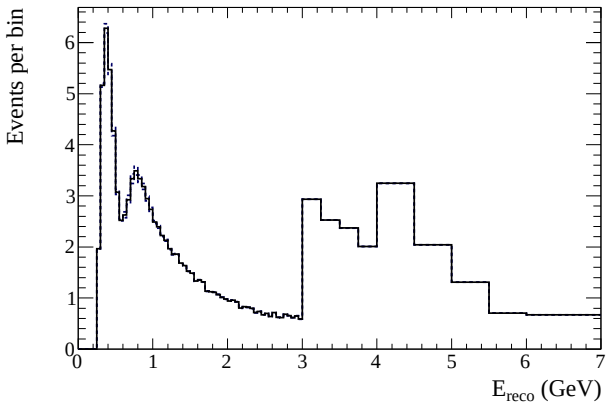


Figure 14: As in Fig. 10, but considering only *non-23-sector* oscillation parameter uncertainties.

6. Results of 6.3933×10^{20} POT fitter validation

The performance of the fitter was evaluated using an ensemble of 5.1k toy MC experiments, generated with an exposure of 6.3933×10^{20} POT, and input oscillation parameters $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4 and $\sin^2\theta_{23} = 0.37$. A further 5.1k toy MC experiments were created for each of the two cases $\sin^2\theta_{23} = 0.513$ and $\sin^2\theta_{23} = 0.65$. A further set of toy MC experiments with the same input oscillation parameters and an exposure of 7.8×10^{21} POT (the T2K ultimate POT goal) were also used. These simulated experiments included statistical fluctuations and fluctuations of all 44 systematic parameters and δ_{CP}. The toy MC experiments were fitted using MINUIT and the MINUIT HESSE errors were computed. Residual and pull[4] distributions were studied for oscillation and systematic parameters.

Residual distributions for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ are shown in Figs. 15, 17, and 19 for 6.3933×10^{20} POT. A clear three peak structure is seen in $\sin^2\theta_{23}$, with peaks occurring at the input point, the mirror point, and the point where maximum disappearance occurs[5]. Figs. 16, 18, and 20 are the corresponding plots for 7.8×10^{21} POT. The three-peak structure is lost with high statistics; there are peaks only at the input point and the mirror point.

Residual distributions for the systematic parameters are shown in Figs. 21-24 for $\sin^2\theta_{23} = 0.37$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, Figs. 29-32 for $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, and Figs. 37-40 for $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4. The corresponding pull plots are shown in Figs. 25-28, Figs. 33-36, and Figs. 41-44.

No problem was identified with the fitter performance, which performed as expected. There are biases in the fitted - true values for 3 systematic parameters, namely spectral function, π-less Δ decay and the NC bin in the SuperK detector efficiency + FSI + SI matrix, but these biases are understood. The bias in the spectral function systematic is due to its true input values being drawn from a uniform distribution between 0 and 1 rather than from a Gaussian distribution. The bias in π-less Δ decay is due to the method of calculation of event weights when this parameter is varied. By default, 20% of Δs decay in this way in NEUT, and the uncertainty is 20%. This means that it is unphysical to tweak this parameter to be more negative than -1σ, and consequently there is an asymmetry in weights between positive tweaks (which change continuously as the tweak increases) and negative tweaks (for which all tweaks more negative than -1σ are assigned the same weight as that for a -1σ tweak). The bias in the NC bin in the SuperK detector efficiency + FSI + SI bin matrix comes from the low statistics at current POT. When Poisson fluctuations are made of the bin contents of the E_{reco} distributions, a downward fluctuation is more likely than an upward fluctuation (see replies to comments and revision history for TN141 [4]). Adding the fact that the NC modes are distributed around the oscillation maximum, the (most likely) downwards Poisson fluctuations translate into a (most likely) negative value of the NC efficiency, explaining a bias to negative values for this parameter. As can be seen from the plots, when moving away from the point of maximal disappearance as input (and so reducing the depth of the oscillation dip), the NC bin in the SuperK detector efficiency + FSI + SI matrix bias is reduced.

[4] pull $= (f_{bf} - f_{true})/\sigma_{fit}$

[5] In the 3-flavour oscillation formula[6] $P(\nu_\mu \to \nu_\mu) \approx 1 - \left(\cos^4(\theta_{13})\sin^2(2\theta_{23}) + \sin^2(\theta_{23})\sin^2(2\theta_{13})\right)\sin^2\left(\frac{\Delta m^2_{31} L}{4E}\right)$, the maximum amount of disappearance occurs not at $\sin^2(\theta_{23}) = 0.5$, but it is offset slightly due to the next-to-leading order term. The extremum occurs at $\sin^2(\theta_{23}) = \frac{1}{2(1-\sin^2(\theta_{13}))} \approx 0.513$. In addition, the oscillation probability is approximately symmetric about this point, meaning that for every input value $!= 0.513$, there exists a mirror point with a similar oscillation probability. More discussion of this effect can be found in Appendix E of [5].

[6] In the fits, we use full 3-flavour oscillation formulae, and not this approximation. See and Appendix A in [4] for full details.

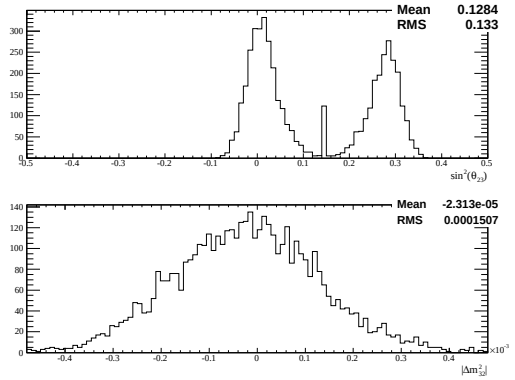


Figure 15: Residual distributions for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.37$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

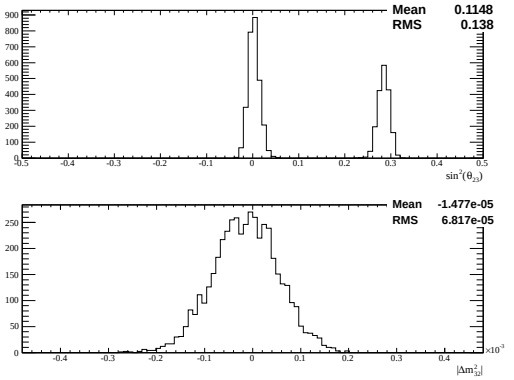


Figure 16: Residual distributions for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ 5.1k toy MC experiments at 7.8×10^{21} POT (the T2K ultimate POT goal), generated at $\sin^2\theta_{23} = 0.37$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

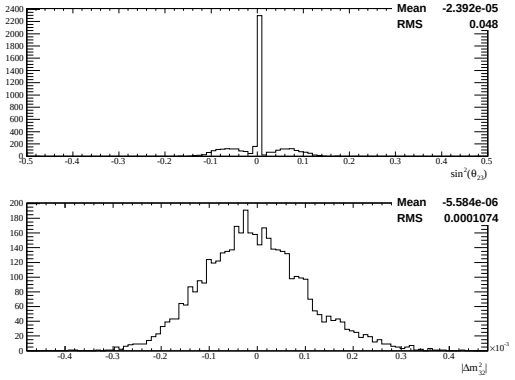


Figure 17: Residual distributions for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

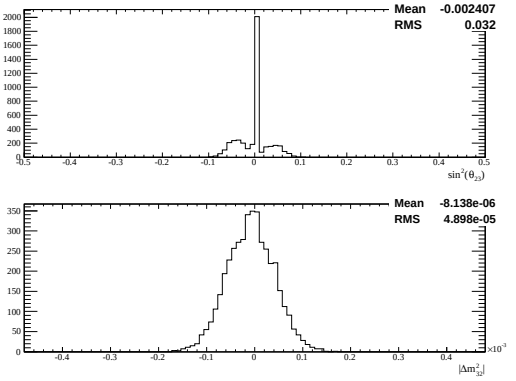


Figure 18: Residual distributions for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ 5.1k toy MC experiments at 7.8×10^{21} POT (the T2K ultimate POT goal), generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

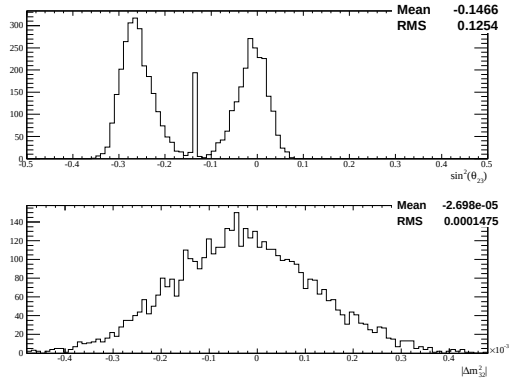


Figure 19: Residual distributions for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

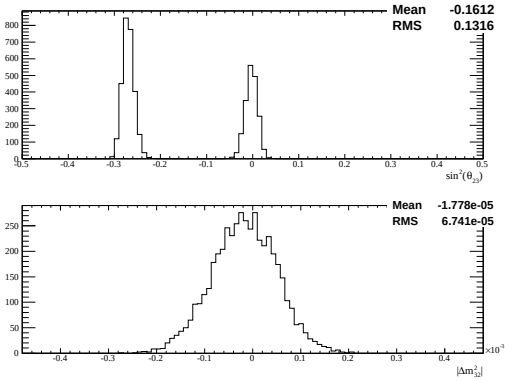


Figure 20: Residual distributions for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ 5.1k toy MC experiments at 7.8×10^{21} POT (the T2K ultimate POT goal), generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

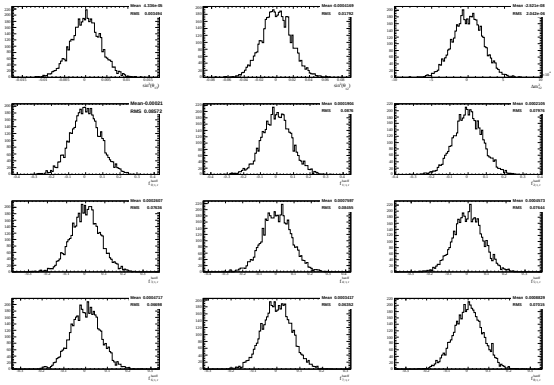


Figure 21: Residual distributions for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21}, and $f^{banff}_{0;t,r}$-$f^{banff}_{8;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.37$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

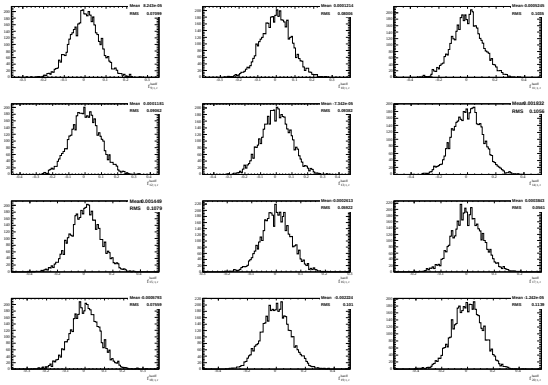


Figure 22: Residual distributions for $f^{banff}_{9;t,r}$-$f^{banff}_{20;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.37$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

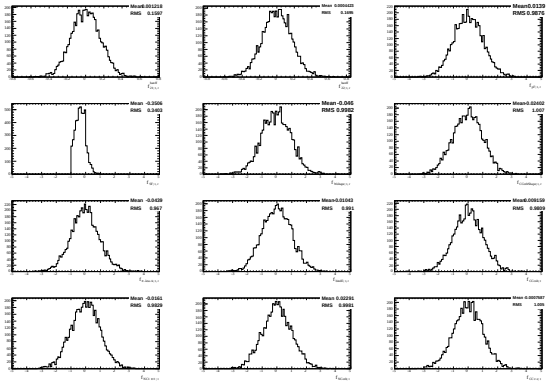


Figure 23: Residual distributions for $f^{banff}_{21;t,r}$-$f^{banff}_{22;t,r}$, and the first 10 uncorrelated cross section parameters, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.37$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

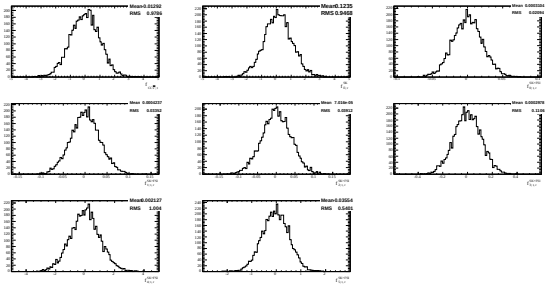


Figure 24: Residual distributions for $f_{CC\bar{\nu};t}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{0;r}$-$f^{SK+FSI}_{5;r}$ computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.37$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

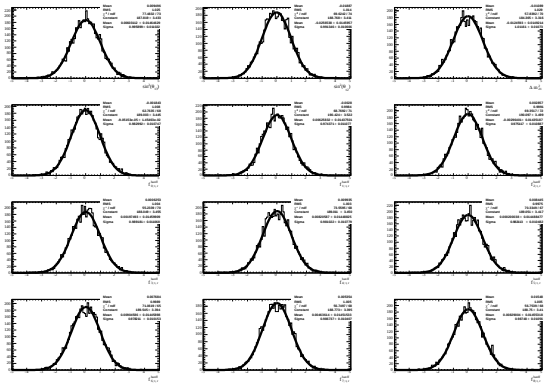


Figure 25: Pull distributions for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21}, and $f^{banff}_{0;t,r}$-$f^{banff}_{8;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.37$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

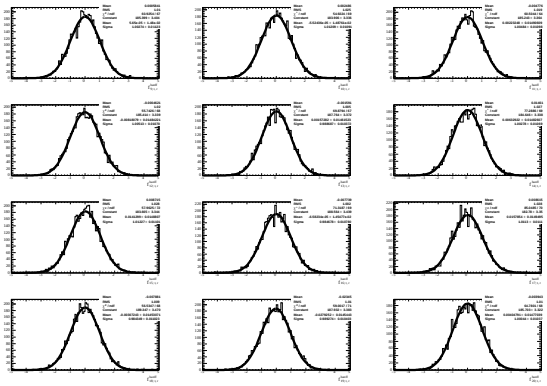


Figure 26: Pull distributions for $f^{banff}_{9;t,r}$-$f^{banff}_{20;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.37$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

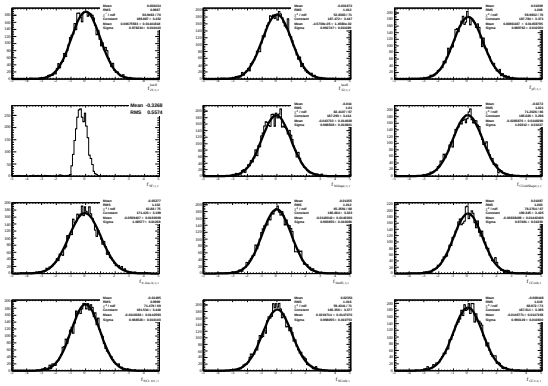


Figure 27: Pull distributions for $f^{banff}_{21;t,r}$-$f^{banff}_{22;t,r}$, and the first 10 uncorrelated cross section parameters, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.37$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

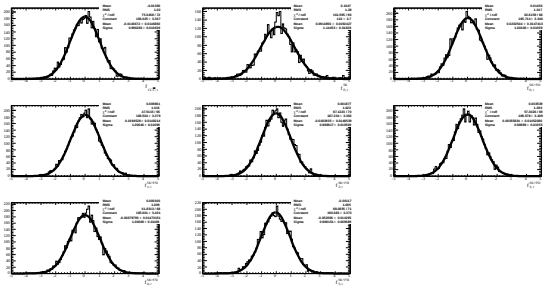


Figure 28: Pull distributions for $f_{CC\bar{\nu};t}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{0;r}$-$f^{SK+FSI}_{5;r}$ computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.37$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

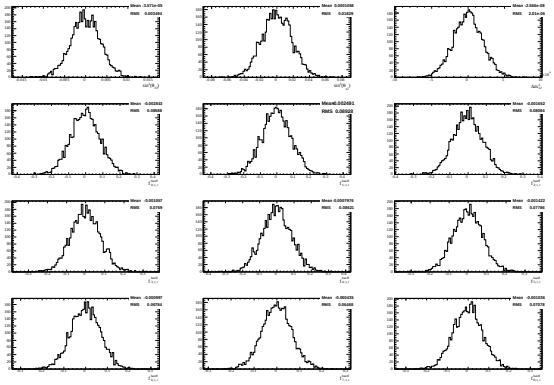


Figure 29: Residual distributions for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21}, and $f^{banff}_{0;t,r}$-$f^{banff}_{8;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

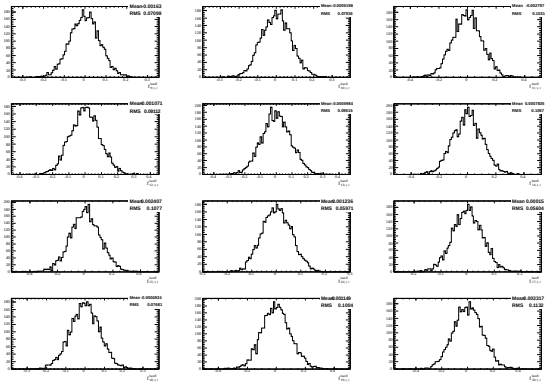


Figure 30: Residual distributions for $f^{banff}_{9;t,r}$-$f^{banff}_{20;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

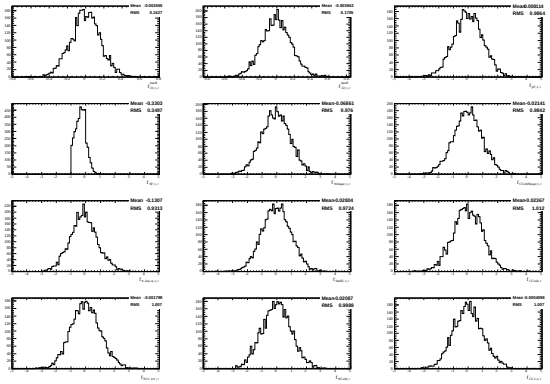


Figure 31: Residual distributions for $f^{banff}_{21;t,r}$-$f^{banff}_{22;t,r}$, and the first 10 uncorrelated cross section parameters, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

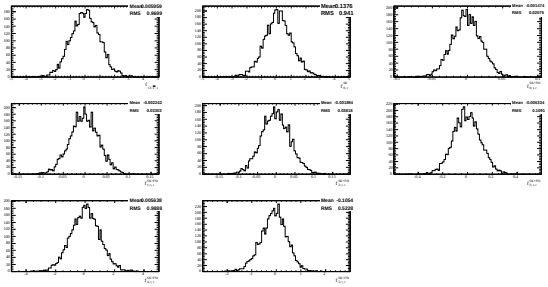


Figure 32: Residual distributions for $f_{CC\bar{\nu};t}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{0;r}$-$f^{SK+FSI}_{5;r}$ computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

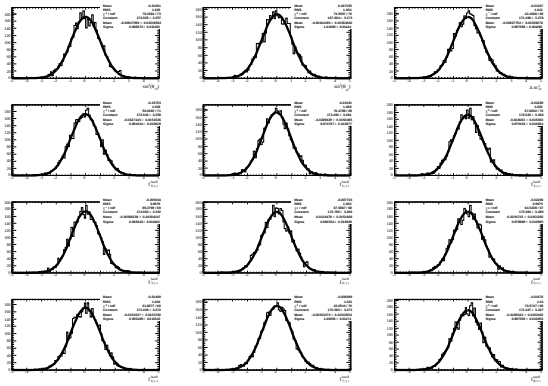


Figure 33: Pull distributions for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21}, and $f^{banff}_{0;t,r}$-$f^{banff}_{8;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

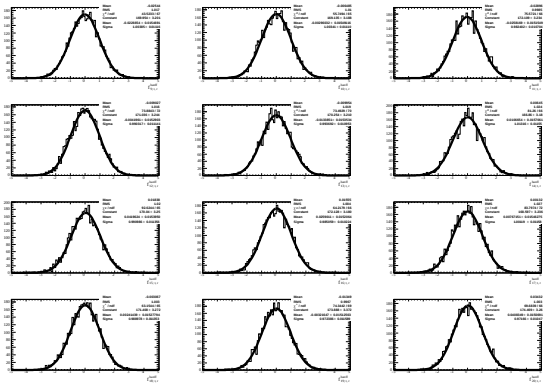


Figure 34: Pull distributions for $f^{banff}_{9;t,r}$-$f^{banff}_{20;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

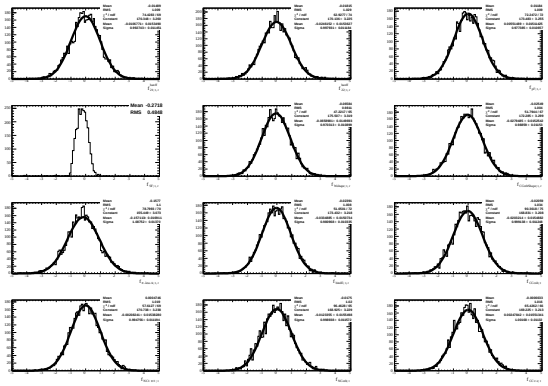


Figure 35: Pull distributions for $f^{banff}_{21;t,r}$-$f^{banff}_{22;t,r}$, and the first 10 uncorrelated cross section parameters, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

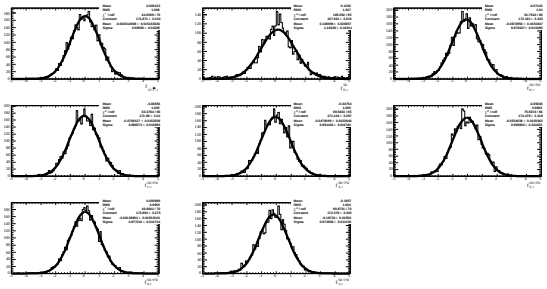


Figure 36: Pull distributions for $f_{CC\bar{\nu};t}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{0;r}$-$f^{SK+FSI}_{5;r}$ computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

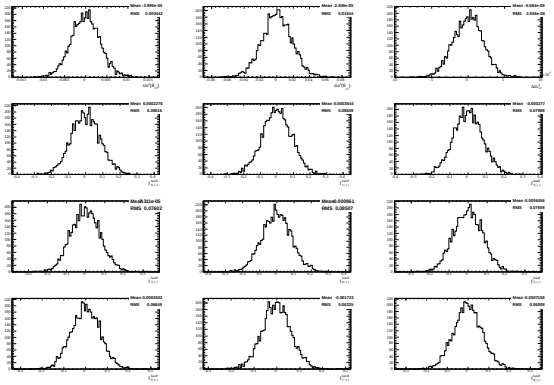


Figure 37: Residual distributions for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21}, and $f^{banff}_{0;t,r}$-$f^{banff}_{8;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

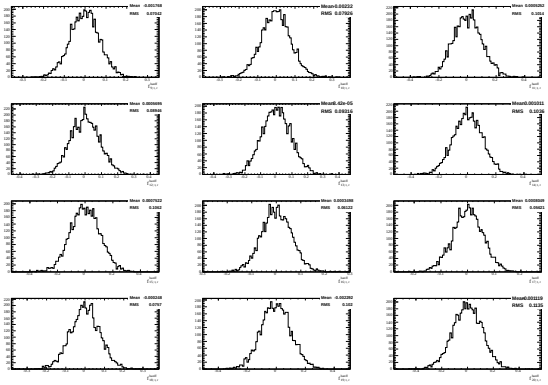


Figure 38: Residual distributions for $f^{banff}_{9;t,r}$-$f^{banff}_{20;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

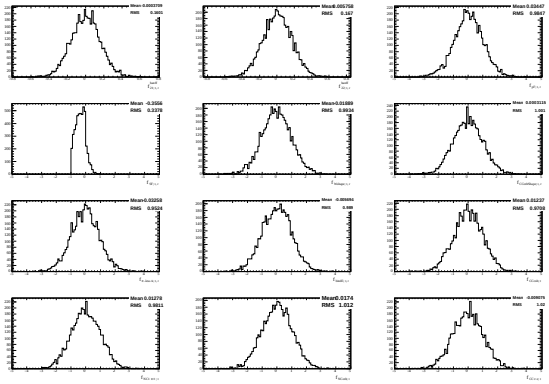


Figure 39: Residual distributions for $f^{banff}_{21;t,r}$-$f^{banff}_{22;t,r}$, and the first 10 uncorrelated cross section parameters, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

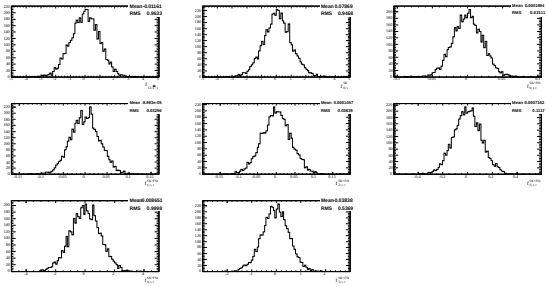


Figure 40: Residual distributions for $f_{CC\bar{\nu};t}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{0;r}$-$f^{SK+FSI}_{5;r}$ computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

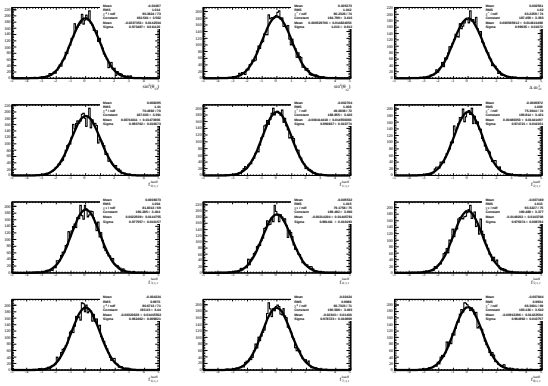


Figure 41: Pull distributions for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21}, and $f^{banff}_{0;t,r}$-$f^{banff}_{8;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

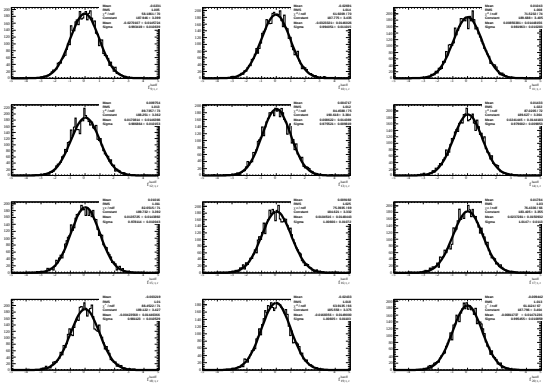


Figure 42: Pull distributions for $f^{banff}_{9;t,r}$-$f^{banff}_{20;t,r}$, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

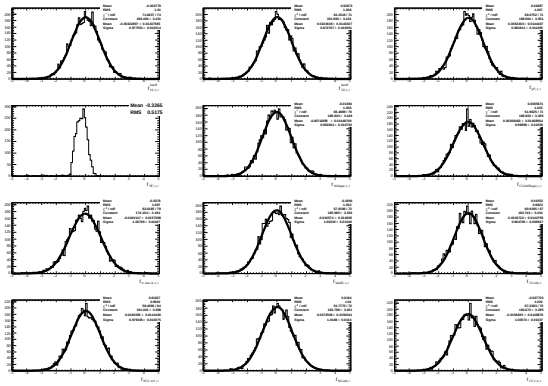


Figure 43: Pull distributions for $f^{banff}_{21;t,r}$-$f^{banff}_{22;t,r}$, and the first 10 uncorrelated cross section parameters, computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

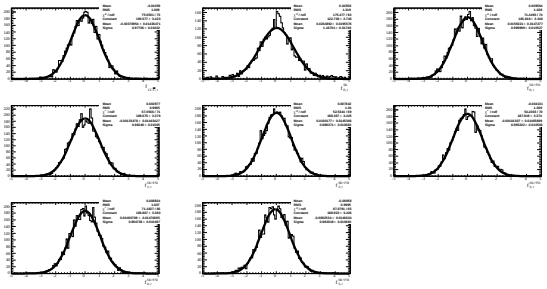


Figure 44: Pull distributions for $f_{CC\bar{\nu};t}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{0;r}$-$f^{SK+FSI}_{5;r}$ computed from fits of 5.1k toy MC experiments at 6.3933×10^{20} POT, generated at $\sin^2\theta_{23} = 0.65$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

Analysis	Run period	$\sin^2\theta_{23}$	Δm^2_{32}	N_{obs}	N_{exp}	χ^2_{bf}/ndf	p-value
2013 3-flavour Normal	Run 1+2+3+4a	0.514	$2.463 \cdot 10^{-3}$	115	115.77	64.88/73	0.88
2013 3-flavour Inverted	Run 1+2+3+4a	0.511	$2.530 \cdot 10^{-3}$	115	115.72	64.87/73	0.89

Table 8: Summary of best-fit parameters for the oscillation fits to the Run 1+2+3+4a dataset. The quoted best-fit χ^2 values (χ^2_{bf}) were computed from Eq. 5 in Sec. 4.1.5 in Ref. [4] with the reconstructed energy binning used in each fit. The table also shows the p-values obtained from the goodness-of-fit test described in Sec. 4.2 in [4] for all fits.

7. Results of the ν_μ-disappearance analysis with the Run 1+2+3+4a (6.3933×10^{20} POT) dataset

7.1. Best-fit oscillation parameters and systematic parameter pulls

A 3-flavour ν_μ-disappearance fit was performed on the combined 6.3933×10^{20} POT Run 1+2+3+4a dataset, using the method described in Sec. 4.1 of Ref. [4], in which all 44 nuisance parameters considered in this work are allowed to float.

This analysis found $|\Delta m^2_{32}| = 2.463\times10^{-3}$ eV^2/c^4 and $\sin^2\theta_{23} = 0.514$ to be the best-fit values for the Run 1+2+3+4a dataset for a fit assuming the normal mass hierarchy. A similar best-fit value was obtained from a separate fit, assuming the inverted mass hierarchy, of $|\Delta m^2_{32}| = 2.530\times10^{-3}$ eV^2/c^4 and $\sin^2\theta_{23} = 0.511$.

The results of these fits are summarized in Tab. 8. In this table are the best-fit oscillation parameters, the observed and expected number of single-ring μ-like events, and the values of χ^2 per degree of freedom at the best-fit.

The pull for each systematic parameter f included in the fit, is calculated as

$$\frac{f_{bf} - f_{nom}}{\sigma_{bf}}, \tag{2}$$

where f_{bf} is the best-fit value of the systematic parameter, f_{nom} is the nominal value of the parameter (corresponding to no systematic variation), and σ_{bf} is the Hesse error output from Minuit.

The pulls of all 44 nuisance parameters allowed to float in the fit are shown in Figs. 45 and 46 for the fit to the Run 1+2+3+4a dataset assuming the normal and inverted mass hierarchy respectively. From these figures, it is apparent that most systematic parameters barely move from their nominal values. We note that, although all 44 systematic parameters are allowed to float in the fit, only 2(2) of them move by more than $\pm0.1\sigma$ for the normal(inverted) fit. When a systematic parameter that is free to move stays close to its nominal value, this means that the reduction in χ^2 from moving that parameter is less than the corresponding increase in the penalty term (χ^2 is defined in equation 3 in Sec. 4.1.5 in Ref. [4]). Only certain systematics are able to change the predicted reconstructed energy spectrum to a sufficient extent to offset the penalty term.

7.2. Goodness-of-fit tests

Goodness-of-fit tests were performed as described in Sec. 4.2 of Ref. [4], and the p-values for the fits to the Run 1+2+3+4a dataset are shown in Tab. 8. The χ^2 distribution from which the p-value was calculated is shown for the Run 1+2+3+4a data in Figs. 47 and 48 for the normal and inverted mass hierarchy respectively. Both fits have a high degree of plausibility.

7.3. Best-fit single-ring μ-like reconstructed energy spectra

The reconstructed neutrino energy distribution of the single-ring μ-like events in the Run 1+2+3+4a dataset assuming the normal mass hierarchy is shown in Fig. 49 along with the best-fit prediction and the prediction made under the no-oscillation hypothesis. Also shown is the ratio between the data and the no-oscillation spectrum and the ratio between the best-fit spectrum and the the no-oscillation spectrum; in both of these ratios, the characteristic energy-dependent deficit can be clearly seen. The corresponding spectra for the fit assuming the inverted mass hierarchy is shown in Fig. 50. All these distributions are shown in the 73-bin scheme in reconstructed neutrino energy that was used in the fitting procedure (see Sec. 2.2.5 in Ref. [4]).

Since the reconstructed energy binning used in the fit is relatively fine, it can be beneficial to see the above distributions using a coarser binning. The best-fit spectra obtained from the fits to the Run 1+2+3+4a dataset is shown using a coarser binning scheme[7] in Figs. 51 and 52 for the normal and inverted mass hierarchy respectively. It is also interesting to see the 32 components of the best-fit spectrum, each of which is calculated separately in this analysis. These components are shown for the Run 1+2+3+4a fit in Figs. 53 and 54 for the normal and inverted mass hierarchy respectively. In these plots, the components of the spectrum are grouped into 5 categories: ν_μ CCQE, ν_μ CCnonQE, $\bar{\nu}_\mu$ CC, $\nu_e/\bar{\nu}_e$ CC and NC.

Finally, the normal and inverted mass hierarchy best-fit predictions are compared for the Run 1+2+3+4a dataset only in Fig. 55, using the 73 reconstructed energy bins used in the fit, and in Fig. 56, using the coarser binning described above. The two best-fit spectra are nearly identical.

[7] 16 reconstructed energy bins arranged as follows: 1 0.3-GeV bin from 0 - 0.3 GeV, 8 0.1-GeV bins from 0.3 - 1.1 GeV, 1 0.15-GeV bin from 1.1 GeV - 1.25 GeV, 1 0.25-GeV bin from 1.25 - 1.5 GeV GeV, 1 0.5-GeV bin from 1.5 - 2 GeV, 2 1-GeV bin from 2-4 GeV, 1 2-GeV bin from 4-6 GeV and 1 bin above 6 GeV.

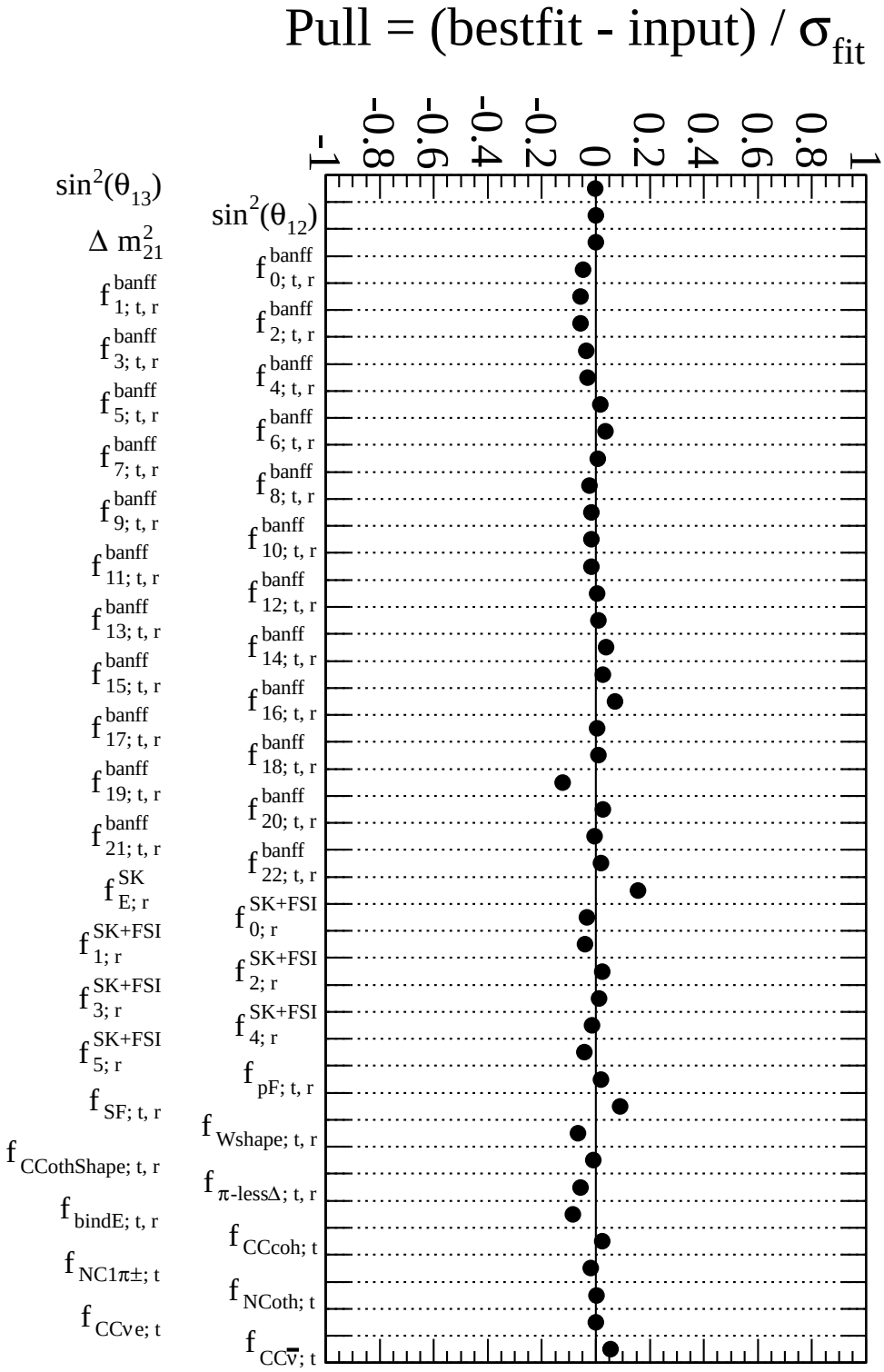


Figure 45: Systematic parameter pulls for the fit to the Run 1+2+3+4a dataset. The normal mass hierarchy is assumed.

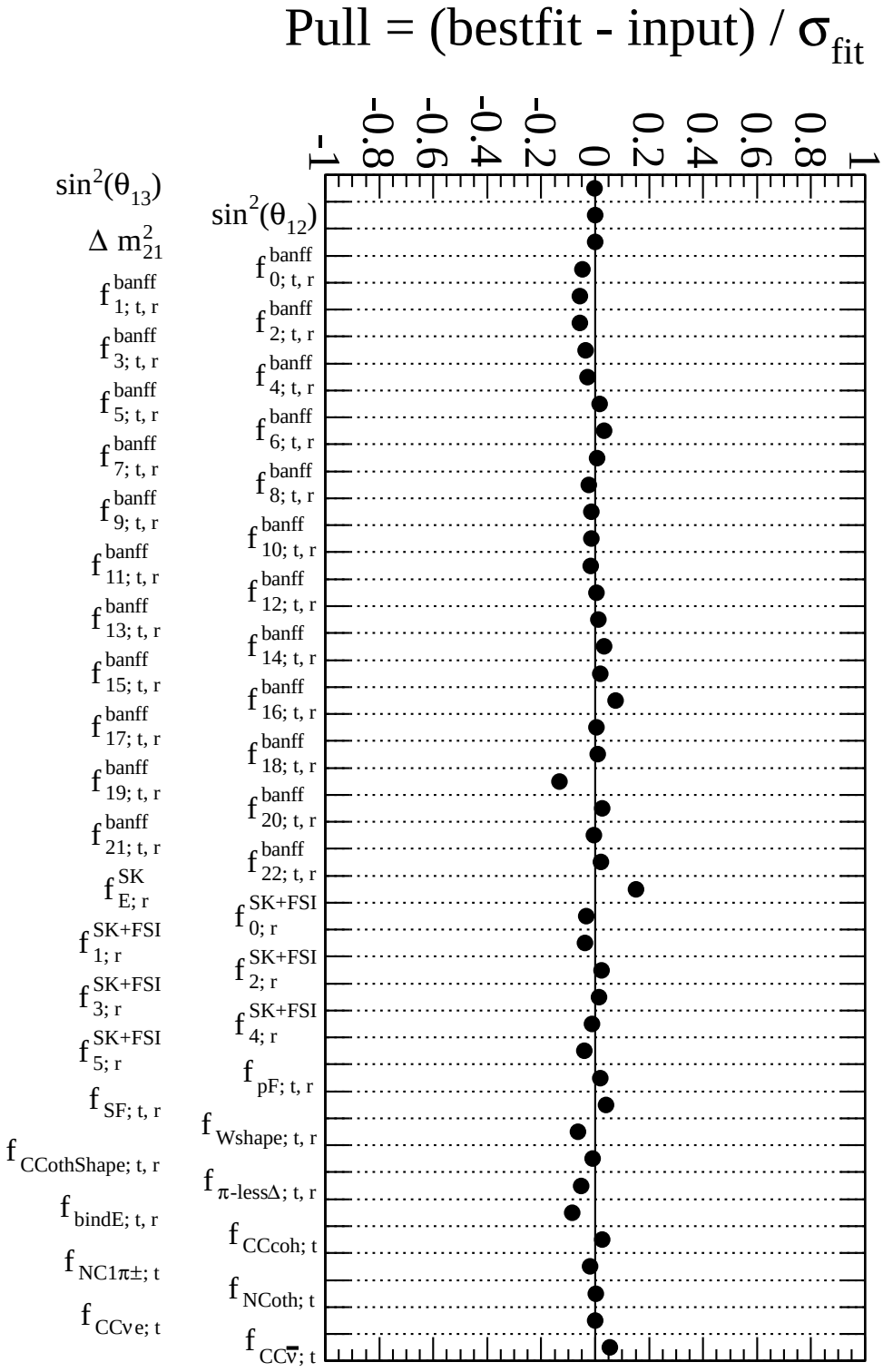


Figure 46: Systematic parameter pulls for the fit to the Run 1+2+3+4a dataset. The inverted mass hierarchy is assumed.

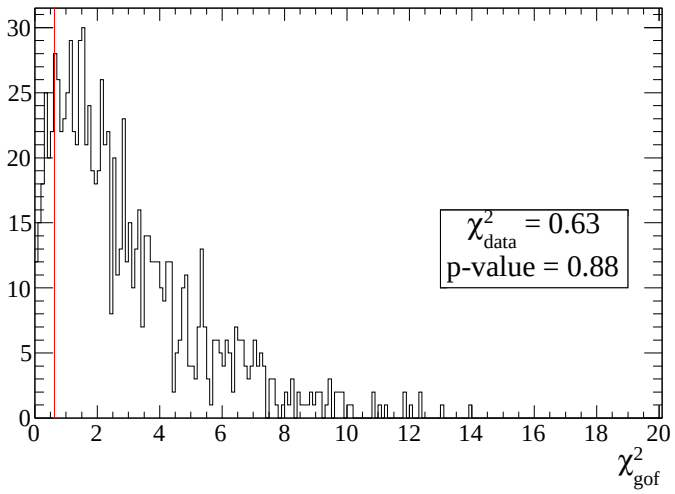


Figure 47: Distribution of χ^2_{gof}, the goodness-of-fit (gof) χ^2, from 1k toy MC experiments generated at the Run 1+2+3+4a best-fit oscillation point. The χ^2_{gof} value for the Run 1+2+3+4a data (χ^2_{data}) is highlighted. The normal mass hierarchy is assumed.

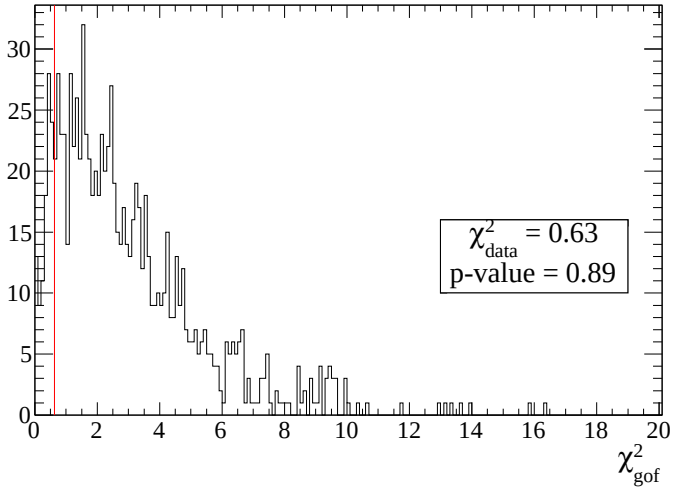


Figure 48: Distribution of χ^2_{gof}, the goodness-of-fit (gof) χ^2, from 1k toy MC experiments generated at the Run 1+2+3+4a best-fit oscillation point. The χ^2_{gof} value for the Run 1+2+3+4a data (χ^2_{data}) is highlighted. The inverted mass hierarchy is assumed.

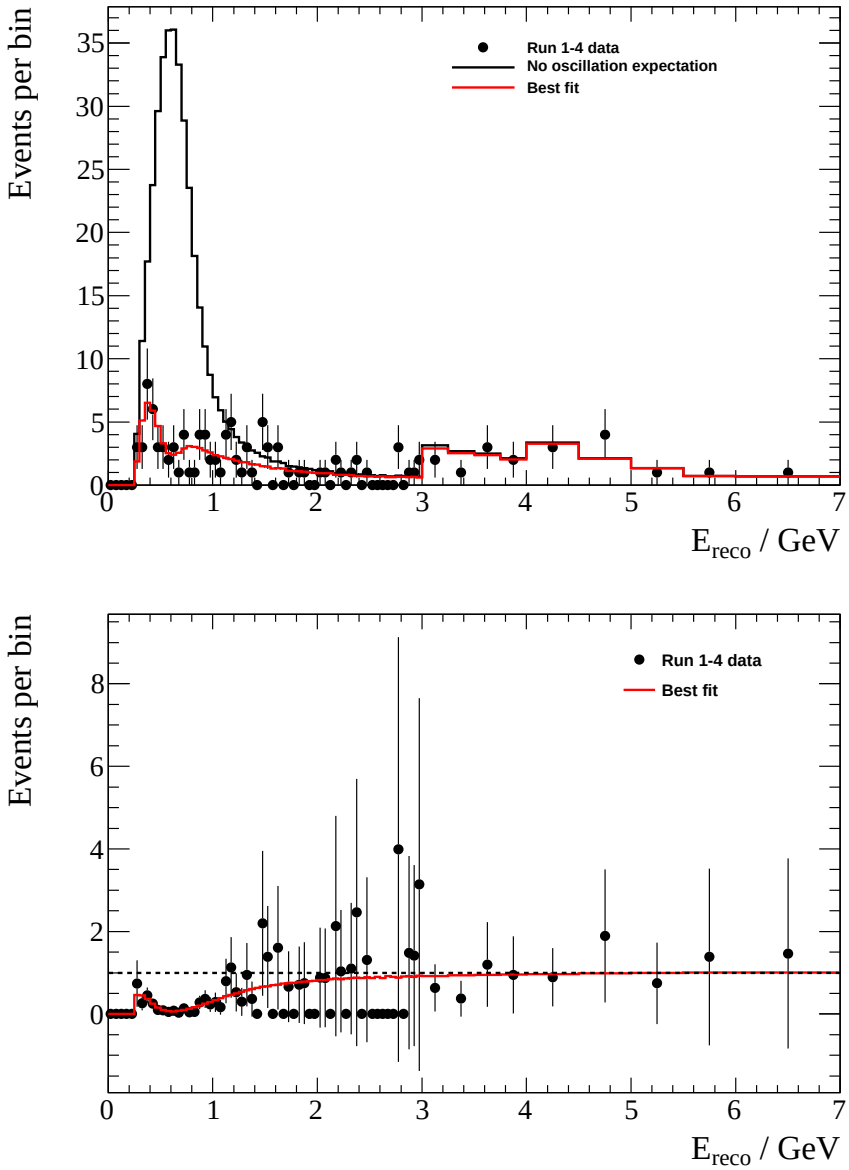


Figure 49: Best-fit reconstructed energy spectrum (top) obtained from the fit to the combined Run 1+2+3+4a dataset. The Run 1+2+3+4a dataset and the prediction obtained from the no-oscillation hypothesis are also shown. The spectra are shown using the 73 bins used in the fit (see Sec. 2.2.5 in Ref. [4]). The ratio of the data and best-fit distribution to the no-oscillation hypothesis is shown in the bottom plot. The characteristic energy-dependent deficit can be clearly seen. The normal mass hierarchy is assumed.

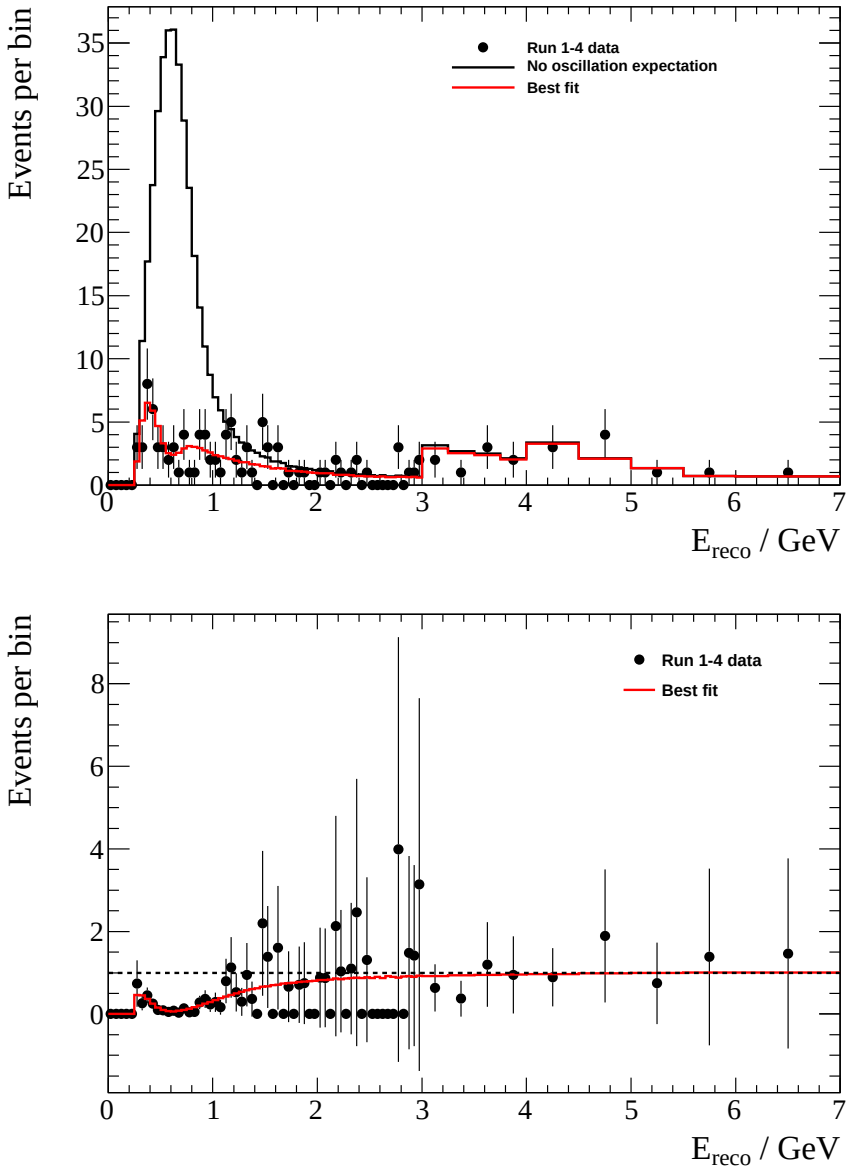


Figure 50: Best-fit reconstructed energy spectrum (top) obtained from the fit to the combined Run 1+2+3+4a dataset. The Run 1+2+3+4a dataset and the prediction obtained from the no-oscillation hypothesis are also shown. The spectra are shown using the 73 bins used in the fit (see Sec. 2.2.5 in Ref. [4]). The ratio of the data and best-fit distribution to the no-oscillation hypothesis is shown in the bottom plot. The characteristic energy-dependent deficit can be clearly seen. The inverted mass hierarchy is assumed.

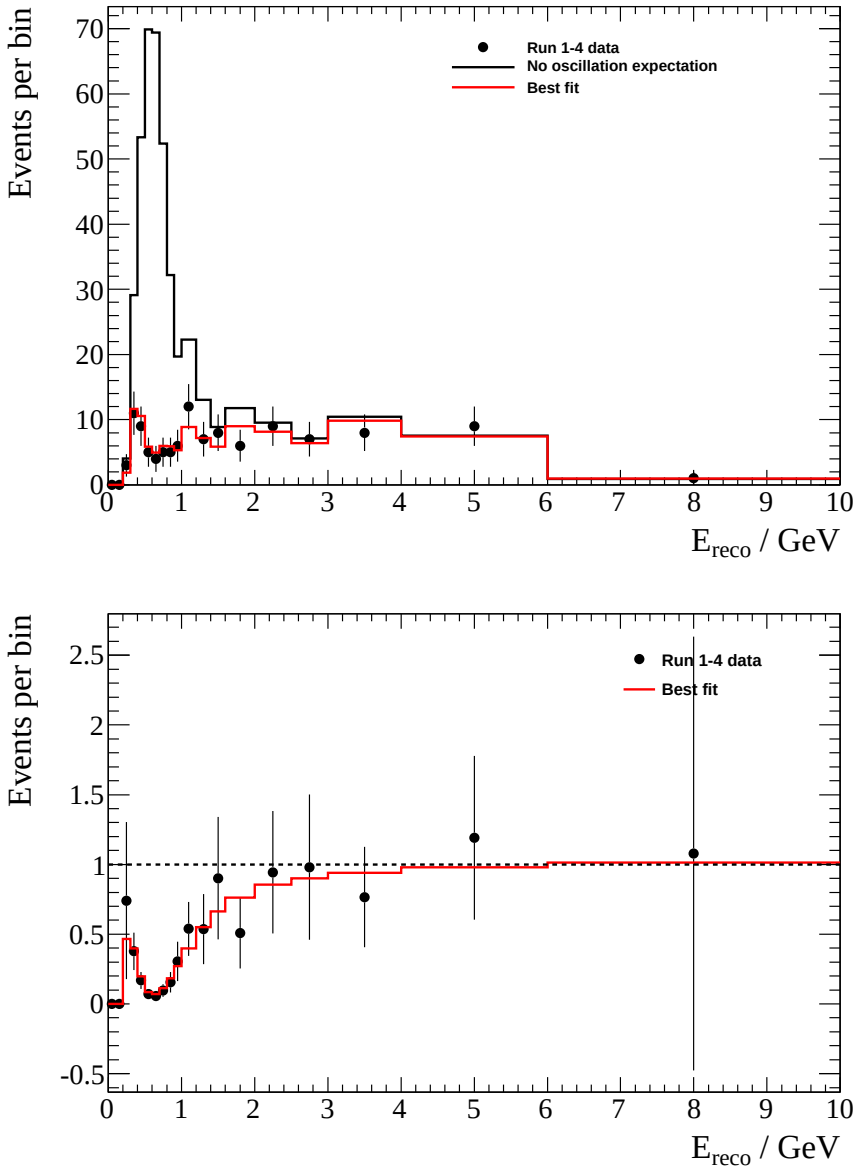


Figure 51: Best-fit reconstructed energy spectrum (top) obtained from the fit to the combined Run 1+2+3+4a dataset. The Run 1+2+3+4a dataset and the prediction obtained from the no-oscillation hypothesis are also shown. The spectra are shown using the coarse binning scheme discussed in the text (see Sec. 7.3). The characteristic energy-dependent deficit can be clearly seen. The normal mass hierarchy is assumed.

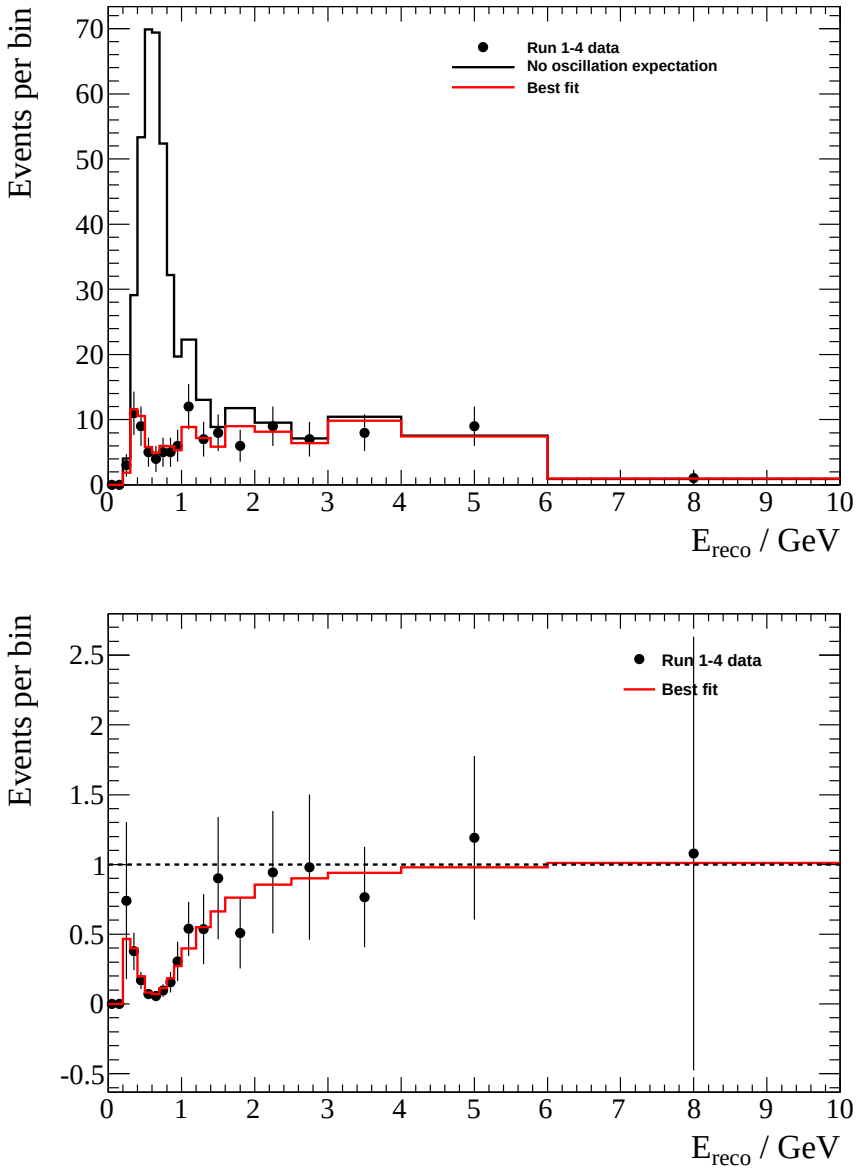


Figure 52: Best-fit reconstructed energy spectrum (top) obtained from the fit to the combined Run 1+2+3+4a dataset. The Run 1+2+3+4a dataset and the prediction obtained from the no-oscillation hypothesis are also shown. The spectra are shown using the coarse binning scheme discussed in the text (see Sec. 7.3). The characteristic energy-dependent deficit can be clearly seen. The inverted mass hierarchy is assumed.

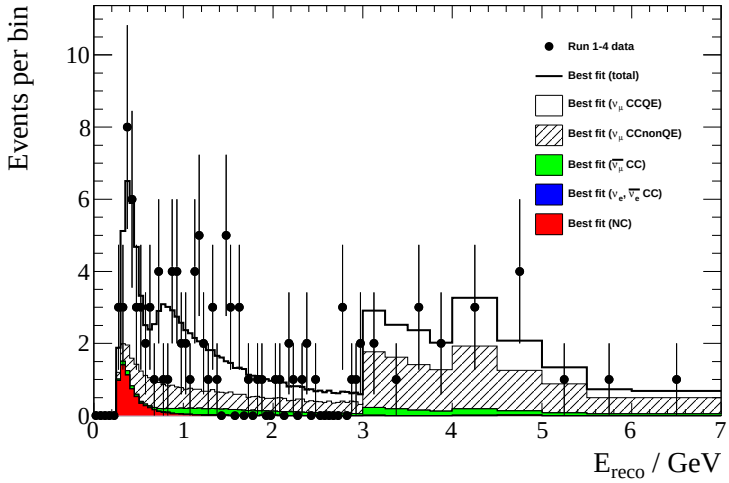


Figure 53: Best-fit reconstructed energy spectrum in the 73-bin scheme used in the fit (see Sec. 2.2.5 in Ref. [4]) and its components obtained from the fit to the Run 1+2+3+4a dataset. The 32 components, each of which is calculated separately, are grouped into just 5 categories: ν_μ CCQE, ν_μ CCnQE, $\bar{\nu_\mu}$ CC, $\nu_e/\bar{\nu_e}$ CC and NC. The normal mass hierarchy is assumed.

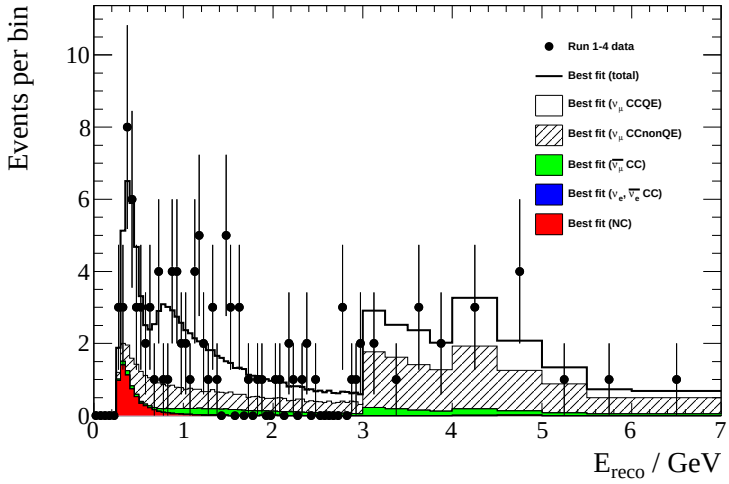


Figure 54: Best-fit reconstructed energy spectrum in the 73-bin scheme used in the fit (see Sec. 2.2.5 in Ref. [4]) and its components obtained from the fit to the Run 1+2+3+4a dataset. The 32 components, each of which is calculated separately, are grouped into just 5 categories: ν_μ CCQE, ν_μ CCnQE, $\bar{\nu_\mu}$ CC, $\nu_e/\bar{\nu_e}$ CC and NC. The inverted mass hierarchy is assumed.

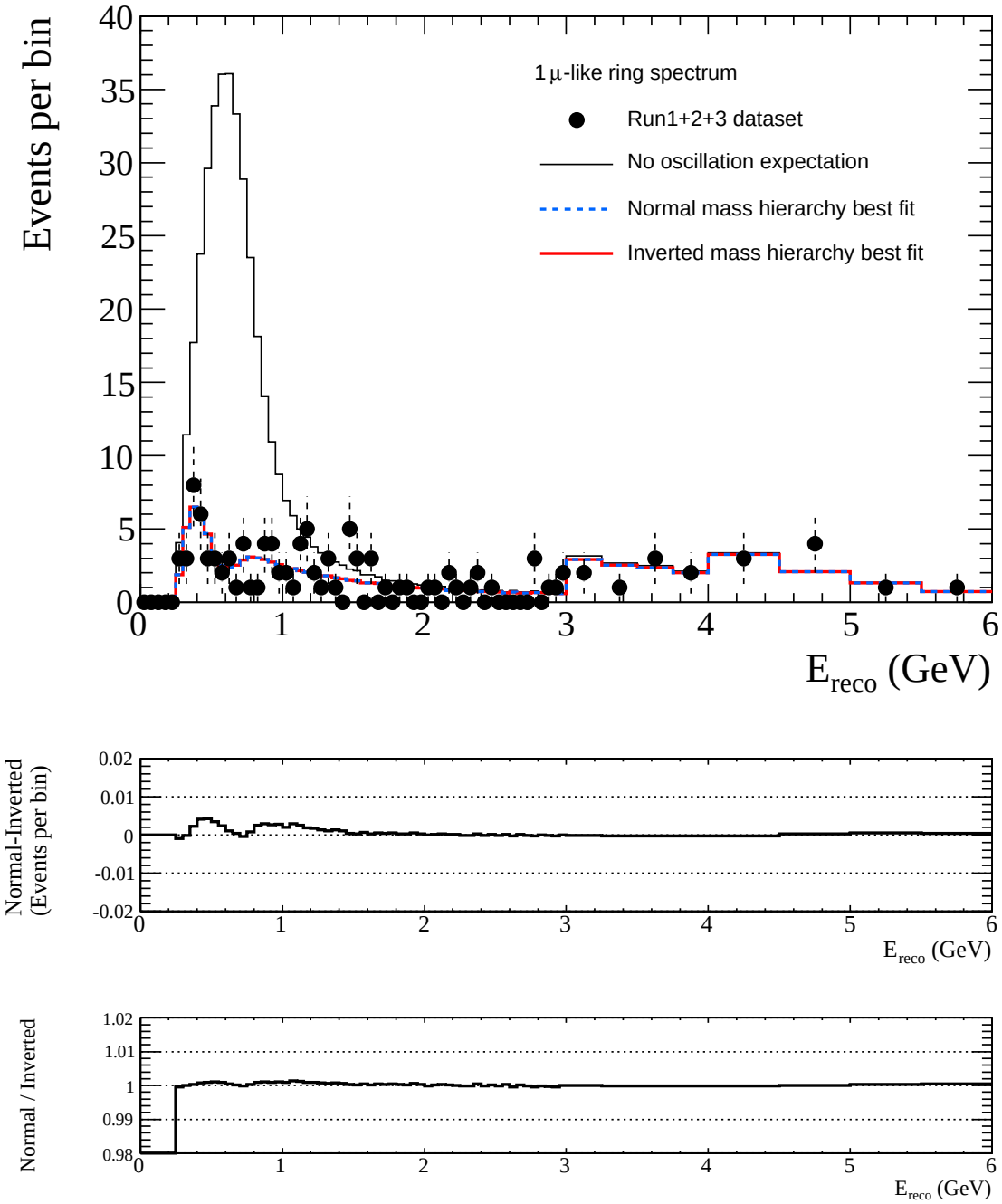


Figure 55: Comparison between best-fit reconstructed energy spectra obtained from the normal and inverted mass hierarchy fits to the Run 1+2+3+4a datasets. The spectra are shown using the 73 bins used in the fit (see Sec. 2.2.5 in Ref. [4]). Also shown are the absolute difference between the normal and inverted mass hierarchy best-fit spectra and the ratio between them.

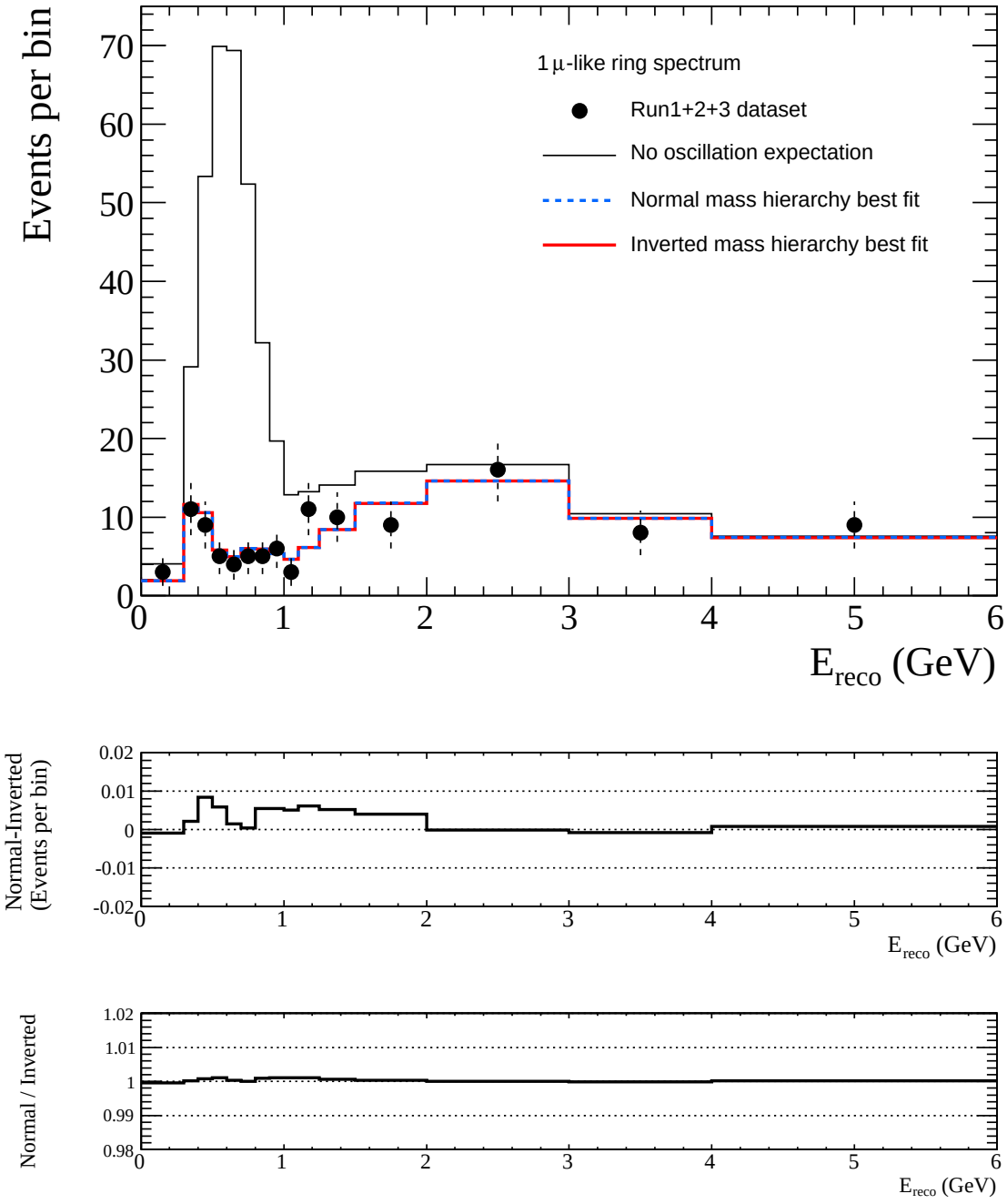


Figure 56: Comparison between best-fit reconstructed energy spectra obtained from the normal and inverted mass hierarchy fits to the Run 1+2+3+4a datasets. The spectra are shown using the coarse binning scheme discussed in the text (see Sec. 7.3). Also shown are the absolute difference between the normal and inverted mass hierarchy best-fit spectra and the ratio between them.

7.4. Confidence regions

The first step in constructing confidence contours in the 2-dimensional ($\sin^2\theta_{23}$, $|\Delta m^2_{32}|$) oscillation parameter space is to construct the $\Delta\chi^2$ surface in that space. The oscillation parameter space is divided into a rectangular grid of width $\sin^2\theta_{23} = 0.005$, $|\Delta m^2_{32}| = 0.05\times10^{-3}$ eV^2/c^4, with a grid point at the centre of each rectangle. Two separate fits are made of the data at each grid point: in one fit, the values of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ are fixed to the true values of the grid point and the 44 systematic parameters are allowed to float, while in the other fit $\sin^2\theta_{23}$, $|\Delta m^2_{32}|$ and the 44 systematic parameters are allowed to float. δ_{CP} is fixed at 0 in both fits. $\Delta\chi^2$ is then calculated as the difference between the minimum values of χ^2 from the two fits (χ^2 is defined in Eq. 5 in Sec. 4.1.5 in Ref. [4]).

The 2-dimensional $\Delta\chi^2$ surface for the fit to the Run 1+2+3+4a dataset is shown in Figs. 57 and 58 as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, for the normal and inverted mass hierarchies respectively. The same $\Delta\chi^2$ range is used in both colour plots to allow comparison. Profile $\Delta\chi^2$ distributions are also shown as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ for the fit to the Run 1+2+3+4a dataset in Figs. 59 and 60 respectively. Using these plots and the constant-$\Delta\chi^2$ method described in Sec. 5.4 of Ref. [4], 1-dimensional limits on the parameters can be obtained[8]. These limits are presented in Tab. 9.

2-dimensional confidence regions were constructed for the fit to the Run 1+2+3+4a dataset using the constant-$\Delta\chi^2$ method described in Sec. 4.5 of Ref. [4]. The 68% and 90% CL regions for the fit to the Run 1+2+3+4a dataset are shown in Figs. 61 and 62 for the normal and inverted mass hierarchies respectively. All confidence regions presented include the effect of systematic uncertainties.

To construct 2-dimensional confidence regions using the method of Feldman and Cousins [31], described in Sec. 4.5 of Ref. [4], one needs to construct $\Delta\chi^2_{critical}$ surfaces, in the ($\sin^2\theta_{23}$, $|\Delta m^2_{32}|$) oscillation parameter space, for the 6.3933×10^{20}-POT 3-flavour ν_μ-disappearance analysis. At each grid point, an ensemble of toy MC experiments is created with $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ set to the true values of that grid point and all 44 systematic parameters randomized (δ_{CP} is also randomized uniformly between $-\pi$ and $+\pi$). Each toy experiment is fitted twice: again in the first fit the values of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ are fixed to the true values of the grid point and the 44 systematic parameters are allowed to float, while in the second fit $\sin^2\theta_{23}$, $|\Delta m^2_{32}|$ and the 44 systematic parameters are allowed to float. This means that χ^2 is *independently* minimised for *each* toy MC experiment with respect to the systematic parameters. For each toy MC experiment, $\Delta\chi^2$ is given by the difference between the minimum values of χ^2 from the two fits (the first fit gives the "true" χ^2 and the second fit gives the "best-fit" χ^2). The value of $\Delta\chi^2_{critical}$ for a grid point is the value of $\Delta\chi^2$ such that α of the toy experiments at that grid point have $\Delta\chi^2 < \Delta\chi^2_{critical}$, where α is the desired confidence level.

In Fig. 63, we show a Monte Carlo estimation using the Feldman-Cousins method of the 2-dimensional $\Delta\chi^2_{critical}$ values used for calculation of the 68% and 90% CL confidence regions, assuming the normal mass hierarchy. The surfaces were generated using an ensemble of toy MC experiments which included both randomized systematic variations and statistical fluctuations. Slices of the 2-dimensional distributions shown in Fig. 63, are shown in Figs. 64 and 65. From these $\Delta\chi^2_{critical}$ distributions, we anticipate that the confidence regions produced using the Feldman-Cousins method will be narrower along $|\Delta m^2_{32}|$ at maximal mixing than those made with the constant-$\Delta\chi^2$ method. The Feldman-Cousins contours will also be wider in $\sin^2\theta_{23}$ for values of $|\Delta m^2_{32}|$ near the best fit. This is the same pattern that was seen by our previous analyses [2, 4] that fit for $\sin^2(2\theta_{23})$.

Using the $\Delta\chi^2$ and $\Delta\chi^2_{critical}$ surfaces presented here, the 2-dimensional Feldman-Cousins 68% and 90% CL regions were constructed and they are shown in Fig. 66. Due to the large amount of CPU time needed to generate the contours with the complete 2013 VaLOR analysis, the Feldman-Cousins confidence regions are currently shown only for the normal mass hierarchy; the inverted mass hierarchy confidence regions will be prepared at a later date. All confidence regions presented include the effect of systematic uncertainties.

Confidence intervals were also constructed using the method of Feldman and Cousins in each of $\sin^2\theta_{23}$ and Δm^2_{32} separately. To make the 1-dimensional confidence interval in $\sin^2\theta_{23}$, an ensemble of toy MC experiments is generated, and each experiment is fitted twice. In the first fit, Δm^2_{32} and all 44 systematic parameters are allowed to float, while $\sin^2\theta_{23}$ is fixed to the true input value for that experiment. In the

[8] For 1 fit parameter, the critical values of $\Delta\chi^2$ used are 1.00 (68% CL), 2.71 (90% CL), 3.84 (95% CL), 6.63 (99% CL). For 2 parameters fitted simultaneously, the critical values of $\Delta\chi^2$ used are 2.30 (68% CL), 4.61 (90% CL), 5.99 (95% CL), 9.21 (99% CL).

second fit, $\sin^2\theta_{23}$, Δm^2_{32} and all 44 systematic parameters are allowed to float. Again this means that χ^2 is independently minimised with respect to the systematic parameters for each toy MC experiment. The value of $\Delta\chi^2$ is given by the difference between the minimum values of χ^2 from the two fits for each experiment.

One issue in constructing these 1-dimensional confidence intervals concerns the choice of the true input value(s) of the "other" oscillation parameter in the toy MC experiments. For example, true input values of Δm^2_{32} must be chosen when making the toy experiments for the confidence interval in $\sin^2\theta_{23}$. One option in this case would be to use the T2K Run 1+2+3+4a best-fit value of Δm^2_{32} as the true input value for all the toy MC experiments, but this fails to account for the uncertainty in Δm^2_{32}. Alternatively, a uniform distribution of true input values of Δm^2_{32} could be used in some range, say from 2-3$\times10^{-3}$eV^2/c^4, but this ignores the fact that some values of Δm^2_{32} are preferred by the T2K Run 1+2+3+4a data to others.

In this analysis, the true input values of Δm^2_{32} are chosen using the likelihood from the 2-dimensional fit of the T2K Run 1+2+3+4a data in $\sin^2\theta_{23}$ and Δm^2_{32} and the 44 nuisance parameters. Since the value of χ^2 for this fit, as defined in equation 3 in Sec. 4.1.5 in Ref. [4], is -2 ln(likelihood), the likelihood is $e^{-\frac{\chi^2}{2}}$; this likelihood is equal to a constant times $e^{-\frac{\Delta\chi^2}{2}}$ as the best-fit value of χ^2 is constant. At each true value of $\sin^2\theta_{23}$, toy MC experiments are made with different true input values of Δm^2_{32}, and the numbers of experiments with each input value of Δm^2_{32} are chosen to be proportional to $e^{-\frac{\Delta\chi^2}{2}}$, where $\Delta\chi^2$ is the value of the 2-dimensional $\Delta\chi^2$ surface in Fig. 57 or Fig. 58 at those values of $\sin^2\theta_{23}$ and Δm^2_{32}.

The critical value of $\Delta\chi^2$ at each true value of $\sin^2\theta_{23}$ is found by combining all the toy MC experiments with that input value of $\sin^2\theta_{23}$ and different true input values of Δm^2_{32}. It is the value of $\Delta\chi^2$ such that α of the toy experiments have $\Delta\chi^2 < \Delta\chi^2_{critical}$, where α is the desired confidence level.

The 1-dimensional confidence intervals in Δm^2_{32} are constructed in a similar way. At each true value of Δm^2_{32}, the numbers of toy MC experiments with each input value of $\sin^2\theta_{23}$ are proportional to $e^{-\frac{\Delta\chi^2}{2}}$, while all of them have that value of Δm^2_{32} as input. Again each experiment is fitted twice: in the first fit, $\sin^2\theta_{23}$ and all 44 systematic parameters are allowed to float with Δm^2_{32} fixed at its true input value, whereas, in the second fit, Δm^2_{32}, $\sin^2\theta_{23}$ and all 44 systematic parameters are allowed to float. For each experiment, the value of $\Delta\chi^2$ is the difference between the values of χ^2 from the two fits. The critical value of $\Delta\chi^2$ at each true value of Δm^2_{32} is found by combining all the experiments with that input value of Δm^2_{32} and different input values of $\sin^2\theta_{23}$.

In the next section, comparisons are presented of the confidence regions obtained using different construction methods. We also compare our contours from the fit to the combined Run 1+2+3+4a dataset with recent measurements from MINOS and SuperK.

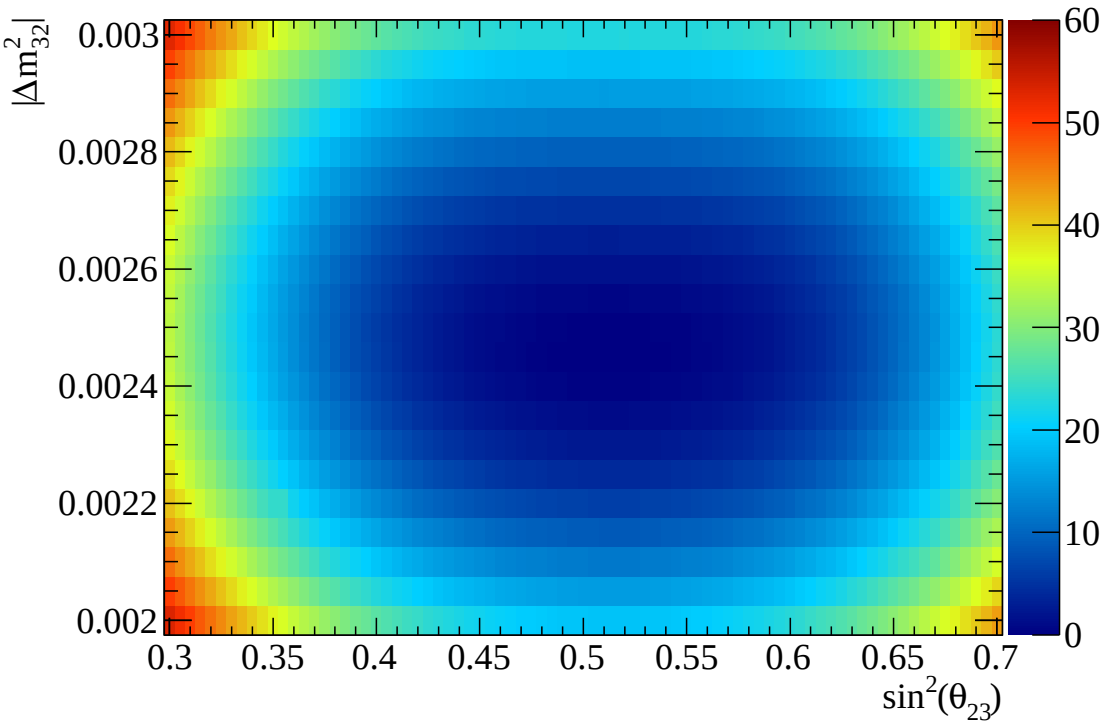


Figure 57: $\Delta\chi^2$ surface for the fit to the Run 1+2+3+4a dataset. The normal mass hierarchy is assumed.

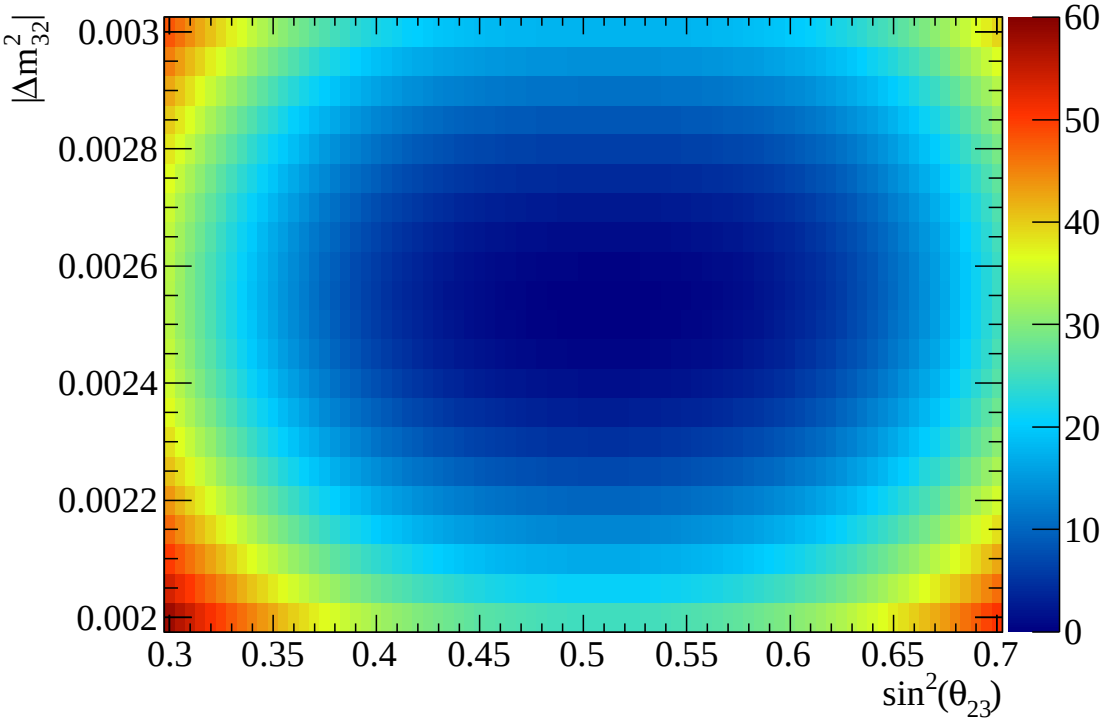


Figure 58: $\Delta\chi^2$ surface for the fit to the Run 1+2+3+4a dataset. The inverted mass hierarchy is assumed.

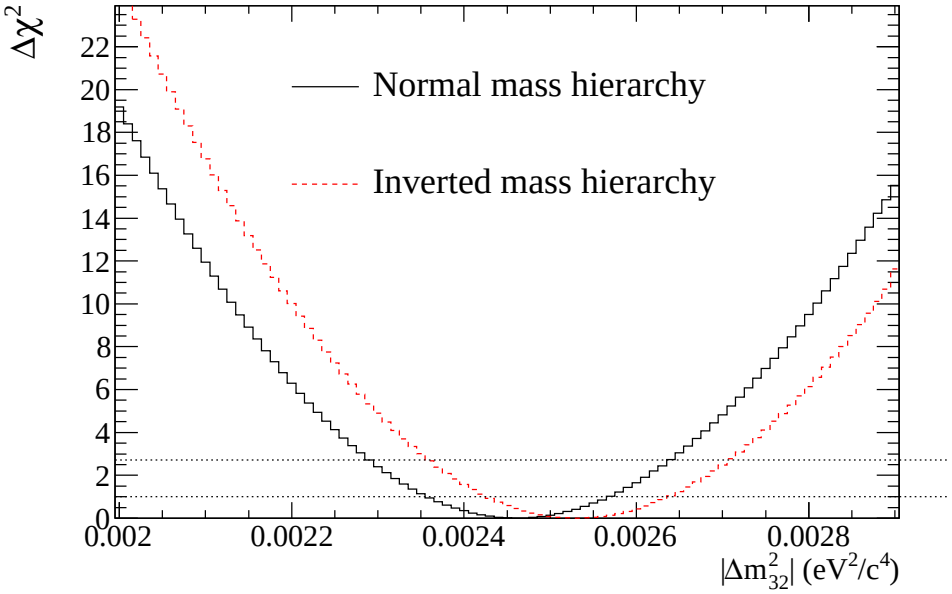


Figure 59: Profile $\Delta\chi^2$, as a function of $|\Delta m^2_{32}|$, for the fits to the Run 1+2+3+4a dataset assuming normal and inverted mass hierarchy. The 68% and 90% constant-$\Delta\chi^2$ values are shown as horizontal dotted lines.

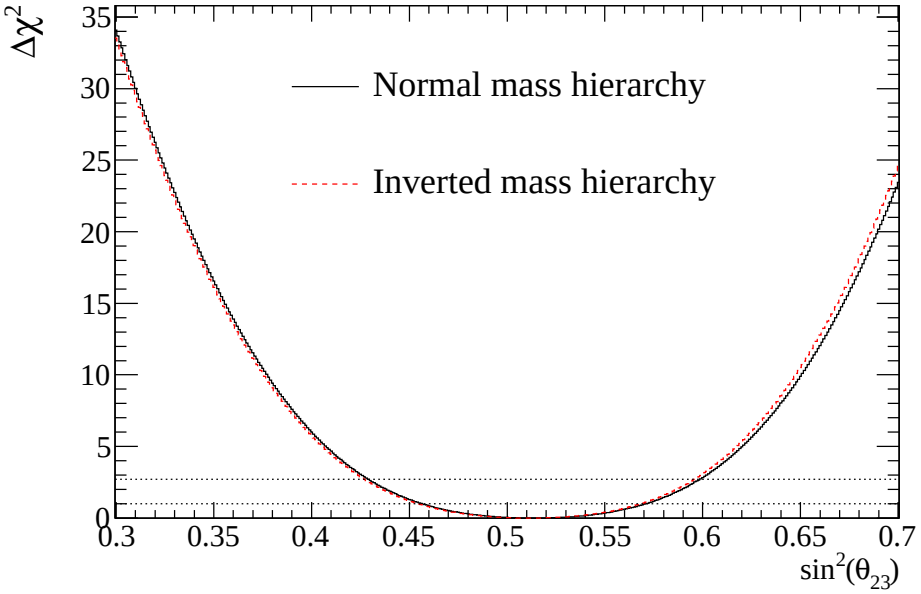


Figure 60: Profile $\Delta\chi^2$, as a function of $\sin^2\theta_{23}$, for the fits to the Run 1+2+3+4a dataset assuming normal and inverted mass hierarchy. The 68% and 90% constant-$\Delta\chi^2$ values are shown as horizontal dotted lines.

Analysis	Run period	Limit	$\sin^2\theta_{23}$		Δm^2_{32} / $\cdot 10^{-3}$ eV^2/c^4	
2013 3-flavour Normal	Run 1+2+3+4a	68%	> 0.456	< 0.572	> 2.35	< 2.57
2013 3-flavour Normal	Run 1+2+3+4a	90%	> 0.428	< 0.599	> 2.29	< 2.64
2013 3-flavour Inverted	Run 1+2+3+4a	68%	> 0.454	< 0.569	> 2.42	< 2.64
2013 3-flavour Inverted	Run 1+2+3+4a	90%	> 0.426	< 0.596	> 2.35	< 2.71

Table 9: Summary of one-dimensional limits on $\sin^2\theta_{23}$ and Δm^2_{32}, for the oscillation fits to the Run 1+2+3+4a dataset. The limits are calculated using the constant-$\Delta\chi^2$ method.

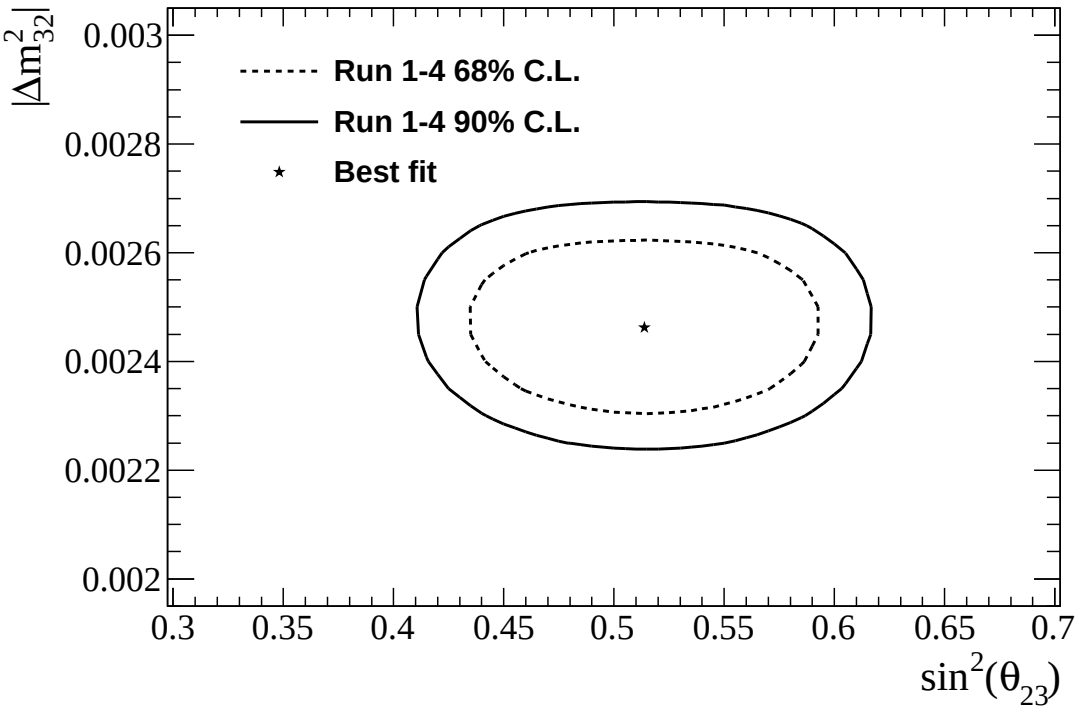


Figure 61: 68% CL and 90% CL allowed region (including systematics), constructed with the constant-$\Delta\chi^2$ method, for the fit to the Run 1+2+3+4a dataset. The best-fit point is also shown. The normal mass hierarchy is assumed.

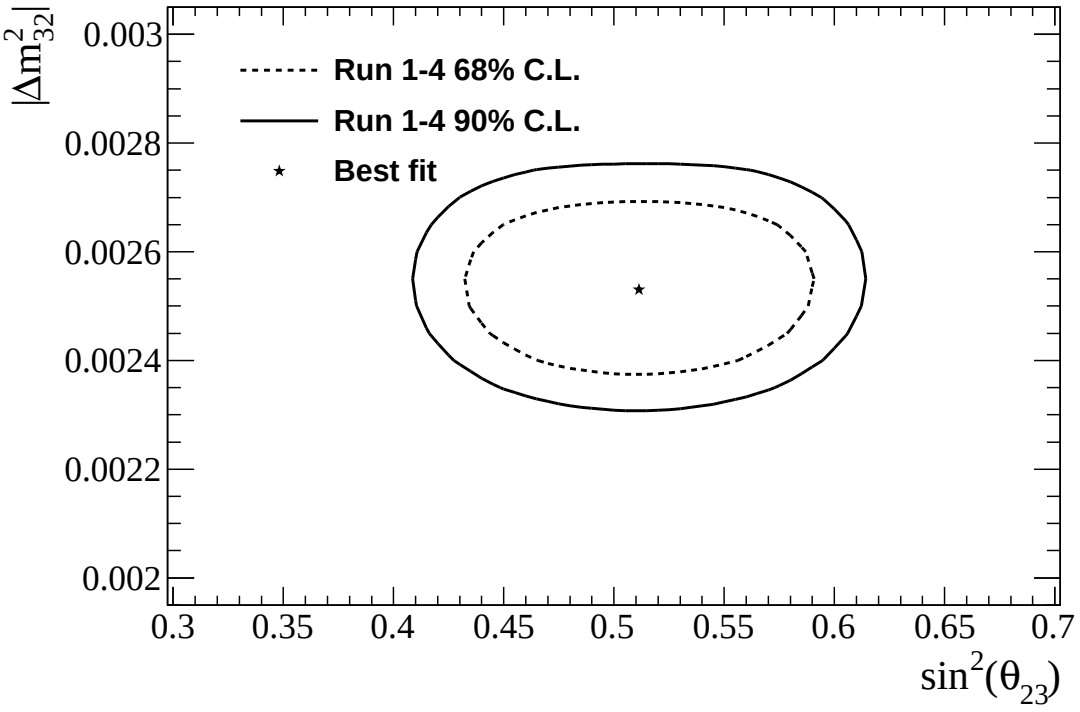


Figure 62: 68% CL and 90% CL allowed region (including systematics), constructed with the constant-$\Delta\chi^2$ method, for the fit to the Run 1+2+3+4a dataset. The best-fit point is also shown. The inverted mass hierarchy is assumed.

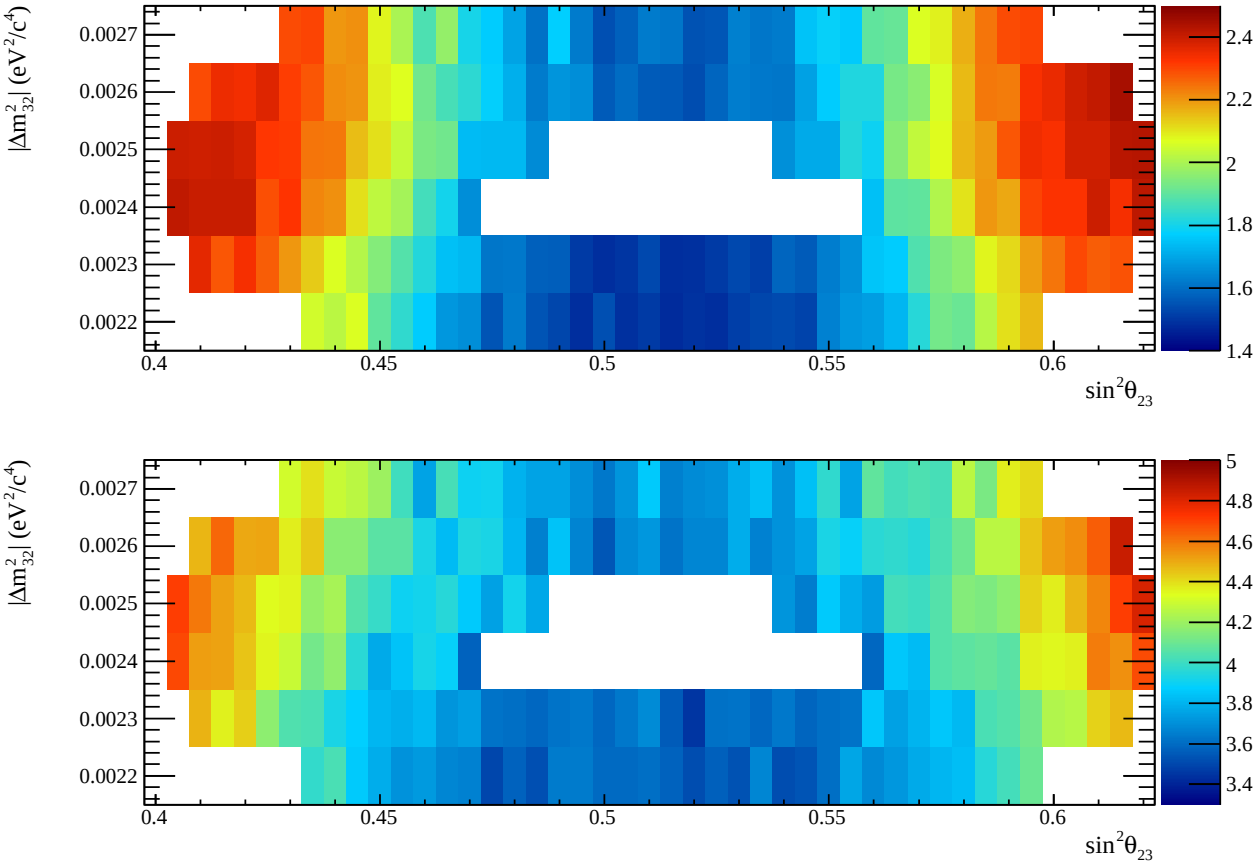


Figure 63: Monte Carlo estimation, according to the method of Feldman and Cousins [31], of $\Delta\chi^2_{critical}$ values $\Delta\chi^2_{68}$ (left) and $\Delta\chi^2_{90}$ (right) as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, for calculation of the 68% and 90% CL confidence regions for the 6.3933 $\times 10^{20}$-POT 3-flavour ν_μ-disappearance analysis. The surfaces were generated from an ensemble of toy MC experiments including both randomized systematic variations and statistical fluctuations. As expected from [31], the values of $\Delta\chi^2_{critical}$ decrease as the limit of the model at $\sin^2\theta_{23} \approx 0.513$ is approached. The normal mass hierarchy is assumed. It should be noted that $\Delta\chi^2_{critical}$ is not calculated for every combination of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, due to us being efficient with computing time.

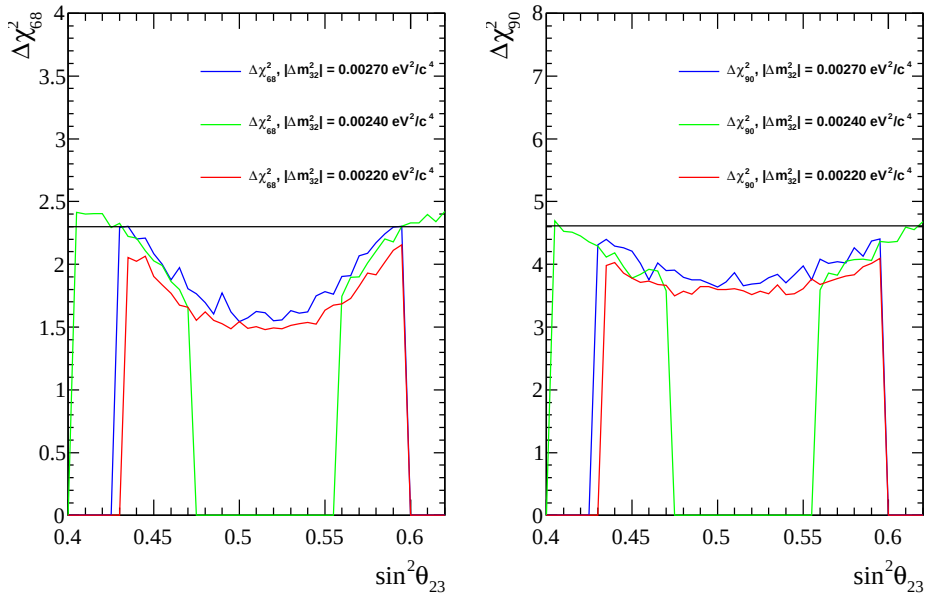


Figure 64: Monte Carlo estimation, according to the method of Feldman and Cousins, of $\Delta\chi^2_{critical}$ values $\Delta\chi^2_{68}$ (left) and $\Delta\chi^2_{90}$ (right), as a function of $\sin^2\theta_{23}$ for 3 different values of $|\Delta m^2_{32}|$ (2.2×10^{-3} eV$^2/c^4$, 2.4×10^{-3} eV$^2/c^4$ and 2.7×10^{-3} eV$^2/c^4$). The $\Delta\chi^2_{critical}$ values were calculated from an ensemble of toy-MC experiments, for an exposure of 6.3933×10^{20} POT, including both randomized systematic variations and statistical fluctuations. For comparison, the canonical value used in the constant-$\Delta\chi^2$ method is also shown as a horizontal line. The normal mass hierarchy is assumed. It should be noted that $\Delta\chi^2_{critical}$ is not calculated for every combination of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, due to us being efficient with computing time.

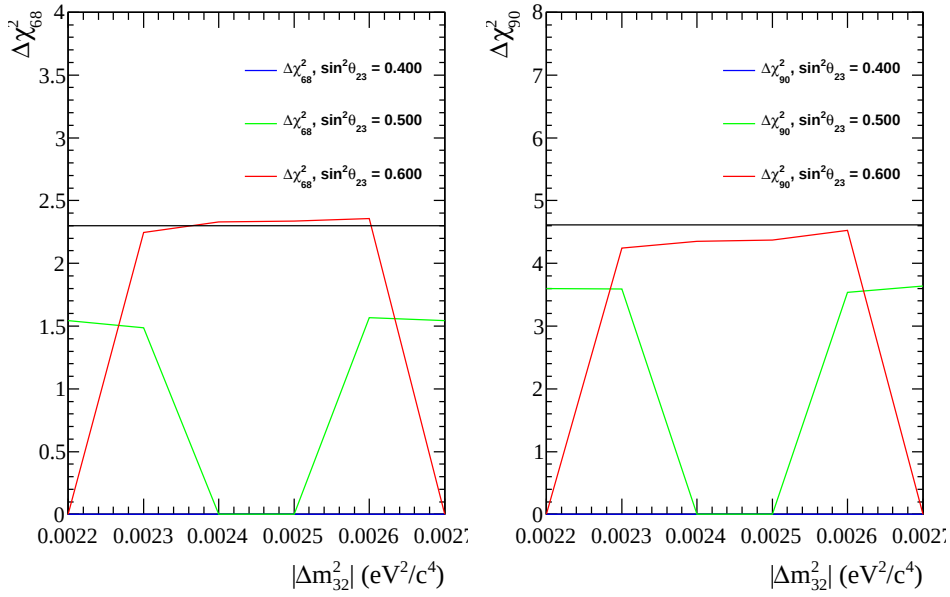


Figure 65: Monte Carlo estimation, according to the method of Feldman and Cousins, of $\Delta\chi^2_{critical}$ values $\Delta\chi^2_{68}$ (left) and $\Delta\chi^2_{90}$ (right), as a function of $|\Delta m^2_{32}|$ for 3 different values of $\sin^2\theta_{23}$ (0.40, 0.50 and 0.60). The $\Delta\chi^2_{critical}$ values were calculated from an ensemble of toy MC experiments, for an exposure of 6.3933×10^{20} POT, including both randomized systematic variations and statistical fluctuations. For comparison, the canonical value used in the constant-$\Delta\chi^2$ method is also shown as a horizontal line. The normal mass hierarchy is assumed. It should be noted that the *holes* in the surface are due to us being efficient with computing time.

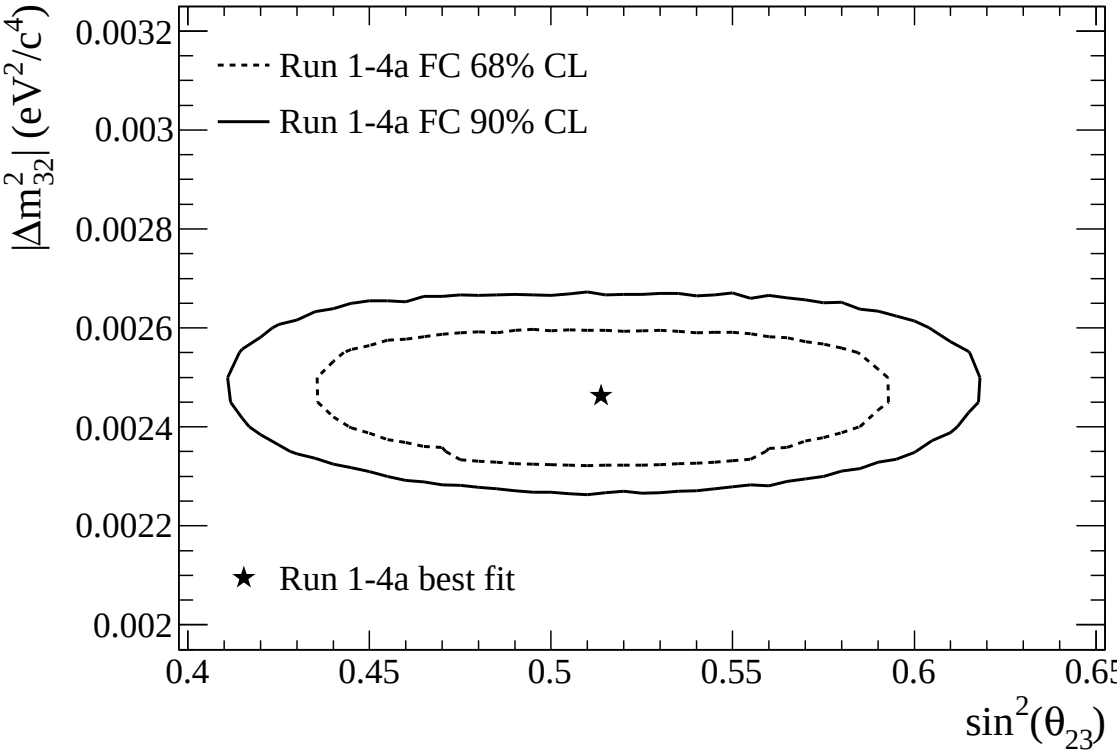


Figure 66: 68% CL and 90% CL allowed region (including systematics), constructed with the method of Feldman and Cousins, for the fit to the Run 1+2+3+4a dataset. The best-fit point is also shown. The normal mass hierarchy is assumed. Smoother FC contours we be prepared for the final result.

7.5. Confidence region comparisons

The Run 1+2+3+4a 2-dimensional confidence regions obtained using the constant-$\Delta\chi^2$ method, for the normal and inverted mass hierarchy assumptions, are compared in Fig. 67. The effect of changing the mass hierarchy is large, and so it is important that both contours be presented together.

The Run 1+2+3+4a 2-dimensional confidence regions obtained using the constant-$\Delta\chi^2$ and Feldman-Cousins methods are compared in Figs. 68 for the normal mass hierarchy assumption. The confidence regions produced using the Feldman-Cousins method are narrower along $|\Delta m^2_{32}|$ at maximal disappearance than those made with the constant-$\Delta\chi^2$ method. The Feldman-Cousins contours are also wider in $\sin^2\theta_{23}$ for values of $|\Delta m^2_{32}|$ near the best fit.

Comparisons were produced between our Run 1+2+3+4a result MINOS (2013) [36], SuperK atmospherics (3-flavour zenith-angle analysis) (2012) [37], and T2K (Run 1+2+3) [8]. The comparison is shown in Fig. 69. A second comparison with other results is shown in Fig. 70, between our Run 1+2+3+4a result MINOS (2013) [35], SuperK atmospherics (both the 2-flavour L/E and 3-flavour zenith-angle analyses) (2012) [37], and T2K (Run 1+2+3) [8]. In this plot, the T2K Run 1+2+3+4a fit result has been converted from $\sin^2\theta_{23}$ to $\sin^2(2\theta_{23})$ by profiling the octant. This is the same method used by SuperK in their zenith angle contour, and, if this method had been applied in the T2K Run 1+2+3 result, it should have closely matched the T2K Run 1+2+3 $\theta_{23} \geq \pi/4$ contour. T2K now has the world's best limit on θ_{23}.

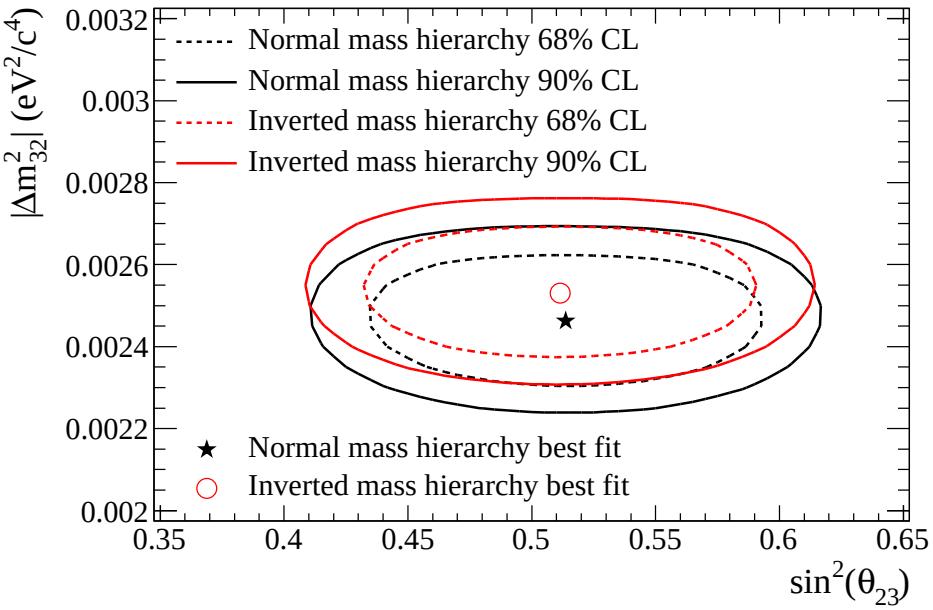


Figure 67: Comparison between the 68% CL and 90% CL allowed regions obtained using the constant-$\Delta\chi^2$ method for the fit to the Run 1+2+3+4a dataset, for normal and inverted hierarchy assumptions.

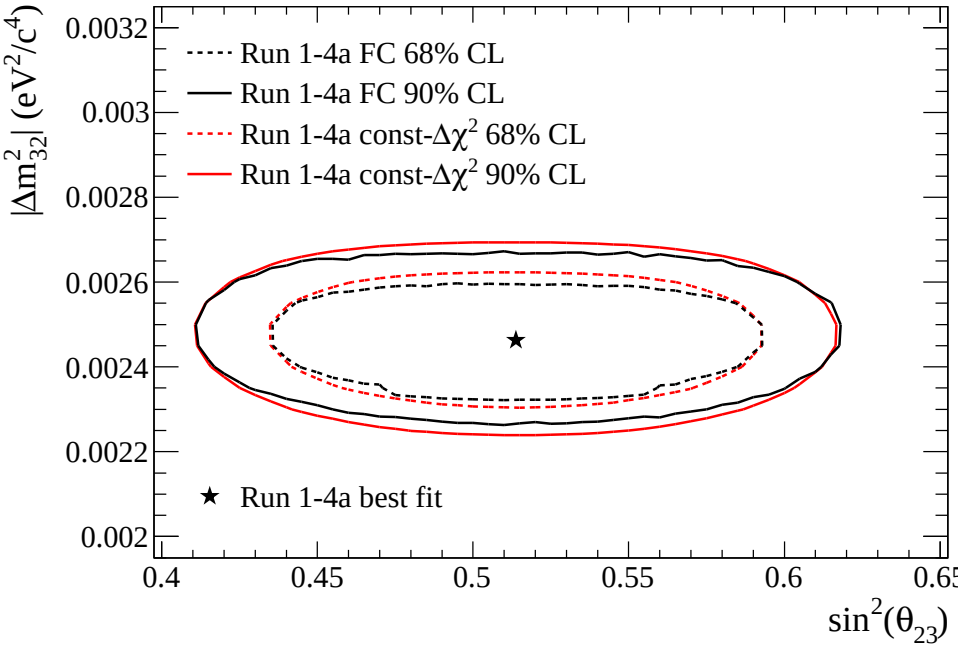


Figure 68: Comparison between the 68% CL and 90% CL allowed regions obtained using the constant-$\Delta\chi^2$ and Feldman-Cousins methods for the fit to the Run 1+2+3+4a dataset. The normal mass hierarchy is assumed. Smoother FC contours we be prepared for the final result.

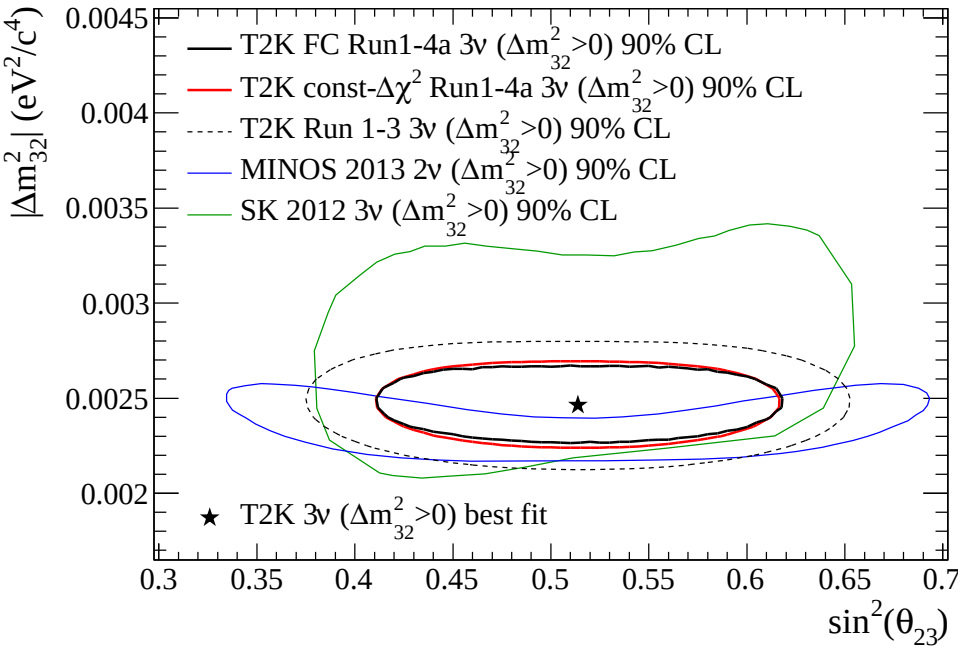


Figure 69: Comparison between 90% CL allowed regions from MINOS [36], SuperK atmospherics [37], T2K Run 1+2+3 [8], and the 90% CL regions obtained from the VaLOR fit to the Run 1+2+3+4a dataset. Contours obtained using both the constant-$\Delta\chi^2$ and Feldman-Cousins methods are shown. The normal mass hierarchy is assumed for all results.

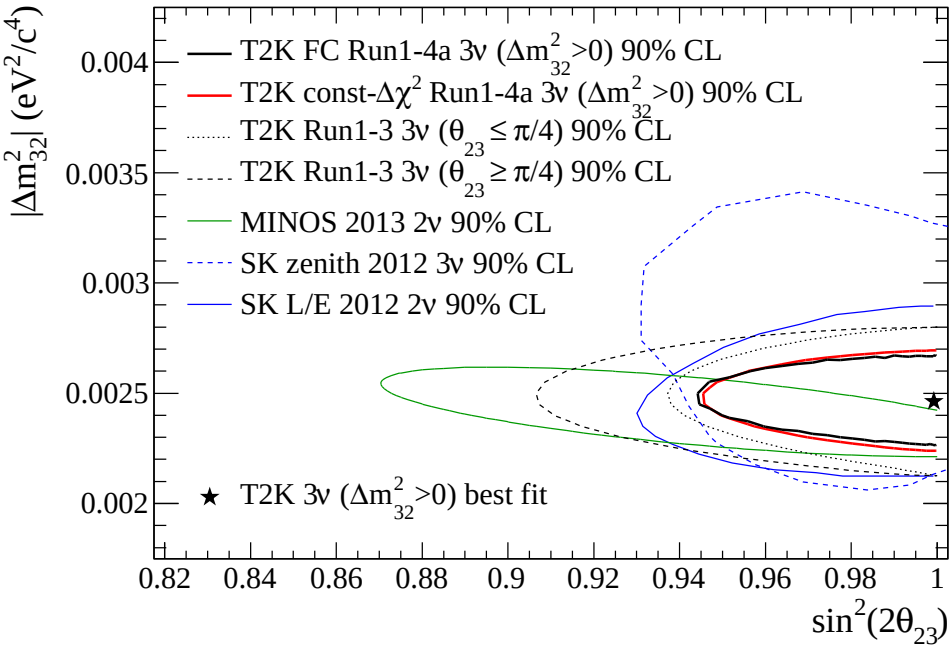


Figure 70: Comparison between 90% CL allowed regions from MINOS [35], SuperK atmospherics [37], T2K Run 1+2+3 [8], and the 90% CL regions obtained from the VaLOR fit to the Run 1+2+3+4a dataset. Contours obtained using both the constant-$\Delta\chi^2$ and Feldman-Cousins methods are shown. The normal mass hierarchy is assumed for all results.

8. Summary

In this technical note, we have presented the results of a 3-flavour ν_μ-disappearance analysis, performed by the Valencia-Lancaster-Oxford-Rutherford (VaLOR) group, on the combined Run1+2+3+4a (6.3933×10^{20} POT) dataset. This analysis is an update of the work presented in Refs. [1], [2], [3] and [4] with improved methods and an enlarged dataset.

This analysis predicts 421.76 ± 23.33 (syst) 1 μ-like ring events in SuperK in the absence of any oscillation, but only 115 were observed. The observed deficit has a strong energy dependence; the ratio of observed to expected, under the no oscillation hypothesis, is $\sim$27% $<$ 0.5 GeV, $\sim$10% between 0.5 and 1 GeV and $\sim$73% $>$ 1 GeV.

A ν_μ-disappearance analysis was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter. A total of 44 energy scale, neutrino flux, neutrino cross section, efficiency, initial-state nuclear environment and final-state rescattering, and oscillation parameter systematics were taken into account. The effect of these systematic uncertainties on the expected number of 1 μ-like ring events, on their reconstructed energy spectrum and on our oscillation measurement were presented in detail.

A 3-flavour ν_μ-disappearance fit to the observed reconstructed energy spectrum of 1 μ-like ring events of the combined Run 1+2+3+4 dataset, allowing all nuisance parameters to float yields $|\Delta m^2_{32}| = 2.463\times10^{-3}$ eV^2/c^4 and $\sin^2\theta_{23} = 0.514$ as the best-fit values (p-value = 0.88). The fit assumed δ_{CP}=0 and the normal mass hierarchy. A fit assuming the inverted mass hierarchy yields $|\Delta m^2_{32}| = 2.530\times10^{-3}$ eV^2/c^4 and $\sin^2\theta_{23}$ = 0.511 as the best-fit values (p-value = 0.89).

Confidence regions obtained using both the constant-$\Delta\chi^2$ method and the method of Feldman and Cousins were presented. Using the profile $\Delta\chi^2$ distributions obtained from the fits, the 90%(68%) CL allowed region from this analysis can be summarized as 2.30×10^{-3} (2.35×10^{-3}) eV^2/c^4 $< |\Delta m^2_{32}| <$ 2.63×10^{-3} (2.57×10^{-3}) eV^2/c^4 and 0.433 (0.456) $< \sin^2\theta_{23} <$ 0.594 (0.572) for the normal mass hierarchy, and 2.37×10^{-3} (2.42×10^{-3}) eV^2/c^4 $< |\Delta m^2_{32}| <$ 2.70×10^{-3} (2.64×10^{-3}) eV^2/c^4 and 0.431 (0.454) $< \sin^2\theta_{23} <$ 0.591 (0.569) for the inverted mass hierarchy.

Numerous comparisons were made of the best-fit values and spectra for different running periods. Comparisons were presented of the confidence regions obtained by using different construction methods and by fits to different datasets. Finally, comparisons of our contours with recent measurements from MINOS and SuperK were also made.

References

[1] C.Andreopoulos, T.Dealtry, J.Dobson, N.Grant, M.Haigh, J.Ilic, G.Pearce and A.Weber, T2K 3.23×10^{19}-POT Muon-Neutrino Disappearance Analysis, T2K-TN-036(v4), http://www.t2k.org/docs/technotes/036

[2] C.Andreopoulos, T.Dealtry, J.Dobson, N.Grant, M.Haigh, J.Ilic, G.Pearce and A.Weber, T2K 1.431×10^{20}-POT Muon-Neutrino Disappearance Analysis T2K-TN-064(v3), http://www.t2k.org/docs/technotes/064

[3] C.Andreopoulos, T.Dealtry, L.Escudero and N.Grant, T2K 1.431×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-087(v1), http://www.t2k.org/docs/technotes/087

[4] C.Andreopoulos, T.Dealtry, L.Escudero and N.Grant, T2K 3.010×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-141(v8), http://www.t2k.org/docs/technotes/141

[5] C.Andreopoulos et al., T2K 3.010×10^{20}-POT Joint 3-Flavour Oscillation Analysis T2K-TN-154(v3), http://www.t2k.org/docs/technotes/154

[6] C.Andreopoulos, T.Dealtry, L.Escudero and N.Grant, T2K 6.3933×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis: Supplementary material on the effect of systematics, T2K-TN-155 supplement, http://www.t2k.org/docs/technotes/155

[7] T2K Collaboration (K.Abe et al.), First Muon-Neutirno Disappearance Study with an Off-Axis Beam, Phys.Rev.D85:031103, 2012; arXiv:1201.1386

[8] T2K Collaboration (K.Abe et al.), Measurement of Neutrino Oscillation Parameters from Muon Neutrino Disappearance with an Off-axis Beam, arXiv:1308.0465 [hep-ex], 2013.

[9] Y.Hayato, A neutrino interaction simulation program library NEUT, Acta Phys.Polon.B40:2477-2489,2009.

[10] Beam WG, Flux tuning (11b v3.2), http://www.t2k.org/beam/NuFlux/FluxRelease/11brelease/11btuned-v3.2

[11] M.Otani and J.Kameda, Super-K events and Updated systematic errors for ν_μ disappearance analysis with T2K 1.431×10^{20} POT T2K-TN-065(v6), http://www.t2k.org/docs/technotes/065

[12] P. de Perio et al., NEUT nuclear effects (FSI) T2K-TN-033(v2.0), http://www.t2k.org/docs/technotes/033

[13] P. de Perio et al., Cross section parameters for the 2012a oscillation analysis T2K-TN-108(v1.3), http://www.t2k.org/docs/technotes/108

[14] P. de Perio et al., Implementation of the NIWG cross-section parametrization, T2K-TN-113(v1.2)

[15] P. de Perio et al., Pion Hadronic Secondary Interactions in Super-Kamiokande, T2K-TN-105(v1.1) http://www.t2k.org/docs/technotes/105

[16] Y. Ashie et al., Phys. Rev. Lett. 93, 101801 (2004); Y. Ashie et al., Phys. Rev. D71, 112005 (2005); J. Hosaka et al., Phys. Rev. D 74, 032002 (2006).

[17] T2KSK, http://www.t2k.org/t2ksk/code/sk11c_mc/sampleinfo.

[18] Beam group, http://www.t2k.org/beam/NuFlux/FluxRelease/11arelease/11atuned-v2.3.

[19] M. Wilking, Status of the 2013 ND280 ν_μ analysis, http://www.t2k.org/asg/meeting/20130411/nd280numustatus/vie

[20] M. Hartz, A. Kaboth, and K. Mahn, Constraining the Flux and Cross Section Models with Data from the ND280 Detector for the 2013 Oscillation Analysis, T2K-TN-166(v3.0) http://www.t2k.org/docs/technotes/166.

[21] J. Kameda, Updated study of the systematic error in ν_μ disappearance analysis from Super-Kamiokande, T2K-TN-159(v2), http://www.t2k.org/docs/technotes/159.

[22] J. Hignight et al., Super-Kamiokande events and data quality studies for T2K Run4 T2K-TN-148(v2r3), http://www.t2k.org/docs/technotes/148.

[23] C. Bronner et al., Run1-4 p-theta electron neutrino appearance analysis, T2K-TN-161(v2.6), http://www.t2k.org/docs/technotes/161

[24] Y. Nishimura, J. Hignight, and S. Nakayama ν_e appearance analysis of RUN1-4 data with neutrino energy spectrum, T2K-TN-162(v2.2), http://www.t2k.org/docs/technotes/162

[25] N Abgrall et al. Measurements of Cross Sections and Charged Pion Spectra in Proton-Carbon Interactions at 31 GeV/c. Phys.Rev., C84:034604, 2011; N. Abgrall et al. Measurement of production properties of positively charged kaons in proton-carbon interactions at 31 GeV/c. arXiv:hep-ex/1112.0150, 2011.

[26] J. Beringer et al. (Particle Data Group), Phys. Rev. D86, 010001 (2012)

[27] Lorena Escudero Sanchez for the VaLOR group, http://www.t2k.org/asg/oagroup/meeting/20121219/VaLOR

[28] Tom Dealtry for the VaLOR group, http://www.t2k.org/asg/meeting/2012/20121015/valornumudisapp/view

[29] G. Cowan (1998) Statistical data analysis, Oxford University Press, ISBN 0-19-850155-2.

[30] F.James, M.Roos, Minuit: A System for Function Minimization and Analysis of the Parameter Errors and Correlations, Comput.Phys.Commun. 10:343 (1975)

[31] G.J. Feldman and R.D. Cousins, A Unified approach to the classical statistical analysis of small signals, Phys.Rev.D57:3873-3889,1998; physics/9711021

[32] R.D. Cousins and V.L.Highland, Nucl.Instr. and Meth. A320:331035, 1992.

[33] K. McFarland for the T2K ν_μ-disappearance contour committee (C.Andreopoulos, N.Grant, J.Kameda, D.Karlen, E.Kearns, K.McFarland, M.Otani, M.Zito) Report to ASG conveners meeting, 17 May 2011, http://www.t2k.org/asg/contour/ASG-Conveners-Report/view

[34] R.Nichol, Talk at Neutrino 2012

[35] P. Adamson, et al., Measurement of Neutrino and Antineutrino Oscillations Using Beam and Atmospheric Data in MINOS, arXiv:1304.6335 [hep-ex], 2013

[36] A. Radovic, Talk at DPF 2013

[37] Y.Itow, Talk at Neutrino 2012

[38] G.L.Fogli, E.Lisi et al., Evidence of theta(13)>0 from global neutrino data analysis, Phys.Rev.D84:053007, 2011; arXiv:1106.6028 [hep-ph]